

When Two Threads Are Better Than One: How Matching-Engine Transactions Cause Long Tails in Market-Data Handlers

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Abstract

The prevailing view in high-frequency-trading (HFT) system design is that the entire hot path — market-data decode, order-book application, strategy evaluation, order construction — should run on a single-threaded event loop, because each additional thread hop adds wall-clock latency. This paper puts two questions to that view on CME MDP3 market data (NQ front-month, channel 318, regular trading hours, 2025–2026: 276 sessions, 3512 thirty-minute windows, 3.3×10^9 packets). First, what produces the queueing tail behind a single-threaded receiver: the packet rate, the clustering of arrivals, or the number of messages per packet? Second, is a servicing chain cut into two or more stages on separate threads better than one thread, and when?

On the first question: the tail is produced by the timing of the matching engine’s transactions and, at the service times production receivers run at, by the number of messages each packet carries. It is not produced by the rate: utilisation is below 0.02 at the service times an HFT hot path occupies, and a window with several times the packet rate has the same tail relative to the service time. The message count per packet matters where production receivers run. At about $7\mu\text{s}$, just under the publisher’s packet spacing, decoding the messages of a long packet one after another supplies two thirds of the simulated p_{99} excess at $8\mu\text{s}$; on the live receiver the decode cost of a message’s position in its packet reproduces 71–80% of the observed p_{99} on four of six streams; and a long packet holds the thread while the packets behind it queue; from $16\mu\text{s}$ up it adds a few percent. Service times are also not constant: on a live receiver, messages with nothing queued ahead of them still reach twice the floor at p_{99} , and with packets arriving $7.5\mu\text{s}$ apart any such slow service can start a queue; the simulation does not model this and its source is left open. The tail is not produced by the exchange’s packetisation: grouping messages by matching-engine transaction shows that 98.3% of transactions occupy one packet, that the packets of a split transaction arrive $15\mu\text{s}$ apart, and that 99.9% of the gaps shorter than the feed’s $7.5\mu\text{s}$ tight-gap floor separate different transactions. Redrawing transaction times uniformly removes 75–98% of the tail; from $16\mu\text{s}$ up, decoupling transaction sizes from their timing, merging each transaction into one packet, or spreading its packets leaves the tail within $\pm 15\%$. The timing acts through two channels whose weight depends on the service time. At $16\mu\text{s}$ the tail is carried mainly by consecutive transactions that the publisher sends back to back, about $7.5\mu\text{s}$ apart, — a stream with the same gaps in random order keeps 79% of it — and at $64\mu\text{s}$ and above by runs of transactions, which that permutation destroys (37% and 23% survive). The runs are the self-excitation a Hawkes fit measures (branching ratio 0.795 on transaction starts). One stage upstream, the exchange’s own timestamps show where the tight gaps come from: on the matching engine’s clock 17% of consecutive transactions are less than

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$7.5\mu\text{s}$ apart, against 1% of packets on the publisher’s clock, and the delay from engine to packet grows about tenfold during engine bursts, from a median of 86 to $912\mu\text{s}$. CME’s market-data publisher is a queue with a service time of about $7.5\mu\text{s}$ per packet, the feed’s tight-gap floor is its output spacing, and a receiver is the second stage of a tandem that starts at the exchange. Why the engine bursts — the engine gaps reach $0.18\mu\text{s}$, which fits many participants racing on the same signal — is a question about the market, not about the receiving software, and is left open. All results are for NQ.

On the second question: yes above a threshold and no below it, and the threshold is a property of the feed (measured on NQ). Below the tight-gap floor there is no queue to shorten and a cut adds one $1.7\mu\text{s}$ hop to every message; above it, one balanced cut removes most of the p_{99} excess for one hop on the median, and pays on 99.7% of the 3512 windows at $10\mu\text{s}$ and on every window from $16\mu\text{s}$. Only the slowest stage matters: a cut that does not shorten the largest stage gains nothing. With the same N cores and every cost charged, a pool that dispatches whole packets to N servers has median $T + 2h$ at any N and resequences almost nothing, so for a stage that carries no state it is level with a two-stage cut and better than a cut into three or more; the order-book stage carries state and cannot be dispatched. At the production operating point of the reference receiver, about $7\mu\text{s}$, the chain should not be split; there the levers are the per-message decode cost and the spread of service times, because a stage cut does not shorten the decode of a long packet. We give the analytic framework — a burst-limit throughput identity and an exact pathwise reduction of the deterministic-service tandem to a single bottleneck server — test it on the corpus, and compare the arrival statistics against an independent measurement on a live production receiver seven months later, on three instruments; the clustering statistics agree and the production decoder’s service floor coincides with the service time at which the simulated tail appears. Code is in the KASPAR-HFT repository; the packet captures are not redistributable.

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1 Introduction

1.1 The single-thread event loop against the multi-actor tandem

The design of the ingress-to-order pipeline in a high-frequency trading system is a choice between two archetypes: the single-threaded event loop that processes every step of every message on one core (SBE decode, book application, strategy evaluation, order construction, all under one call chain, taking a total service time we denote T), and the multi-actor pipeline — a *tandem* of N servers in series — in which those steps are split across N separate actors, each on its own thread with its own mailbox, performing its part of the work and handing the message to the next across one of $N - 1$ explicit hops whose per-hop wall-clock cost we denote h . The low-latency HFT community is predominantly in the single-thread camp, and single-threaded is generally regarded as the safe choice. The argument cites the cost of a mailbox hop (a mailbox enqueue, a cache-line transfer between the sender’s core and the receiver’s core, and the cost of scheduling or spinning the receiver) and holds that $(N - 1)h$ adds to every message’s end-to-end latency, so that each additional stage is a fixed cost that no throughput or tail benefit recovers.

That argument holds when the single-server throughput $1/T$ exceeds the peak arrival rate. On CME MDP3 the arrival process is strongly self-exciting: packets arrive in Hawkes-clustered bursts [24, 27, 28]. The exchange publishes each matching-engine event — a transaction — as, almost always, one packet, so the packet stream relays the timing of transactions without adding clustering of its own (Section 4.3). During such bursts the instantaneous arrival rate exceeds the single-thread event loop’s service rate $1/T$, and cluster-induced queueing produces a tail one to two orders of magnitude larger than h . The paper’s central empirical observation, on 276 sessions (3512 thirty-minute windows) of NQ front-month packet arrivals, is that this cluster tail dominates end-to-end latency: the 99th-percentile latency (p_{99}) of a constant-service queue driven by real packet arrivals is 2.2 times the service floor at the corpus-median window for a $16\mu\text{s}$ task and 4.6 times for a $32\mu\text{s}$ task, and 2.5 and 6.1 times at the 95th-percentile window. Under a rate-matched Poisson null of the same arrival count the p_{99} still equals the median at $16\mu\text{s}$ and is only 1.3 times it at 32, against the Hawkes arm’s 4.6; over this band the tail is almost entirely a clustering effect.

The question is then whether splitting the pipeline into $N > 1$ stages, each an independent actor on its own core with per-stage service T/N , reduces the tail, and whether the additional median cost $(N - 1)h$ is small relative to the tail reduction. Neither architecture is better unconditionally. Both conditions hold above a service time that depends on the feed’s arrival statistics and fail below it; the paper measures that threshold on one feed, and its recommendation is that designers measure it on theirs.

1.2 Prior recommendations for single-thread hot paths

The single-thread rule has a lineage in the low-latency-systems community, summarised here in reverse chronological order. It has three layers: the HFT-specific codification of the last decade, the general low-latency-systems and event-driven-server culture from which HFT inherited it, and one prominent published counter-argument that the field discarded.

Modern HFT-community codification (2010–present). The founding statement of the current HFT-community formulation is Thompson’s Single Writer Principle [38] — “a single-writer design consistently outperforms a ‘properly’ lock-free or wait-free multi-writer design on most realistic workloads” — motivated by and applied in the LMAX Exchange architecture [39, 40]. Rigtorp’s widely cited low-latency tuning guide [41] formalises the operational rule (“run a single thread in

SCHED_OTHER per core and use busy waiting/polling”), and Chronicle Software’s public engineering materials [42] adopt the same architecture for market-data connectors (“within a single event loop . . . busy-spinning and affinity locks”), pinning Chronicle Queue’s design to a single writer per queue. The Aeron messaging library [43] inherits the same lineage. The academic side is thinner, but Kwan et al. [44] codify the pattern as the standard low-latency-HFT design pattern, and recent practitioner posts [45] restate it explicitly (“all critical trading logic runs in a single-threaded event loop, eliminating context switches and lock contention entirely”). Inside HFT specifically, the same recommendation has been reinforced by two decades of private prop-trading arms-race practice among the Chicago desks (Getco, Jump, Optiver, DRW; late 2000s), reported second-hand in industry retrospectives but without an open-literature reference.

Older event-driven-server lineage (1995–2010). HFT did not invent this rule; it inherited it from the event-driven-server culture. The single-threaded event loop as an architectural pattern dates at least to Schmidt’s Reactor pattern [46] in the mid-1990s. It was sharpened by Kegel’s C10K essay [47], which named the OS thread-per-connection model as the bottleneck of scalable network servers and argued for event-driven designs. The pattern went mainstream in production with Sysoev’s nginx web server [48], whose single-threaded worker process is explicitly justified as context-switch avoidance, and later with Dahl’s Node.js runtime [49], which pushed the single-threaded event loop into ordinary application code.

Pro-pipelining minority — outside HFT. A legitimate multi-decade pro-pipelining tradition exists in the datacenter and networking-systems literature; it has been the field’s non-single-thread minority throughout the period covered by the previous two paragraphs. The earliest prominent published argument is the Staged Event-Driven Architecture (SEDA) of Welsh et al. [50], which advocated exactly the multi-stage pipeline of independently scheduled event handlers that the present paper’s tandem design revives. Welsh’s 2010 retrospective [51] is often cited as a walk-back to single-thread, but it is actually a hedged pro-pipelining recommendation: it advocates decoupling stages from thread pools, grouping multiple stages within a single thread-pool domain where latency is critical, and putting a separate thread pool and queue in front of any stage with long or nondeterministic runtime — effectively the present paper’s design rule in different vocabulary. Beyond SEDA the pro-pipelining tradition includes the Click Modular Router [52] for staged packet processing; Naiad [53] for low-latency dataflow; Snap [54] for Google’s multi-engine network stack; and the datacenter microsecond-tail-latency scheduler line — Shinjuku [55], Shenango [56], Caladan [57], Perséphone [58] — which argues for centralised multi-core dispatch specifically to control the microsecond-scale tail. Perséphone in particular is targeted at “wide service-time distributions,” the closest existing framing to our arrival-law claim; but the datacenter line frames the problem as heavy-tailed *service*, not clustered *arrivals*. DPDK explicitly supports both run-to-completion and pipeline modes and its documentation recommends pipeline mode “if some set of packets require longer to process” [59]. Inside HFT specifically the pro-pipelining position is rare in the open literature; a Chinese patent [60] describing a pipelined FAST market-data decoder is one commercial instance.

None of the pro-pipelining sources above ties the design recommendation to Hawkes / self-exciting / near-critical arrival dynamics. The closest theoretical hooks are the MMPP-input tandem work of Kim and Chydzinski [16], discussed further in Section 1.4, and the batch-arrival tandem phase-transition analysis of Vinnikov [61]. The present paper’s mechanism, that the recommendation reverses when arrivals are near-critically self-exciting, is one we have not found in prior work.

The present author’s earlier work belongs on this list with a qualification.¹ An earlier paper by the present author [36] did not argue for the single-thread principle in its own right. That paper addresses a different objection: the actor model had been considered unsuitable for HFT because of asynchronous-dispatch overhead, and **fast_send** is presented there as the mechanism that removes that overhead for co-located actors, making the actor model usable inside a servicing chain that the prevailing design placed on one thread. Its reference architecture (reader to buffer by asynchronous **send**; decode, book and signal chained by **fast_send** on one thread; signal to order manager by asynchronous **send**) takes the single-thread servicing chain as given and does not compare it against more asynchronous topologies. The present paper revises that design context, not the **fast_send** mechanism: on Hawkes-clustered feeds the servicing chain should be split into asynchronous stages once its service time exceeds the threshold established in Part II. **fast_send** remains the mechanism for chaining actors placed on one thread; what changes is where the thread boundaries inside the chain should be. The design rule is stated in Section 12.

1.3 Plan of the paper

The paper asks two questions and answers each in a named place.

1. *What produces the queueing tail behind a single-threaded receiver?* Three candidates are tested. The packet rate: no, at HFT service times (Section 4.9). The clustering of arrivals: yes (Section 4.2), and Section 4.3 identifies the clustered process as the stream of matching-engine transactions, acting through consecutive transactions sent back to back at the publisher’s spacing at short service times and through runs of transactions, the self-exciting component, at long ones. Why the matching engine processes transactions within a microsecond of each other, which is what backs up the publisher, is a question about the market, outside the scope of this paper; Section 12.3 states what can and cannot be claimed about it. The number of messages per packet: at the service time a production receiver runs at, about $7\mu\text{s}$, it carries most of the tail, through the decode of a long packet message by message and the queue that packet opens behind it; from $16\mu\text{s}$ up it adds a few percent (Sections 7.6 and 7.7). Service times that vary from packet to packet, which the simulation does not model, add a tail of their own on the live receiver.
2. *Are two queues better than one?* Yes above a service-time threshold set by the feed and no below it (Sections 4.1 and 4.6); only the slowest stage matters (Theorem 1); with equal cores and every cost charged, a dispatch pool is level with a two-stage cut and better than a cut into three or more for a stateless stage, and the stateful order-book stage cannot be dispatched (Section 4.4); at the production operating point the answer is no (Section 8).

Section 5 looks one stage upstream, at the exchange’s matching engine and market-data publisher through the timestamps the feed carries, and shows that the publisher is a queue whose output spacing is the floor the receiver sees.

The paper proceeds in four parts.

- **Theory.** Two closed-form results (Section 2) predict the pipeline behaviour. Proposition 1 shows that under any arrival process an N -stage tandem has an N -fold throughput speedup

¹The tail compression reported here had been observed in production before its mechanism was understood: splitting a long task into shorter asynchronously connected stages was known to reduce the tail on a per-strategy basis, and the working assumption at the time was that single-thread implementations were simply faster in the inner loop, not that the recommendation depended on the arrival law. The present paper is the explanation of that observation.

on isolated bursts. Theorem 1 shows that a deterministic-service tandem reduces exactly to a single server with the largest stage’s service, and that the end-to-end wait is bounded pathwise by $1/N$ of the single-stage wait plus $(N - 1)h$, at every quantile and under every arrival sequence. Corollary 2 shows that under a rate-matched Poisson null the single-stage wait at the q -quantile is already zero whenever the utilisation satisfies $\rho < 1 - q$ — which at this feed’s rate covers p_{99} for every task up to $16\mu\text{s}$ — so over that band there is nothing to compress and splitting costs exactly $(N - 1)h$. The Hawkes–Oakes cluster representation explains why the single-stage wait on a real feed is one to two orders of magnitude above h . A self-exciting process is equivalent to a Poisson stream of immigrant arrivals, each of which generates a cluster of offspring. Cluster size is governed by the branching ratio n , the mean number of direct offspring per arrival: the mean cluster holds $1/(1 - n)$ arrivals and the largest clusters reach a scale of $(1 - n)^{-2}$ arrivals (the cluster-size law is the Borel distribution; Appendix A). At the fitted $n \approx 0.8$ the mean is about five and the model-implied scale about twenty-five transactions. The last member of a run of K arrivals closer together than the service time waits behind $K - 1$ predecessors on a single server (Proposition 1), so the single-stage wait grows with run length. At $8\text{--}16\mu\text{s}$ the run is typically two consecutive packets sent back to back at the publisher’s spacing; from $64\mu\text{s}$ many members of one cluster fall within one service time. This is the wait the tandem reduces.

- **Simulation.** We build an N -stage tandem Lindley recursion on real CME MDP3 packet-arrival streams at a hop cost $h = 1.7\mu\text{s}$ (the one-way async mailbox cost derived by halving the round-trip figure in Table 4 of Maciejewski [36]) and a service floor swept across $T \in \{2, 4, 8, 16, 32, 64, 128\}\mu\text{s}$, from tasks below the hop cost to tasks far above it. We sweep $N \in \{1, 2, 4, 8\}$ per scenario on each of the 3512 thirty-minute RTH windows of the corpus. Each window is also simulated under a uniform Poisson null P that redraws the same number of arrivals uniformly over the same support (Section 3.5); the null isolates the contribution of clustering, and is not a model of what any designer assumes. Section 3.6 adds seven further streams that remove the ingredients of the clustering one at a time, at the packet level and at the level of matching-engine transactions.
- **Design equation for practitioners.** From the empirical result we extract design rules for a latency-critical servicing chain (Sections 8 and 9): split a stage only when its service time is above the feed’s tight-gap floor, about $7.5\mu\text{s}$ on this corpus, where a queueing tail first appears, and in practice from about $10\mu\text{s}$, where the tail excess first exceeds the hop cost on nearly every window; choose the partition with the smallest achievable largest stage $s_{\max}$, since the tail depends only on $s_{\max}$ and a cut that does not lower it adds a hop and nothing else; place the largest stage first, at ingress; keep sub-tasks below the bottleneck together on one thread via `fast_send`, using an asynchronous `send` only at boundaries that lower $s_{\max}$; and settle each boundary by end-to-end measurement against a representative feed rather than by local reasoning at a single call site.
- **Live cross-validation.** The corpus statistics are compared with an independent measurement on a production receiver taken seven months after the corpus ends, on three instruments (Section 7); the clustering statistics agree, the measured decoder floor coincides with the service time at which the simulated tail appears, and the sweep is rerun with the measured per-message decode cost (Section 7.6). The live ZN stream’s far tail is traced to the message count of a few large packets (Section 7.7).

1.4 Contribution and prior-art positioning

A literature search covering foundational tandem theory [6–10], tandem theory under bursty arrivals [13–17], Hawkes single-queue and adjacent work [5, 18–22], and empirical and applied HFT and RPC-pipeline work [31–35] together with the datacenter tail-latency literature found no direct prior art on any of the paper’s three claims. The nearest neighbour on the exchange side is Yoon [30], who reports that a matching engine’s tail latency under load is set by queueing at the serialised match loop during bursts; the present paper is the receiver-side counterpart, with the arrival process measured rather than assumed. The nearest on the arrival-process side is the Hawkes literature on order flow [24–28], which models order-book events for microstructure and forecasting and does not connect them to a receiver’s service queue or to the packet process that carries them. The first claim is theoretical: that the choice between a single-queue (single-thread) architecture and an N -stage tandem (multi-actor pipeline) is not fixed but *reverses with the arrival law*. At the service times an HFT hot path occupies, single-digit to low-tens of microseconds, Poisson arrivals make the single queue the better choice, because at that utilisation there is no wait to compress and each additional stage adds hop latency only; Hawkes-clustered arrivals at the same mean rate make the tandem better on the tail, because the cluster wait it removes exceeds the hop cost it adds. The reversal is between arrival laws at fixed rate and service time. A Poisson queue also develops a tail at sufficiently long service times, but later and more mildly. We have found no prior work stating that the architectural recommendation reverses between these two arrival laws. The second is empirical: a tandem-queue latency sweep over service floors and stage counts, driven by CME MDP3 packet timestamps, controlled against a rate-matched Poisson null, with a hop cost calibrated from the low-load round trip measured on a production HFT actor framework. We have found no prior study that combines a real exchange-calibrated tandem sweep with both that Poisson counterfactual and a measured framework hop cost. The third is the identification of the clustered process. Grouping 3.55×10^9 messages by matching-engine transaction shows that the packet stream is the transaction stream relayed one packet per event, that the transaction stream is self-exciting with the same branching ratio as the packets, and — through seven null streams that remove one ingredient of the clustering at a time (Section 3.6) — that from $16 \mu\text{s}$ up the receiver’s tail is produced by the timing of transactions and not by their size, their division into packets, the packet rate, or second-scale load, acting through transactions sent back to back at the publisher’s spacing at short service times and through runs of them at long ones; at the $7 \mu\text{s}$ of a production receiver the decode of long packets message by message carries most of it. We have found no published measurement of the packet process of an exchange feed at this level, and no prior test of whether an exchange’s packetisation or the market’s self-excitation is what a receiver queues on. The full reference list appears in Section 14.

1.5 Note on the use of AI in this paper

This paper was produced with Anthropic’s Claude (via Claude Code) under the author’s direction for code generation (the analysis pipeline in the KASPAR-HFT repository), exploratory analysis, mathematical drafting, prior-art search, prose editing, and figure preparation. The analytical results of Part I, including the proof of Theorem 1, were drafted with the model and then independently checked by the author. The author independently reviewed all mathematical arguments, experimental methodology, numerical results, and citations, and takes sole responsibility for the contents of the paper.

1.6 Notation

The following symbols are used throughout.

Symbol	Meaning
T	total service time per message on a single-stage pipeline, in μs
N	number of stages in the tandem pipeline (one actor per stage)
T/N	per-stage deterministic service time in the N -stage tandem
h	per-hop wall-clock cost of the async mailbox handoff between adjacent stages
k	size of an arrival burst (number of packets in one cluster)
$\tau_k^{(N)}$	exit time of the last message of a k -burst under N stages
$s_i, s_{\max}$	per-stage services of an unequal tandem, $\sum_i s_i = T$, and their maximum
$w_j(s)$	Lindley waiting time of message j at a single deterministic server with service s
$W_j^{(N)}, W_q^{(N)}$	end-to-end wait of message j (latency minus T , hops included) under N stages, and its q -quantile
p_q	q -th percentile of the end-to-end latency distribution (p_{50}, p_{99} , etc.)
$\Delta(N)$	clustering tail excess $p_{99}(N) - p_{50}(N)$ (§4.6)
$\Delta_{\text{single}}(s)$	single-server clustering tail excess at deterministic service s ; $\Delta(N) = \Delta_{\text{single}}(T/N)$ (§4.6)
γ	fitted exponent in $\Delta(N) \approx \Delta(1)N^{-\gamma}$
κ	operator's tail-versus-median weight in the design equation
N^*	stage count minimising the design objective $(N - 1)h + \kappa \Delta(N)$ (§4.6)
μ, α, β	Hawkes background rate, jump size, and decay rate (§3.2)
$n = \alpha/\beta$	Hawkes branching ratio
K	Borel-distributed offspring cluster size (§2.2)
$\bar{\lambda}_{\text{pkt}}$	mean packet-arrival intensity per 30-min window, packets per second
n_{pkt}	branching ratio of the packet-arrival Hawkes fit per window
n_{tx}	branching ratio of the transaction-start Hawkes fit per window (§4.3)
$n_{\text{pkt}}^{\#}$	per-window packet count
ρ	utilisation, $\rho = \bar{\lambda}_{\text{pkt}} \cdot T$
τ	bin width of the Fano factor, in seconds (distinct from the service time T)
$F(\tau)$	Fano factor at bin width τ : $\text{Var}[C_\tau]/\text{E}[C_\tau]$
H	Hurst exponent, from log-log slope of $F(\tau)$
H, P, G, B	as stream labels: the real stream and the three packet-level null streams (§3.6); the stream label H is distinct from the Hurst exponent
$\sigma_i, \bar{\sigma}$	span of packet i (SBE messages per datagram) and its window mean (§7.6)
S_i	span-dependent service time of packet i (§7.6)

The per-window packet count and the packet-Hawkes branching ratio would both naturally be written n_{pkt} ; the paper reserves n_{pkt} for the branching ratio and writes $n_{\text{pkt}}^{\#}$ for the count.

Part I

Theory

2 Analytical framework

Readers not interested in the formal presentation may skip to Section 2.3, which restates the results in practical terms.

Throughout, the *tandem* is the paper’s model of the multi-actor pipeline: the total per-message service time T is split into N stages placed in series, each stage an independent deterministic-service server running on its own thread that performs its share of the work and passes the message to the next stage across a mailbox hop of cost h . The single-thread event loop is the $N = 1$ case, in which one server performs all of T ; adding stages shortens each server’s service but inserts a hop between neighbours.

A proposition, a theorem and a corollary underlie the simulation. Proposition 1 is the isolated-burst identity. A message’s end-to-end *latency* is its *service time* T plus its *queueing wait*, the time it spends behind predecessors before service begins. Running the N stages concurrently raises throughput from $1/T$ to N/T , so an N -stage tandem clears a back-to-back burst N times faster: the service time is unchanged at T , and the last message’s queueing wait falls by a factor of N . This is the classical pipeline speedup, stated without proof.

For example, split a service- T server into two equal $T/2$ stages and feed it a burst of k back-to-back messages. All of the queueing now happens at the first stage: its last message waits behind $k - 1$ predecessors that each occupy it for only $T/2$, a wait of $(k - 1)T/2$ against $(k - 1)T$ on the single server. The second stage never queues: stage-one departures arrive spaced $T/2$ apart, equal to its service time, so each message passes through in $T/2 + h$. The wait is not spread across two queues; it remains in one bottleneck stage that is now twice as fast. The halving requires an equal split: cutting $T = 10$ as $7 + 3$ leaves a bottleneck of 7, so the wait falls only to $7/10$. In general the tail scales with the largest stage $s_{\max}$ rather than with the number of stages; Theorem 1 makes this exact. Theorem 1 is the general statement: the tandem’s end-to-end wait equals the wait at a single server with the largest stage’s service plus $(N - 1)h$, exactly and on every sample path, and that single-server wait is at most $1/N$ of the wait at service T . Both are elementary and independent of the arrival law; Theorem 1 rests on a reduction theorem of Friedman [11] and on the monotonicity of the Lindley recursion. Corollary 2 treats the Poisson null: the Poisson wait at quantile q is zero until $\rho = 1 - q$, which at the utilisation of a CME packet feed covers p_{99} for every task up to $16\mu\text{s}$; over that band splitting costs $(N - 1)h$ and removes nothing. The arrival law enters only through the size of the single-stage wait to which the factor $1/N$ is applied, and the Hawkes–Oakes cluster representation explains why that wait is large on a real feed. Part II measures the slack in the bound on real arrivals.

2.1 Burst-limit throughput theorem

Proposition 1 (Burst-limit N -factor speedup). *Consider an isolated cluster of k arrivals whose j -th member arrives no later than $(j - 1)T/N$ after the first, so that no server idles during the burst (an instantaneous batch is the canonical case). Let $\tau_k^{(1)}$ denote the exit time of the last arrival under a single-stage deterministic-service queue with service T , and let $\tau_k^{(N)}$ denote the exit time under an N -stage tandem with per-stage deterministic service T/N and per-stage hop delay h . Then*

$$\tau_k^{(1)} = kT, \quad \tau_k^{(N)} = T + (k - 1)\frac{T}{N} + (N - 1)h, \quad (1)$$

and in particular

$$\lim_{k \rightarrow \infty} \frac{\tau_k^{(1)}}{\tau_k^{(N)}} = N. \quad (2)$$

Remark 2.1. *Proposition 1 is the classical pipelined-tandem throughput result [8, 9]; we state it without proof and use it only to fix the baseline against which the arrival-dependent tail reduction of Theorem 1 is measured. What is paper-specific is the numerical calibration of T and h on a production HFT actor framework and the observation that under real CME MDP3 arrivals the ratio $\tau_k^{(1)}/\tau_k^{(N)}$ is what governs the observed tail-latency compression.*

Reading Equation (1). In the paper’s HFT setting we take $T = 10\mu\text{s}$ as a typical HFT task on a single-thread event loop, and k is the size of a Hawkes-cluster burst that hits the pipeline back-to-back. The single-stage formula $\tau_k^{(1)} = kT$ says that in an event-loop design the last message of a k -burst waits behind $k - 1$ predecessors, each of which spends the full T on the one server; a 10-packet cluster on a $10\mu\text{s}$ server therefore has a last-message latency of $100\mu\text{s}$. The tandem formula $\tau_k^{(N)} = T + (k - 1)(T/N) + (N - 1)h$ decomposes the last-message latency into three physical pieces: (i) T , the total service work every message must traverse regardless of N ; (ii) $(k - 1)(T/N)$, the pipelined wait behind $k - 1$ predecessors at throughput N/T rather than $1/T$; and (iii) $(N - 1)h$, the one-time hop-cost tax paid to traverse $N - 1$ mailbox boundaries on the way through. The single message ($k = 1$) pays only (i) and (iii). Under Hawkes arrivals term (ii) is large on a single stage and falls by a factor of N under a tandem, while term (iii) is an additive cost of $(N - 1)h$.

Reading Equation (2). The ratio $\tau_k^{(1)}/\tau_k^{(N)}$ tends to N as the cluster size k grows, meaning that in the burst-saturation limit a tandem of N actors clears the burst N times faster than an event loop. This is the classical pipeline throughput speedup restated for HFT: a 4-actor pipeline drains a large burst in one-quarter of the wall-clock time that a single-thread event loop would take, and a 12-actor pipeline in one-twelfth. The speedup is obtained with the same total service work per message: the pipeline does not reduce the work per event, it overlaps the work across cores so that the queue behind a cluster clears in parallel. The arrival law does not enter Equation (2); what the arrival law controls is *how often* the burst regime is entered, and Theorem 1 shows the same factor governs every message on every sample path.

On what T is, and on the natural stage boundaries. The paper adopts $T = 10\mu\text{s}$ as a ballpark typical HFT single-thread event loop. Any real HFT pipeline of that duration is already a sequential chain of internally atomic sub-tasks (SBE decode, order-book application, strategy touch, order construction on CME MDP3, per Section 9), so the N stages of the tandem are placed at the existing boundaries between those sub-tasks; the paper is not subdividing any single sub-task, and in particular is not subdividing the SBE decode, which is a linear pass over the wire bytes of a single message and cannot be split. Ten microseconds is chosen for exposition; none of the paper’s qualitative results depend on it. The relevant knob for the design equation is the ratio T/h : doubling T at fixed h doubles the cluster-wait terms of Equation (1) and leaves the hop tax unchanged. The floor below which splitting cannot pay (Section 4.6) turns out to be set by the feed rather than by h : it is the arrival stream’s tight-gap floor, and on this corpus it binds well above $2h$. The paper writes “per-stage service T/N ” as a simplifying modelling convention that assumes equal-sized stages; in reality the per-stage services s_i sum to T but are unequal, and the achievable pipeline throughput is $1/\max_i s_i$ rather than N/T . Theorem 1 handles the unequal case exactly: the factor becomes

$s_{\max}/T$ in place of $1/N$, the tail still contracts under Hawkes, and the median still grows linearly at slope h per hop.

The mechanism applies to any deterministic-service tandem regardless of the arrival law; the arrival law determines how often large- k clusters occur. The next subsection makes the per-burst factor a pathwise bound.

2.2 Exact reduction of the tandem to one server, and the scaling bound

Proposition 1 treats one isolated burst. The result that carries the paper's design claim is that the same N -factor holds pathwise for every message under every arrival sequence, at the level of the end-to-end wait. Fix an arrival sequence $a_1 \leq a_2 \leq \dots$ (any point process, clustered or otherwise, with batches allowed as coincident times) and write $\delta_j = a_{j+1} - a_j \geq 0$. For a single FIFO server with deterministic service s fed by this sequence, the waiting time of message j obeys the Lindley recursion [12]

$$w_1(s) = 0, \quad w_{j+1}(s) = (w_j(s) + s - \delta_j)^+. \quad (3)$$

Define the end-to-end wait of message j in the N -stage tandem as $W_j^{(N)} = (\text{exit time}) - a_j - T$, i.e. latency in excess of the total service work; under $N = 1$ this is $W_j^{(1)} = w_j(T)$.

Theorem 1 (Reduction and scaling bound). *Consider an N -stage tandem of FIFO servers with deterministic per-stage services $s_1, \dots, s_N$ summing to T , deterministic hop delay h between consecutive stages, unbounded buffers, and an arbitrary arrival sequence. Let $s_{\max} = \max_i s_i$. Then for every message j and every sample path,*

$$W_j^{(N)} = w_j(s_{\max}) + (N - 1)h, \quad (4)$$

$$w_j(s') \leq \frac{s'}{s} w_j(s) \quad \text{for all } 0 < s' \leq s, \quad (5)$$

and consequently

$$W_j^{(N)} \leq \frac{s_{\max}}{T} W_j^{(1)} + (N - 1)h = \frac{W_j^{(1)}}{N} + (N - 1)h \quad \text{under equal stages } s_i = T/N. \quad (6)$$

Since Equation (6) holds pathwise, it holds for every quantile of the stationary distribution whenever one exists: $W_q^{(N)} \leq W_q^{(1)}/N + (N - 1)h$ for all q . The same inequality holds for the empirical (sample) quantiles computed over any finite trace, directly from the pathwise bound and without invoking stationarity; the sample-quantile form is what Part II measures, and it should be read as a statement about the observed windows rather than about a stationary-distribution quantile or a finite-sample maximum.

Proof. Equation (4), equal stages. Departures from a FIFO server with deterministic service s are spaced at least s apart, because each departure requires s of uninterrupted service after the previous one. A FIFO server with deterministic service s whose arrivals are spaced at least s apart never queues: by induction, message j departs at its arrival time plus s , which is at or before the arrival of message $j + 1$. Stage 1 is a single server with service T/N fed by the original sequence, so message j leaves it at $a_j + w_j(T/N) + T/N$. The arrival sequence at stage 2 is the stage-1 departure sequence shifted by h , spaced at least T/N apart, so stage 2 adds exactly $h + T/N$ and its departures are again spaced at least T/N apart; the same holds at each subsequent stage. Summing, the exit time is $a_j + w_j(T/N) + N \cdot T/N + (N - 1)h$, and subtracting $a_j + T$ gives Equation (4) with $s_{\max} = T/N$.

Equation (4), unequal stages. By the order-invariance theorem of Friedman [11], for a tandem of FIFO stages with constant service times, unlimited queues, and any arrival sequence, each customer's time in system is independent of the order of the stages. Deterministic hop delays preserve this: a hop is a constant shift of one stage's departure sequence into the next stage's arrival sequence, and constant shifts commute with the Lindley recursion, so reordering stages together with their hops leaves every exit time unchanged. Reorder so that the stage with service $s_{\max}$ comes first. Its departures are spaced at least $s_{\max}$ apart, and every subsequent stage has service $s_i \leq s_{\max}$, so by the no-queueing argument above each of them adds exactly $s_i + h$ to every message. The exit time is $a_j + w_j(s_{\max}) + \sum_i s_i + (N - 1)h$, and subtracting $a_j + T$ gives Equation (4).

Equation (5). By induction on j with $w_1(s') = 0 = (s'/s)w_1(s)$. The map $x \mapsto (x + s' - \delta_j)^+$ is nondecreasing, so

$$w_{j+1}(s') = (w_j(s') + s' - \delta_j)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \delta_j\right)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \frac{s'}{s}\delta_j\right)^+ = \frac{s'}{s}w_{j+1}(s),$$

where the second inequality uses $\delta_j \geq (s'/s)\delta_j$, which holds because $\delta_j \geq 0$ and $s' \leq s$. Combining Equations (4) and (5) with $s' = s_{\max}$, $s = T$ gives Equation (6). $\square$

Remark 2.2 (What the theorem says and what it does not). *Equation (4) says all queueing in an equal-stage deterministic tandem happens at stage 1; stages $2, \dots, N$ contribute exactly $T/N + h$ each to every message. Proposition 1 is the isolated-burst special case, where Equation (6) holds with equality. Equation (6) is a bound rather than an identity because the tandem runs its bottleneck stage at utilisation ρ/N while the single stage runs at ρ : the induction step in the proof of Equation (5) gives away $(1 - 1/N)\delta_j$ at every message j that the single server is still serving when message $j + 1$ arrives, so the tandem clears each inter-arrival gap N times more effectively than the scaled bound credits it for. The paper's empirical γ (Section 4.6), defined by $W_q^{(N)} - (N - 1)h \approx N^{-\gamma} W_q^{(1)}$, therefore satisfies $\gamma \geq 1$ in the model, and the excess $\gamma - 1$ measures that slack. Nothing in the theorem depends on the arrival law; what the arrival law determines is the size of $W_q^{(1)}$ that the factor $1/N$ is applied to, which is the subject of the next remark. The hop cost h enters every quantile as exactly $(N - 1)h$, in the median and in the tail alike.*

Remark 2.3 (Unequal stages: only the largest stage matters). *Real cut points do not give equal stages. Equation (4) states the consequence: the end-to-end wait of any deterministic tandem equals the wait at a single server with service $s_{\max}$, plus h per hop. Three consequences follow.*

1. *The tail factor is $s_{\max}/T$, and the stage count enters only through the hop tax. Splitting a $10 \mu\text{s}$ task as $7 + 3$ compresses the tail excess to at most 70% of single-stage, against 50% for $5 + 5$, and both pay one h .*
2. *The order of the stages is irrelevant to end-to-end latency: $7 + 3$ and $3 + 7$ give identical latencies on every sample path. Order determines only where the queue physically sits (below).*
3. *A cut that does not lower $s_{\max}$ adds a hop and nothing else. Refining $7 + 3$ into $7 + 2 + 1$ leaves every quantile of the wait unchanged and raises end-to-end latency by h . Conversely, sub-tasks can be merged onto one thread (via **fast_send**) without any tail cost as long as the merged service does not exceed $s_{\max}$. The design problem is therefore to choose the partition with the smallest $s_{\max}$ reachable at the natural cut points, and then the fewest hops that realise it.*

Appendix C checks all three numerically.

Remark 2.4 (Where the burst goes: the bottleneck absorbs the clustering). *The proof of Equation (4) rests on one fact: departures from a deterministic server with service s are spaced at least s apart. Applied to a Hawkes input, that fact says what the bottleneck stage does to the clustering. Upstream of the bottleneck the stream is the raw feed, with intra-cluster inter-arrival times far below $s_{\max}$ and instantaneous intensity far above $1/s_{\max}$. Downstream of the bottleneck, every cluster still leaves as the same group of K messages, so the count-level branching structure is unchanged (the Borel cluster sizes, and the Fano factor at time scales beyond $Ks_{\max}$, are exactly those of the input), but the intra-cluster spacing has been regularised to $s_{\max}$ and the instantaneous intensity is capped at $1/s_{\max}$. The clustering is not removed; it is regularised to a spacing that any consumer at least as fast as the bottleneck can absorb, which is why stages with service $s_i \leq s_{\max}$ never queue. If the bottleneck is stage 1, all of the waiting in the pipeline sits in its mailbox and every downstream mailbox holds at most one message; if the bottleneck is stage $b > 1$, the stages before it queue partially (the raw burst is still bursty relative to their smaller services) and stage b absorbs the remainder, with the same total by Equation (4).*

Four practical consequences follow. First, place the largest stage first, at ingress: end-to-end latency is unchanged by the order, but this confines all queueing to one mailbox, keeps the downstream mailboxes at depth one, and places the burst-absorbing queue where its receiver is continuously busy during a cluster (the regime in which the load-dependent hop cost of Section 4.10 is smallest). Second, any burst-aware logic, such as load shedding, coalescing of superseded book updates, or a cancel-on-burst risk rule, can only observe the raw burst upstream of the bottleneck; downstream it sees a rate- $1/s_{\max}$ stream and cannot tell a cluster from steady flow except by counting. Third, a packet-Hawkes fit run on the bottleneck's departure stream would report a lower intensity peak and a longer effective kernel while recovering the same branching ratio in counts; operators instrumenting a pipeline should fit on the ingress stream. Fourth, deterministic hop delay is what keeps downstream stages queue-free; hop jitter of amplitude ϵ lets a downstream stage queue by at most ϵ per message, so the reduction holds up to that jitter in practice.

Fact (Hawkes–Oakes 1974 cluster representation [2]). *A stationary Hawkes process with background rate $\mu > 0$ and branching ratio $n \in [0, 1)$ is stochastically equivalent to a Poisson-cluster process: a rate- μ Poisson process of immigrant arrivals, each of which independently spawns an offspring cluster whose total size K (including the immigrant) follows the Borel distribution*

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!}, \quad k = 1, 2, 3, \dots \quad (7)$$

with mean $E[K] = 1/(1 - n)$ and variance $\text{Var}[K] = n/(1 - n)^3$ [3].

Remark 2.5 (Where Hawkes enters: the size of the single-stage wait). *Under Poisson arrivals at the utilisation of a CME packet feed ($\rho = \bar{\lambda}_{\text{pkt}}T \ll 1$), the single-stage wait is zero for almost every message, so $W_q^{(1)} = 0$ for every quantile q with $\rho < 1 - q$ and Equation (6) has nothing to compress; that is Corollary 2, and at this corpus's rate it holds at p_{99} for all $T \leq 16 \mu\text{s}$. Under Hawkes arrivals the cluster representation puts K near-coincident messages behind one immigrant, and in the instantaneous-batch picture the last of them waits $(K - 1)T$ on a single stage. Stirling's formula applied to Equation (7) gives, for large k ,*

$$\Pr(K = k) \approx \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k, \quad (8)$$

and $ne^{1-n} < 1$ for every $n < 1$, with $-\log(ne^{1-n}) \approx (1 - n)^2/2$ near criticality. The Borel batch size is therefore light-tailed at every fixed $n < 1$: its survival function follows the critical $k^{-1/2}$ power

law only in the window $k \ll k^* = (1 - n)^{-2}$ and decays geometrically beyond it (Appendix A). The consequences for the paper are, first, that the largest clusters, and hence the single-stage cluster wait they produce, have a characteristic scale of order $Tk^* = T(1 - n)^{-2}$, which at the corpus-median $n \approx 0.8$ (Section 3.2) is $\approx 25T$ and at $n = 0.9$ would be $100T$, one to two orders of magnitude above the service floor and the hop cost; second, that heavy-tail results built on subexponential batch sizes, such as the single-big-batch principle for $M^{[K]}/G/1$ [4] and the monotone-separable tail asymptotics of Baccelli et al. [14], do not apply to this batch distribution at any fixed $n < 1$, so no closed-form power-law exponent for the tail probability should be expected; and third, that the design claim does not depend on any of this, because Theorem 1 applies the factor $1/N$ to whatever $W_q^{(1)}$ the feed produces. Part II measures $W_q^{(1)}$ on real arrivals.

Corollary 2 (Poisson null). *Under homogeneous Poisson arrivals at rate $\bar{\lambda}$ with $\rho = \bar{\lambda}T < 1 - q$, the q -quantile of the single-stage wait is $W_q^{(1)} = 0$, because $\Pr(W^{(1)} > 0) = \rho$ for $M/D/1$ by the PASTA property (Poisson arrivals see time averages). By Theorem 1, $W_q^{(N)} \leq (N - 1)h$, and by Equation (4) $W_q^{(N)} \geq (N - 1)h$, so*

$$W_q^{(N)} = (N - 1)h \quad \text{for all } N \text{ and all } q \leq 1 - \rho. \quad (9)$$

At the utilisations of a CME packet feed (corpus-median 452 packets per second, so $\rho = 0.0036$ at $T = 8\mu\text{s}$) this covers p_{99} at every $T \leq 16\mu\text{s}$ in the paper's sweep: end-to-end latency is $T + (N - 1)h$ at p_{50} and at p_{99} alike, and splitting costs $(N - 1)h$ with nothing to compress. Where ρ/N exceeds $1 - q$, Equation (6) still holds and the tandem reduces the Poisson wait as well; splitting then helps under Poisson for the ordinary utilisation reason rather than for clustering. The sweep contains six such p_{99} cells, all at the long-service end: $(T, N) = (32, 1), (64, 1), (64, 2), (128, 1), (128, 2)$ and $(128, 4)$, spanning $\rho/N = 0.0144$ to 0.0578 . It contains one such p_{95} cell ($T = 128, N = 1, \rho = 0.0578 > 0.05$) and many $p_{99.9}$ cells, and Section 4.1 finds the wait in those cells and in no others.

2.3 Informal statement of the two results

This subsection restates Proposition 1, Theorem 1 and Corollary 2 informally. The formal statements above are the reference.

Components of latency. A message's end-to-end latency is the sum of three terms: the work done on it (T in total, however many stages that work is spread over), the hop tax (h for every mailbox boundary it crosses, so $(N - 1)h$ in an N -stage pipeline), and the time it spends sitting in queues waiting for a busy actor. The third term is the wait, W . The work is fixed by the code and the hop cost by the framework; the wait is the only term the architecture changes, and on a bursty feed it dominates the tail.

Proposition 1: one burst. Suppose k messages land at once. On a single thread the last of them waits for the $k - 1$ ahead of it, each costing T : it waits $(k - 1)T$. On an N -stage pipeline the first stage takes only T/N per message, so the last message waits $(k - 1)T/N$ at stage 1 and then flows through the remaining stages without waiting again, because each of them is fed at exactly the rate it can serve. The pipeline clears a burst N times faster for the same total work, because N cores process N different messages at the same instant.

Theorem 1: every message, any arrival sequence. Two facts.

1. In a pipeline where each stage does a fixed amount of work per message, only the slowest stage ever builds a queue. Every other stage receives messages no faster than it can process them, because the slowest stage upstream (or, if the slowest stage is downstream, the stages before it) meters them out. The pipeline therefore behaves as one thread whose per-message work is $s_{\max}$, plus the hop cost. With equal stages, $s_{\max} = T/N$.
2. Take any recording of arrival times, real or synthetic, and replay it through two single-threaded servers: one doing T of work per message and one doing T/N . The second server’s queue wait for message number j is never more than $1/N$ of the first server’s wait for the same message. The reason is that in the faster server every backlog is N times smaller and every idle gap between arrivals clears at least as much of it.

Combined: for every message in every replay, the pipeline’s wait is at most the single-thread wait divided by N , plus $(N - 1)h$. Because this holds message by message, it holds for any summary statistic: median, p_{99} , $p_{99.9}$, or maximum. The bound is met with equality on an isolated instantaneous burst and is strict whenever arrivals are spread out, because the pipeline’s first stage is idle more often than the single thread and drains more of its backlog in each idle gap. The measured compression is written $N^{-\gamma}$; the theorem guarantees $\gamma \geq 1$, and Section 4 measures $\gamma = 3.3$ and 2.7 at $T = 64$ and $128 \mu\text{s}$. At shorter service times the tandem reduces the excess to zero and no exponent is defined.

Role of the arrival law. The arrival law does not enter either theorem. It enters only through the size of the single-thread wait to which the factor $1/N$ is applied. Under Poisson traffic at the load of a CME packet feed, the single thread is idle when almost every message arrives, so for tasks up to $16 \mu\text{s}$ its p_{99} wait is zero; there is nothing to divide by N , and the pipeline is simply $(N - 1)h$ slower (Corollary 2). Under the real feed, transactions arrive in self-exciting clusters and the packets relay them, the single thread’s p_{99} is 2.2 to 41.5 times T in the paper’s data for tasks of 16 to $128 \mu\text{s}$, and the pipeline divides the excess by N or better. The Hawkes cluster representation gives a model-implied cluster size: Borel-distributed, with large clusters of order $(1 - n)^{-2}$ transactions, about 25 at the corpus’s $n \approx 0.8$. What the receiver waits behind depends on the service time: at $8\text{--}16 \mu\text{s}$ the wait is usually behind one predecessor sent $7.5\text{--}16 \mu\text{s}$ earlier, and only from about $64 \mu\text{s}$ does a run of cluster members fall within one service time (Sections 4.2 and 6).

Design rules.

- The tail improvement is set by the slowest stage, $s_{\max}/T$, and by nothing else about the decomposition. Balanced stages are the best case for a given number of hops.
- A cut that does not reduce the slowest stage costs h at every quantile and improves nothing. Sub-tasks can be merged onto one thread until their combined work reaches $s_{\max}$ at no cost to the tail.
- The order of the stages does not change any message’s latency. It changes only where the queue physically sits; putting the slowest stage first keeps all queueing in one mailbox.
- Every hop costs h on every message, at the median and in the tail.

- For a stream that must be processed in order, whether to split reduces to one comparison: the single-thread wait at the quantile of interest, times $(1 - s_{\max}/T)$, against h per hop added. On a Poisson feed at HFT service times the left side is zero; on a clustered feed at the same rate it is tens to thousands of microseconds. With the same cores, a stateless stage on a stream that tolerates reordering can instead dispatch whole packets to N servers; Section 4.4 compares the two without charging dispatch its ingress, hop or resequencing cost.

The guarantees above are exact within a specific model. Its assumptions follow.

Fixed work per message. Each stage is assumed to spend the same amount of time on every message. That is what makes only the slowest stage queue: a stage that sometimes takes longer than its upstream neighbour would occasionally fall behind and build its own queue, and then the pipeline’s wait would exceed the single-server wait at $s_{\max}$. The simulator of Part II uses constant per-packet service, so the theorems describe it exactly; Section 7.6 relaxes this for the message count per packet. Real actors also have work that varies with book state, and for them the theorems are a model to be checked by measurement (Section 9).

No drops, no bounds, in-order. Mailboxes are unbounded, messages are never discarded or coalesced, and each stage processes in arrival order. A framework that drops or merges messages under load changes the arithmetic (in the pipeline’s favour, at the cost of lost messages); a priority queue changes which message pays the wait but not the total.

Fixed hop delay. The hop is modelled as a constant h . A jittery hop lets a downstream stage queue by at most the jitter amplitude per message, so the reduction holds up to that amplitude. The paper’s calibration uses the low-load, wake-from-sleep value of h , which Section 4.10 argues is an upper bound inside a burst.

Equal stages, no cross-core cost beyond h . The N stages are taken to split the work evenly and to exchange nothing but the message: moving book state or cache lines between the cores that run adjacent stages is charged nothing beyond h . Unequal stages are handled by Remark 2.3; a cross-core cost would add to h and is a matter for measurement (Section 4.10).

Proposition 1 needs a dense burst. The one-burst identity assumes the k messages arrive faster than the first stage can drain them, i.e. faster than one per T/N . For a burst that trickles in, the ratio in Equation (2) is $kN/(k + N - 1 + N(N - 1)h/T)$, which climbs toward N as the burst grows. Theorem 1 does not need this assumption; it covers every arrival pattern.

Limits of the theory. It does not say how large the single-thread wait is on a given feed. That number depends on the feed’s clustering and utilisation and, as the remark after the Borel fact explains, has no clean closed form at a fixed branching ratio: the Borel cluster size is light-tailed for every $n < 1$, so the heavy-tail formulas that would give a power-law exponent do not apply. The paper measures the single-thread wait on real CME arrivals instead (Part II). Nor does the theory predict how far below the $1/N$ bound the pipeline lands; that slack is also measured. What the theory guarantees is that the pipeline’s wait is at most $1/N$ of the single-thread wait plus the hop cost, on any arrival sequence.

Part II

Empirical experiment

3 Experiment setup

This section describes the empirical experiment that tests the analytical predictions of Section 2: the data corpus, the confirmation that packet arrivals are Hawkes-clustered (the reason the single-stage wait that Theorem 1 divides by N is large at all), the tandem-Lindley simulator, and the sweep design that produces the numbers reported in Section 4. Figure 1 shows the end-to-end data flow of a single 30-minute window; Algorithm 1 states the full per-window procedure in pseudocode.

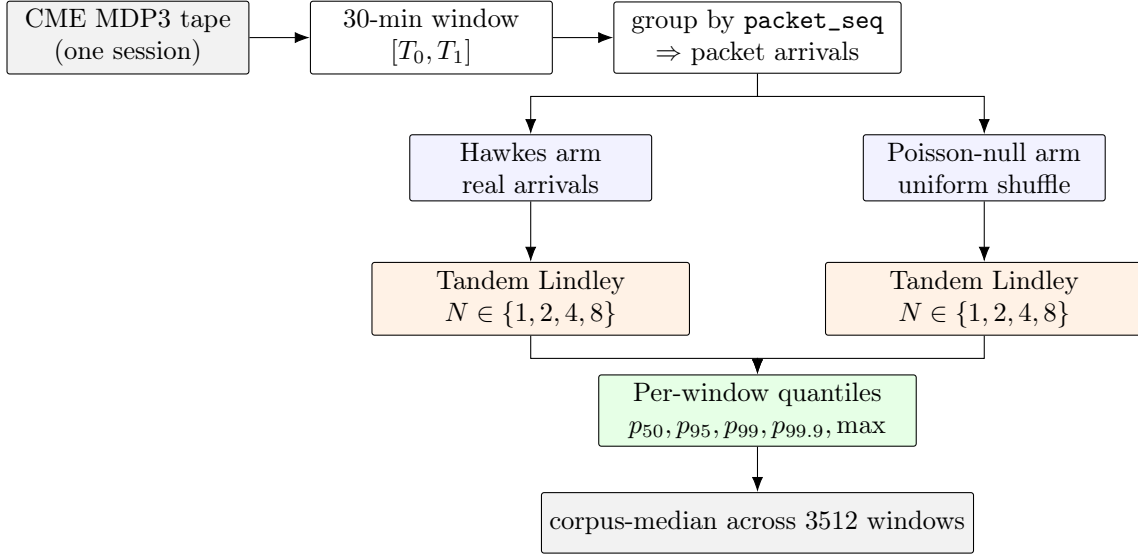


Figure 1. Per-window data flow of the empirical experiment. A single session tape is chopped into 30-minute RTH windows. Each window’s messages are grouped by `packet_seq` into a real packet-arrival stream (Hawkes arm). A rate-matched Poisson-null arm is generated by uniformly shuffling the same arrival count over the same window. Both arms are run through the N -stage tandem Lindley simulator at $N \in \{1, 2, 4, 8\}$. Per-window quantiles are recorded and aggregated by corpus median across all 3512 windows in the 276-session corpus.

Algorithm 1 Per-window procedure of the empirical experiment.

```
1: input: session tape  $S$ , service  $T$ , hop  $h$ , stage counts  $\mathcal{N}$ 
2: for each 30-min RTH window  $W$  in  $S$  with  $[T_0, T_1]$  do
3:    $\{m_i\} \leftarrow$  messages in  $W$ 
4:    $\{(\text{seq}, t)\} \leftarrow \{(\text{packet\_seq}(m_i), \text{sendingTime}(m_i))\}$ 
5:    $\text{arr}_H \leftarrow \text{sort}(\{\min_i t_i : \text{seq}_i = s\}_s)$   $\triangleright$  Hawkes arm (real arrivals)
6:    $n_{\text{pkt}}^\# \leftarrow |\text{arr}_H|$ 
7:    $\text{arr}_P \leftarrow \text{sort}(\text{Uniform}(\min \text{arr}_H, \max \text{arr}_H)^{n_{\text{pkt}}^\#})$   $\triangleright$  Poisson-null arm (seeded per window)
8:   for regime  $\in \{H, P\}$ ,  $N \in \mathcal{N}$  do
9:      $\text{lat} \leftarrow \text{TANDEMLINDLEY}(\text{arr}_{\text{regime}}, N, T, h)$ 
10:    record  $(S, W, \text{regime}, N, \{p_{50}, p_{95}, p_{99}, p_{99.9}, \max\}(\text{lat}))$ 
11:   end for
12: end for
13: return corpus-median of each recorded quantile across all  $(S, W)$  pairs
14: function TANDEMLINDLEY( $\text{arr}, N, T, h$ )
15:    $\text{dep} \leftarrow \text{arr}$ 
16:   for stage  $k = 1, \dots, N$  do
17:     if  $k > 1$  then  $\text{dep} \leftarrow \text{dep} + h$ 
18:   end if
19:    $\text{dep} \leftarrow \text{dep} + \text{LINDLEY}(\text{dep}, T/N)$ 
20: end for
21:   return  $\text{dep} - \text{arr}$ 
22: end function
```

3.1 Data corpus

The corpus is 281 sessions of CME MDP3 channel 318 (NQ) front-month pcap-derived tapes covering 2025-01 through 2026-02, RTH only, holidays and half-days excluded. Each MBO record carries a `packet_seq` (the UDP packet sequence number) and two nanosecond timestamps: `transactTime`, the CME matching-engine event time carried in the MDP3 message body, and `sendingTime`, the gateway’s per-packet UDP header timestamp. Messages are grouped into packets by `packet_seq`, and a packet’s arrival time is its `sendingTime`.

`transactTime` is not a suitable arrival clock. It is the matching engine’s event time, not the time at which anything reaches the receiver: the 1.7% of transactions that the gateway splits across two or more packets (Section 4.3) carry one `transactTime` on packets that leave about $15 \mu\text{s}$ apart, so keying arrivals on it collapses distinct packets onto one instant. `sendingTime` is emitted once per packet at nanosecond resolution and is the receiving pipeline’s input clock; ordered by `packet_seq` it produces no ties and no sequence inversions on the corpus. All results below are keyed on `sendingTime`. Section 4.3 shows that merging each transaction into one packet leaves the tail within $\pm 15\%$, so the choice of clock does not affect the results.

Each session tape is chopped into non-overlapping 30-minute RTH windows; for each window we compute the packet-arrival stream and the per-window quantities listed in Section 1.6. The corpus yields 3512 per-window records from the 276 sessions that contain at least one complete RTH window, roughly 13 windows per session; the remaining five of the 281 tapes are holiday or half-day sessions with none.

3.2 Hawkes clustering of packet arrivals (empirical prerequisite)

The paper’s simulator input is the packet-arrival stream, so the first empirical question is whether that stream is clustered or Poisson. Three model-free and one model-based diagnostic on the 276-session, 3512-window NQ front-month corpus reject Poisson. The fraction of interarrival gaps shorter than one-tenth of the mean gap has a corpus median of 73% per 30-minute window, a 5th–95th percentile range of 54–78% and a full range of 38–85%, versus the Poisson benchmark $1 - e^{-0.1} \approx 9.52\%$ that would obtain for any exponential-gap process regardless of rate; not one of the 3512 windows comes within a factor of three of the Poisson value. The Fano factor $F(\tau) = \text{Var}[C_\tau]/\text{E}[C_\tau]$ of packet counts in bins of width τ rises from $F(0.1\text{ s}) = 51.7$ to $F(60\text{ s}) = 1027$ across the corpus median, against $F(\tau) \equiv 1$ for Poisson at every τ ; the log-log fit gives a Hurst exponent $H = 0.74$, against $H = 0.5$ for Poisson. These are within-window figures, corpus-median over the 3512 windows; pooling a whole session instead raises the Fano level by roughly a factor of four without changing the shuffle ratio, for reasons taken up in Section 7.4. Running the identical estimator over the paper’s own Poisson-null arm returns $F(\tau) = 1.00$ at $\tau = 0.1$ through 5 s, 0.94 at 60 s, and $H = 0.50$, which confirms that the estimator is unbiased at every bin width used here and that the Hawkes figures above are not an artefact of binning or of finite window length. Fitting the standard exponential-kernel Hawkes process [1] $\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)}$ to each window by MLE [23] (Ogata log-likelihood with the standard recursive intensity, closed-form compensator, Nelder–Mead over (μ, α, β)) gives a branching ratio $n = \alpha/\beta$ with a corpus median of 0.798 and a 5th–95th percentile range of 0.761–0.832 across the 3512 windows (extremes 0.566 and 1.000, the latter a single window pinned at the fit boundary), with every fit converged; the kernel’s decay time $1/\beta$, on 254 windows from 20 sessions, has a median of $167\text{ }\mu\text{s}$ (5th–95th percentile $140\text{--}205\text{ }\mu\text{s}$), so the fitted process concentrates its triggering at lags of order a hundred microseconds and places only 4% of it within $7.5\text{ }\mu\text{s}$; the same optimum is recovered by a reparameterised L-BFGS-B fit to four decimals on spot-checked windows. This is the same near-critical regime that Filimonov and Sornette [24] and Hardiman et al. [27] document on ES, at a somewhat lower level than their $n \approx 0.9$; they fit trade arrivals rather than packet arrivals, and on this corpus the fit on matching-engine transaction starts gives the same value as on packets (0.795 against 0.798, Section 4.3). Filimonov and Sornette [25] show that an exponential-kernel fit with a constant background over a fixed window overstates n when the background intensity varies within the window, and that ES estimates fall well below criticality once that variation is modelled; the n reported here is descriptive and subject to the same caveat, and none of the paper’s conclusions rests on its value. The model-free diagnostics above, the Fano-against-shuffle comparison of Section 4.3 and the gap histograms there hold whatever the fitted n ; the nonparametric estimates of Achab et al. [26] on EUREX order books give the same near-critical picture without a kernel assumption. A Fisher–Yates shuffle of the interarrival gap sequence within each window, which preserves the marginal gap distribution exactly and destroys only the ordering, removes most but not all of the excess: the corpus-median Fano at $\tau = 5\text{ s}$ falls from 259.7 to 17.6, and at $\tau = 1\text{ s}$ from 108.4 to 17.5, i.e. 94% and 85% of the excess over the Poisson value of 1. What the shuffle removes is specifically the *scaling*: the shuffled Fano is flat at 16.1–17.6 across every bin width from 0.1 to 60 s, giving $H = 0.50$ against the unshuffled $H = 0.74$. All of the long-range dependence therefore lives in the ordering, which is what distinguishes a self-exciting process from a renewal process. The flat residual of ≈ 17 is a scale-free over-dispersion contributed by the gap marginal itself, which is heavy-tailed; so the corpus is neither a renewal process nor a pure Hawkes process with light-tailed gaps, but a self-exciting process with heavy-tailed gaps. This split bounds what the Hawkes fit describes: the fitted n characterises the ordering component, and the simulator is driven by the recorded arrivals, not by the fitted process. The simulator input is therefore an arrival process far from Poisson, and the matched-rate Poisson null is a strict counterfactual. The Hawkes

fit is used only as a description of the observed clustering; the tandem reduction of Section 2 does not depend on the Hawkes assumption, since any arrival process with the same clustering drives the same single-server queue. The model-free diagnostics above, not the fit, establish the empirical premise.

3.3 Simulator

Packet arrivals $t_1 < t_2 < \dots$ enter stage 1. Each stage k is a single-server FIFO queue with deterministic service T/N (own actor, own thread), and departures from stage k are delayed by h before arriving at stage $k + 1$. Standard Lindley recursion per stage:

$$W_{i+1}^{(k)} = \max\left(0, W_i^{(k)} + \frac{T}{N} - (a_{i+1}^{(k)} - a_i^{(k)})\right), \quad (10)$$

with $a_i^{(k+1)} = a_i^{(k)} + W_i^{(k)} + T/N + h$.

Service is the same for every packet. Section 7.6 makes it grow with the number of messages in the packet; nothing in the paper makes it random. A real receiver’s service time varies from packet to packet for reasons other than message count (Section 7.7), and for a single server, more variable service at the same mean lengthens the mean wait (the Pollaczek–Khinchine formula for Poisson arrivals). The simulated tails are therefore what the arrival timing and the message count produce on their own, not everything a real receiver pays.

3.4 How the simulator models concurrent stages

Each stage’s actor runs on its own thread, so stage 2 can be processing message $k - 1$ while stage 1 is processing message k . The simulator captures this because Lindley is applied separately per stage, each with its own busy-idle timeline. What the split buys is not the overlap as such but the shorter queue in front of each stage: a burst that one server of service T holds in a queue of length k is held by the first stage at T/N and drained N times faster.

Worked example. Three packets arriving at $t = 0, 1, 2 \mu\text{s}$; single-stage total service $T = 8 \mu\text{s}$ (a service floor in the sweep of Table 2); two-stage tandem with per-stage service $T/2 = 4 \mu\text{s}$ and hop $h = 1.7 \mu\text{s}$:

	msg 1	msg 2	msg 3
single-stage arrival	0	1	2
single-stage latency (μs)	8.0	15.0	22.0
tandem arrive stage 1	0.0	1.0	2.0
tandem wait stage 1	0.0	3.0	6.0
tandem leave stage 1	4.0	8.0	12.0
tandem arrive stage 2 (+ h)	5.7	9.7	13.7
tandem wait stage 2	0.0	0.0	0.0
tandem leave stage 2	9.7	13.7	17.7
tandem latency (μs)	9.7	12.7	15.7

At wall-clock $t \approx 7$, stage 1 is processing message 2 while stage 2 is processing message 1. Message 1 pays the $1.7 \mu\text{s}$ hop cost (latency $8.0 \rightarrow 9.7$); message 3, at the end of the burst, falls from 22.0 to $15.7 \mu\text{s}$. The median rises and the tail falls, at the same message count, with throughput $2/T$ rather than $1/T$. This is the mechanism of Proposition 1.

A chain of synchronous `fast_sends` on one thread shortens no queue and pays $(N - 1)h$; only stages on their own threads do (Section 9).

Calibration and cost. The paper reports numerical results at $h = 1.7 \mu\text{s}$ — the one-way async mailbox cost obtained by halving the round-trip figure in Table 4 of Maciejewski [36] — swept across $T \in \{2, 4, 8, 16, 32, 64, 128\} \mu\text{s}$ to cover the sub-hop regime through the slow-servicing regime. The simulator is deterministic given the tape and (T, h, N) ; one $O(n)$ pass per stage per window makes its wall-clock cost negligible next to the tape read.

3.5 Sweep design

The design asks whether an N -stage tandem reduces p_{99} latency under real Hawkes packet arrivals and whether the same architecture achieves the same compression under a rate-matched Poisson null. For each 30-minute window we run the tandem Lindley of Section 3.3 at $N \in \{1, 2, 4, 8\}$ under two arrival regimes: the real packet-arrival timestamps from the tape (Hawkes arm, stream H) and a uniform redraw of the same $n_{\text{pkt}}^\#$ arrivals over the same support (the uniform Poisson null P); the further null streams that separate the ingredients of the clustering are defined in Section 3.6. The support is the observed packet span $[\min \text{arr}_H, \max \text{arr}_H]$ rather than the nominal window $[T_0, T_1]$, so the null is rate-matched to the arrivals it replaces rather than to wall-clock; the two coincide on busy windows and the former is the stricter counterfactual on sparse ones. The null is therefore a *conditional homogeneous-Poisson* process: a homogeneous Poisson process at rate $n_{\text{pkt}}^\# / (\max \text{arr}_H - \min \text{arr}_H)$ conditioned on the observed count over the observed support. Per (window, N , regime) we record $p_{50}, p_{95}, p_{99}, p_{99.9}, \max$ of end-to-end latency. Aggregation is corpus-median across all windows; the two headline curves per scenario are $p_{99}(N)$ and $p_{50}(N)$ under Hawkes and Poisson overlaid.

The paper’s empirical calibration is $h = 1.7 \mu\text{s}$ (one-way async cross-thread mailbox cost, per Table 4 of Maciejewski [36], halved for one-way) swept over $T \in \{2, 4, 8, 16, 32, 64, 128\} \mu\text{s}$ so the scenarios span from the sub-hop regime through the slow-servicing regime. Theorem 1 predicts $p_{99}^H(N) - (N - 1)h - T \leq (p_{99}^H(1) - T)/N$ on every window, i.e. a fitted exponent $\gamma \geq 1$ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$; Corollary 2 predicts $p_{99}^P(N) = p_{50}^P(N) = T + (N - 1)h$ exactly in every cell whose per-stage utilisation satisfies $\rho/N < 0.01$, and a strictly positive Poisson wait in every cell where it does not; the prediction is two-sided, so a Poisson p_{99} sitting on the floor at $\rho/N > 0.01$ would falsify it just as a Poisson p_{99} above the floor at $\rho/N < 0.01$ would. Any of (i) a window on which the Hawkes bound is violated, (ii) a Poisson $p_{99}(N)$ on the wrong side of the $\rho/N = 0.01$ boundary, or (iii) Hawkes $p_{50}(N)$ growing at slope $\gg h$ per stage falsifies the paper’s model of the simulator.

3.6 Null streams and the ingredient each removes

The claim under test is causal: the clustering of arrivals produces the queueing tail. A tail in the Lindley recursion can be built by four distinct properties of an arrival stream, and the real stream has all four at once:

1. the *rate*, the number of packets per second;
2. the *gap distribution*: how often two consecutive packets land closer together than the service time T , regardless of the order in which those tight gaps occur;
3. the *ordering*: whether the tight gaps come in runs (a burst of ten packets $5 \mu\text{s}$ apart) or lie singly among long gaps;

4. *slow rate variation*: busy and quiet stretches at the scale of seconds within a window.

A null stream is a synthetic stream that keeps some of these and destroys the rest. Running the identical simulator on the real stream and on the null, and comparing the tails, measures the share of the tail that the destroyed property was carrying. Each null is a controlled subtraction of one ingredient, and the design is factorial in the sense that the four properties are removed in different combinations by different nulls. The reference quantity throughout is the single-stage p_{99} excess, $p_{99} - T$, and the statistic reported is the fraction of the real stream’s excess that survives on the null, corpus-median over the 3512 windows. All nulls are built per window from that window’s own real stream, with seeds derived from the session and window identifiers so that every run is reproducible. The 3512 windows are nested in 276 sessions, so the corpus medians reported here are point estimates and no interval is attached to them.

Packet-level nulls. The uniform Poisson null P of Section 3.5 redraws every packet time uniformly over the window: it keeps (1) and destroys (2), (3) and (4) together, so a tail that vanishes under it is attributable to clustering in some form but not to any particular form. The *gap shuffle* G takes the window’s sequence of interarrival gaps, permutes it at random (Fisher–Yates, the renewal null already used for the Fano diagnostic of Section 3.2), and re-accumulates arrival times: the same packet count, the same multiset of gaps and therefore the same tight-gap fraction and the same $7.5\mu\text{s}$ floor, with the ordering destroyed. G keeps (1) and (2) and removes (3) and (4). A tail that survives G is set by how many tight gaps the stream contains; a tail that G removes needs the runs. The *1 s-binned Poisson null* B cuts the window into one-second bins, keeps each bin’s real packet count, and redraws the arrivals uniformly inside the bin: it keeps (1) and (4) and removes (2) and (3). A tail that survives B is a busy-second load effect, not a microsecond one. Section 4.2 reports all three.

Transaction-level nulls. The packet nulls cannot distinguish two accounts of the tight-gap surplus: that the exchange splits one matching-engine event across back-to-back packets, or that the events themselves arrive clustered and the packets relay them. Section 4.3 therefore groups every message by its `transactTime`, the identifier of one matching-engine event (a *transaction*), and builds five further streams in which each transaction is a block — its packets, their message counts and their internal spacing — and exactly one ingredient is removed. Service in these runs is charged per packet at its measured cost, $S_i = T(1 + r(\sigma_i - 1))$ with σ_i the message count and r the live decoder’s slope-to-floor ratio, so that a large packet adds work rather than redistributing it; if the size of a transaction matters to the tail, these runs can show it. Table 1 lists all nine streams.

The logic of the two families is the same and they are read together. If P and TP remove the tail, the tail is produced by the timing of the transaction process. If TS, TM and TW leave it, the size of transactions, their division into packets, and the spacing of those packets are not the mechanism, and the exchange is relaying events rather than creating bursts. G and TG remove the same ingredient — the arrangement of events into runs — at the packet and at the transaction level respectively; agreement between them identifies the packet-level ordering channel with the clustering of transactions. What survives G is the surplus of tight gaps over a rate-matched Poisson stream, which Table 5 measures directly. That surplus shows the gap distribution is far from exponential; it does not show what produces it, and G itself is a stream with the surplus and no self-excitation. What G removes, the runs, is the self-exciting component.

stream	kept	destroyed	question answered
H	the real stream	—	reference
P	packet count, support	gap distribution, ordering, slow rate variation	is the tail due to clustering in any form?
G	count; the multiset of gaps, hence the tight-gap fraction and the floor	ordering of gaps into runs; slow rate variation	is the tail set by how many tight gaps there are, or by their arrangement?
B	count per one-second bin	microsecond gap structure and ordering	is the tail busy-second load?
TG	every transaction intact, in its original order	the idle gaps between transactions, permuted	does the clustering of transactions into runs build the tail?
TP	every transaction intact	transaction start times, redrawn uniformly	does transaction timing matter at all?
TS	real start times	which transaction shape sits at which start (shapes permuted across starts)	do large transactions arriving at particular moments matter?
TM	real start times	the split of a transaction into packets (each transaction becomes one packet)	does the exchange's packetisation matter?
TW	real start times, the split into packets	the tight spacing inside a split transaction (packets spread by the median idle gap)	does back-to-back spacing inside a transaction matter?

Table 1. The nine arrival streams. P , G and B act on packet times with constant service and are reported in Table 3; TG – TW act on transaction blocks with span-dependent service and are reported in Table 7. Reading the results together: $P \approx 0$ and $TP \approx 0$ say the tail is produced by the timing of the transaction process; TS , TM , $TW \approx 1$ say sizes, packetisation and intra-transaction spacing are not the mechanism; G and TG , which remove the same thing at two levels of description, give the same profile across T and split the clustering into the two channels named in Section 4.3.

4 Empirical results

4.1 Single-thread and tandem tails on the real stream and on the Poisson null

This section presents the main result. It runs the simulated pipeline over the whole corpus and reports, for each task length and each stage count, how long a typical message takes and how long the slowest one in a hundred takes. Each case is run twice: once on the real packet arrivals, which come in bursts, and once on a control in which the same number of packets is scattered uniformly at random over the window. The comparison isolates the effect of bursts. The findings are these. On randomly scattered arrivals there is almost no slow tail at the task lengths a trading hot path uses, so splitting a task across threads gains nothing and costs a hop per stage. On the real, bursty arrivals a slow tail appears once a task is longer than about seven microseconds, and it grows quickly with task length, until the slowest messages take tens of times longer than the task itself. There, splitting the task into two or more stages cuts the tail sharply for a small, fixed cost on the typical message, and the longer the task, the more stages pay off. In short, it is the bursts, not the average load, that create the tail, and that is why splitting helps on real data and not on the random control.

Table 2 reports corpus-median end-to-end latency quantiles for every (T, N) cell of the sweep at $h = 1.7 \mu\text{s}$, on 276 sessions and 3512 thirty-minute windows (five of the 281 session tapes contain no complete RTH window and are dropped). Figure 2 plots the same numbers. The Poisson-null columns are the uniform Poisson null P of Section 3.5 with the same packet count per window.

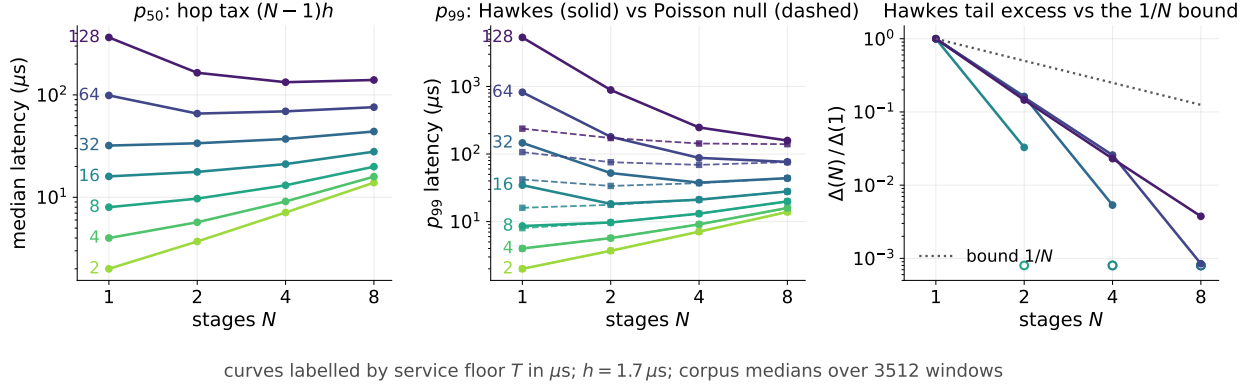


Figure 2. Corpus-median latency versus stage count N for each service floor T (one shade per T , each curve labelled at its $N = 1$ end). Left: median. Centre: p_{99} , Hawkes solid and Poisson null dashed. Right: the Hawkes tail excess ratio $\Delta(N)/\Delta(1)$ against the Theorem 1 bound $1/N$ (dotted); open markers are cells where the tandem drives $\Delta(N)$ to exactly zero, drawn at the panel floor because zero cannot be placed on a log axis. The median rises by exactly h per hop wherever the queue is empty at the median, which is every cell except the three longest-service Hawkes cells. The Hawkes p_{99} falls with N only at $T \geq 16 \mu\text{s}$; at $T \leq 8 \mu\text{s}$ there is no tail to compress and it rises with the hop count, tracking the median. The Poisson p_{99} likewise tracks the median up to $T = 16 \mu\text{s}$ and separates from it at $T \geq 32 \mu\text{s}$, where utilisation queueing begins.

Under the Poisson null the queue is empty when almost every message arrives, so every reported quantile equals the service floor, $p_{50} = p_{99} = T + (N - 1)h$, in every cell with $T \leq 16 \mu\text{s}$. The first exception is $T = 32 \mu\text{s}$, $N = 1$, where Poisson p_{99} is $42.0 \mu\text{s}$ against a floor of 32 : at the corpus-median packet rate of 452 packets per second the utilisation there is $\rho = 452 \times 32 \mu\text{s} = 0.0144$, above the 0.01 at which Corollary 2 says the Poisson wait must first appear at p_{99} , and it does; at $N = 2$ the per-stage utilisation halves to 0.007 and the wait vanishes again. Beyond that the null arm is a perfectly ordinary queue: at $T = 64$ and $128 \mu\text{s}$ the single-stage Poisson p_{99} reaches 106.7 and $237.7 \mu\text{s}$, and splitting reduces it, to 75.7 and $172.4 \mu\text{s}$ at $N = 2$. Under Poisson, splitting is

T	N	Hawkes (real arrivals)			Poisson null		
		p_{50}	p_{99}	$p_{99.9}$	p_{50}	p_{99}	$p_{99.9}$
2	1	2.00	2.00	2.00	2.00	2.00	2.00
2	2	3.70	3.70	3.70	3.70	3.70	3.70
2	4	7.10	7.10	7.10	7.10	7.10	7.10
2	8	13.90	13.90	13.90	13.90	13.90	13.90
4	1	4.00	4.00	4.00	4.00	4.00	5.78
4	2	5.70	5.70	5.70	5.70	5.70	5.70
4	4	9.10	9.10	9.10	9.10	9.10	9.10
4	8	15.90	15.90	15.90	15.90	15.90	15.90
8	1	8.00	8.61	10.14	8.00	8.00	13.79
8	2	9.70	9.70	9.70	9.70	9.70	11.48
8	4	13.10	13.10	13.10	13.10	13.10	13.10
8	8	19.90	19.90	19.90	19.90	19.90	19.90
16	1	16.00	34.59	53.97	16.00	16.00	29.84
16	2	17.70	18.31	19.84	17.70	17.70	23.49
16	4	21.10	21.10	21.10	21.10	21.10	22.88
16	8	27.90	27.90	27.90	27.90	27.90	27.90
32	1	32.00	146.14	259.11	32.00	42.02	62.02
32	2	33.70	52.29	71.67	33.70	33.70	47.54
32	4	37.10	37.71	39.24	37.10	37.10	42.89
32	8	43.90	43.90	43.90	43.90	43.90	45.68
64	1	98.88	820.70	1548.68	64.00	106.73	126.72
64	2	65.70	179.84	292.81	65.70	75.72	95.72
64	4	69.10	87.69	107.07	69.10	69.10	82.94
64	8	75.90	76.51	78.04	75.90	75.90	81.69
128	1	364.45	5316.85	12487.72	128.00	237.65	289.07
128	2	164.58	886.40	1614.38	129.70	172.43	192.42
128	4	133.10	247.24	360.21	133.10	143.12	163.12
128	8	139.90	158.49	177.87	139.90	139.90	153.74

Table 2. Corpus-median end-to-end latency quantiles in μs at $h = 1.7 \mu\text{s}$, for each service floor T (μs) and stage count N ; 276 sessions, 3512 thirty-minute RTH windows of CME MDP3 NQ front-month packet arrivals. The Poisson null is the uniform shuffle of the same arrival count per window. What the table shows: the median column equals $T + (N - 1)h$ in every cell but three (Hawkes $N = 1$ at 64 and 128 μs and $N = 2$ at 128), so each stage costs exactly one hop at the median. Under real arrivals the single-stage p_{99} sits on the floor at $T \leq 4 \mu\text{s}$, is $1.08T$ at $8 \mu\text{s}$, and then grows faster than the service time: $2.2T$, $4.6T$, $12.8T$ and $41.5T$ at $T = 16, 32, 64$ and $128 \mu\text{s}$, with $p_{99.9}$ roughly twice the p_{99} excess again. Under the Poisson null the single-stage p_{99} equals the median up to $16 \mu\text{s}$ and is $1.3T$, $1.7T$ and $1.9T$ at 32, 64 and 128. One cut ($N = 2$) takes the Hawkes p_{99} from 34.6 to 18.3 at $16 \mu\text{s}$, from 146 to 52 at 32, from 821 to 180 at 64 and from 5317 to 886 at 128, for a median cost of $1.7 \mu\text{s}$; further cuts continue to lower the tail until the per-stage service falls below the floor, after which every quantile rises by h per stage. Below about $8 \mu\text{s}$ there is no queue to shorten and splitting is cost only, under either arrival law; from $16 \mu\text{s}$ the real stream carries a tail that the Poisson stream at the same rate does not, and splitting removes most of it for one hop; the Poisson tail that appears from $32 \mu\text{s}$ is ordinary utilisation queueing ($\rho = \bar{\lambda}T$ reaching 0.01–0.05) and is 11 to 47 times smaller than the real one. The reason is the arrival process: a deterministic server queues only when consecutive gaps are shorter than its service time, the real stream has four to seventeen times the Poisson share of such gaps (Table 5) and arranges them in runs, and a cut halves the service each stage must fit into a gap, so the number of gaps that queue, and the length of the runs that compound, both fall with N (Theorem 1).

therefore not useless in general; it is useless in the regime an HFT hot path occupies. The $p_{99.9}$ column shows the same pattern one decade lower: the Poisson wait is visible at $N = 1$ for every $T \geq 4 \mu\text{s}$ (single-stage $\rho > 10^{-3}$) and at $N = 2$ for $T \geq 8 \mu\text{s}$, and absent wherever the per-stage ρ/N falls below 10^{-3} . Every one of these switch-on points is the $\rho = 1 - q$ boundary of Corollary 2, which the corpus reproduces at $q = 0.99$ and 0.999 (Section 6). The null arm is a queue with a small, predictable wait, and over $T \leq 16 \mu\text{s}$ splitting under Poisson costs $(N - 1)h$ at every reported quantile.

Under Hawkes a tail appears only above a threshold. At $T = 2$ and $4 \mu\text{s}$ there is no tail: single-stage p_{99} and $p_{99.9}$ both equal the service floor, the clustered feed is indistinguishable from the Poisson null at p_{99} and $p_{99.9}$, and splitting costs $(N - 1)h$ and removes nothing. The tail appears between $T = 4$ and $8 \mu\text{s}$ and then grows rapidly: single-stage p_{99} is 1.00, 1.00, 1.08, 2.16, 4.57, 12.8 and 41.5 times the service floor at $T = 2, 4, 8, 16, 32, 64, 128 \mu\text{s}$, and single-stage $p_{99.9}$ reaches $97.6T$ at $T = 128$. Splitting into two stages cuts the p_{99} from 34.59 to $18.31 \mu\text{s}$ at $T = 16$, from 146.1 to 52.3 at $T = 32$, from 820.7 to 179.8 at $T = 64$, and from 5316.9 to 886.4 at $T = 128$, in each case for a median cost of $1.7 \mu\text{s}$ per hop; the two-stage p_{99} is below the single-stage p_{99} on 100% of the 3512 windows at every $T \geq 16 \mu\text{s}$. The stage count that minimises p_{99} rises with T : $N = 1$ at $T \leq 8$, $N = 2$ at $T = 16$, $N = 4$ at $T = 32$, $N = 8$ at $T = 64$ and 128. The $T = 2$ and $4 \mu\text{s}$ rows, both below the floor, are included to demonstrate that boundary: splitting there is dominated under both arrival laws. Section 4.8 resolves the $4\text{--}8 \mu\text{s}$ onset to $1 \mu\text{s}$ and Section 4.9 explains what sets its location.

4.2 Gap-shuffle and 1 s-binned nulls

The Poisson null of Table 2 redraws every arrival uniformly over the whole window. That removes three things at once: the microsecond-scale self-excitation, the slower rate variation across the window, and the shape of the gap distribution, including the tight-gap floor of Section 4.9. The tail difference in Table 2 is credited to the first of these, but the mechanism of Section 4.9 runs on the third: a queue forms when consecutive gaps are shorter than the service time, and that is a property of the gap distribution, which a reordering of the same gaps leaves untouched. Two further nulls separate the three. The *gap shuffle* permutes each window's interarrival gaps at random (the renewal null of Section 3.2): same count, same gap distribution, same tight-gap floor, no ordering. The *1 s-binned Poisson* keeps each one-second bin's own packet count and redraws the arrivals uniformly inside the bin: same second-scale rate variation, no microsecond clustering.

T	N	real	uniform	gap shuffle	1 s-binned
2	1	2.00	2.00	2.00	2.00
2	2	3.70	3.70	3.70	3.70
2	4	7.10	7.10	7.10	7.10
2	8	13.90	13.90	13.90	13.90
4	1	4.00	4.00	4.00	4.00
4	2	5.70	5.70	5.70	5.70
4	4	9.10	9.10	9.10	9.10
4	8	15.90	15.90	15.90	15.90
8	1	8.61	8.00	8.54	8.00
8	2	9.70	9.70	9.70	9.70
8	4	13.10	13.10	13.10	13.10
8	8	19.90	19.90	19.90	19.90
16	1	34.59	16.00	30.75	16.00
16	2	18.31	17.70	18.24	17.70
16	4	21.10	21.10	21.10	21.10
16	8	27.90	27.90	27.90	27.90
32	1	146.14	42.02	99.61	47.36
32	2	52.29	33.70	48.45	33.70
32	4	37.71	37.10	37.64	37.10
32	8	43.90	43.90	43.90	43.90
64	1	820.70	106.73	340.62	112.62
64	2	179.84	75.72	133.31	81.06
64	4	87.69	69.10	83.85	69.10
64	8	76.51	75.90	76.44	75.90
128	1	5316.85	237.65	1306.63	246.03
128	2	886.40	172.43	406.32	178.32
128	4	247.24	143.12	200.71	148.46
128	8	158.49	139.90	154.65	139.90

Table 3. Corpus-median p_{99} in μs under the real arrivals and the three nulls of Section 3.6, for every service floor T and stage count N of the sweep. Uniform is the Poisson null of Table 2; the gap shuffle keeps the multiset of gaps and destroys their ordering; the binned null keeps each one-second packet count and destroys the microsecond structure. What the table shows: at $T \leq 4 \mu\text{s}$ every arm sits on the floor $T + (N - 1)h$, because no gap in the stream is shorter than $4 \mu\text{s}$ and no queue can form. At $N = 1$ and $T \geq 8 \mu\text{s}$ the real stream carries an excess over T (0.61, 18.6, 114, 757 and $5189 \mu\text{s}$ at $T = 8, 16, 32, 64, 128$) that the binned null and the uniform null remove almost entirely (both at or within 2–15% of the excess in every cell, and identical to each other at 8 and $16 \mu\text{s}$), while the gap shuffle retains 88%, 79%, 59%, 37% and 23% of it. At $N \geq 2$ the four arms converge on $T + (N - 1)h$ as soon as the per-stage service T/N falls below the floor (at $T = 16, N = 2$: 18.3, 17.7, 18.2, 17.7). The tail is not second-scale load, since a stream with the real busy seconds and random microsecond placement has none of it; at 8 – $32 \mu\text{s}$ the tail is set mainly by how many gaps shorter than T the stream contains, since a stream with the same gaps in random order keeps most of it; and from $64 \mu\text{s}$ the ordering of those gaps into runs carries the majority. The reason is the Lindley recursion itself: a single gap shorter than T costs the next packet one wait, which the following long gap drains, so a short service time is hit by every tight gap whether or not its neighbours are tight, whereas a long service time needs a sustained run of tight gaps before the backlog exceeds one service, and runs are what the shuffle removes. The tandem’s benefit does not depend on which channel carries the tail: a cut that takes T/N below the floor removes the excess under every arm. Session-clustered bootstrap over the 276 sessions (1000 draws), 95% intervals of the gap-shuffle shares: 87.1–88.4%, 76.7–81.3%, 58.2–60.5%, 35.8–37.2% and 21.9–23.6% at $T = 8$ to $128 \mu\text{s}$; of the binned null at 32–128: 12.5–14.2%, 6.2–6.6%, 2.2–2.4%.

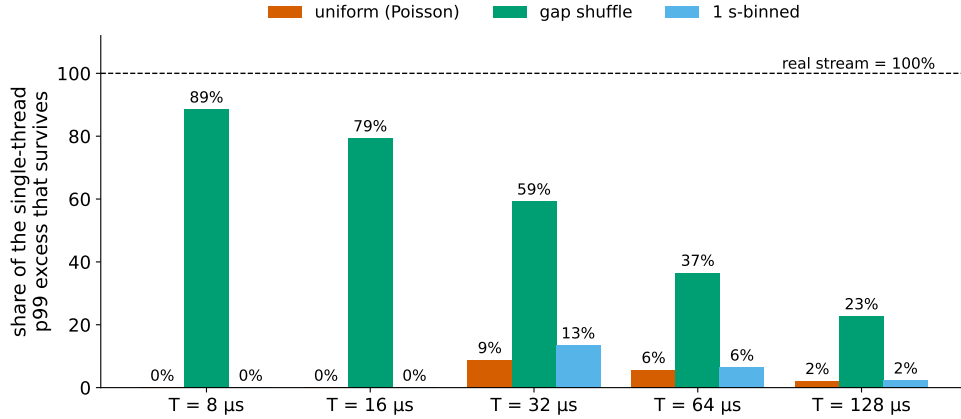


Figure 3. Table 3 as bars: the share of the single-thread p_{99} excess ($p_{99} - T$) that survives each null stream, the real stream being 100%. The gap shuffle (green), which keeps every gap and destroys their order, keeps most of the tail at 8–16 μs and less as T grows: at short service times the tight gaps carry the tail, at long ones their arrangement into runs. The uniform and 1 s-binned streams keep almost none at any T .

Table 3 gives the corpus-median p_{99} under all four arrival arms. The share of the single-stage p_{99} excess over T that survives the gap shuffle is 88% at $T = 8 \mu s$, 79% at 16 and 59% at 32 μs , falling to 37% at 64 and 23% at 128 μs . The 1 s-binned null keeps none of it at 8 and 16 μs and 13%, 6% and 2% at 32, 64 and 128 μs , within a few points of the uniform null. Two readings follow. The tail is not second-scale load: a null that keeps every one-second packet count has no tail where the real stream has one. And at the service times of an HFT hot path the tail is set mainly by the gap *distribution* — the surplus of gaps shorter than the service time over what a Poisson stream at the same rate carries — with the *ordering* of those gaps into runs supplying the remaining fifth at 16 μs and the majority only from 64 μs upward. Section 4.3 traces both to the timing of matching-engine transactions: the surplus to consecutive transactions sent back to back at the publisher’s spacing (Table 12), the ordering to runs of them, which is the self-exciting component. The attribution to clustering stands in that sense; which process places two transactions within a few microseconds of each other is not identified by any stream run here (Section 12.3).

4.3 Grouping packets into matching-engine transactions

The nulls of Section 4.2 act on packet times. A packet is the exchange’s unit, not the market’s. CME MDP3 publishes each matching-engine event — a *transaction*, identified by the common `transactTime` carried by every message it generates — as one or more UDP datagrams, and a datagram may carry the end of one transaction and the start of the next. Two accounts of the tight-gap surplus are therefore open. Either the exchange splits transactions across back-to-back packets, in which case the clustering is an artefact of packetisation and a receiver that reassembled transactions would see none of it; or the transactions themselves arrive clustered and the packets relay them. The corpus decides between the two. Every message in the 3512 windows was grouped by `transactTime` and every packet labelled with the transactions it carries: 3.30×10^9 packets, 3.55×10^9 messages, 3.24×10^9 transactions; 3.0% of packets (median window) carry parts of two transactions — the gateway coalesces transactions that arrive within its emission interval into one datagram, which Section 7.5 takes up.

Consecutive transactions and the gap between them. A *transaction* is one matching-engine event: all messages that carry the same `transactTime`. One incoming order that trades against three resting orders is one transaction, however many messages it produces. Within a session the NQ front-month transactions are put in the order the engine processed them, $A, B, C, D, \dots$; two transactions are *consecutive* when one directly follows the other in that order, with no other NQ front-month transaction between them (A and B , B and C , and so on). The *gap between consecutive transactions*, their inter-arrival time, is the time from one to the next. It is measured on one of two clocks, always named: the matching engine’s (the difference of the two `transactTimes`) or the publisher’s (the difference of the `sendingTimes` of the packets that carry them, zero if one packet carries both). A session with m transactions has $m - 1$ such gaps. Every statement in the paper about how close transactions are is a statement about these gaps; a *tight* gap is one below $16\mu\text{s}$.

	1	2	3	4	5	6–10	11–20	21+
messages per packet	95.897	2.439	0.869	0.401	0.188	0.171	0.027	0.009
packets per transaction	98.295	1.689	0.011	0.002	0.001	0.001	0.000	0.000
messages per transaction	95.362	2.729	0.822	0.558	0.195	0.280	0.038	0.016

Table 4. Size distributions, percent of the pooled corpus (3.30×10^9 packets, 3.24×10^9 transactions). A transaction is one packet 98.3% of the time and two 1.7% of the time; three or more packets is about 1.5 transactions in ten thousand. The message count per transaction is the message count per packet with the split transactions folded back in.

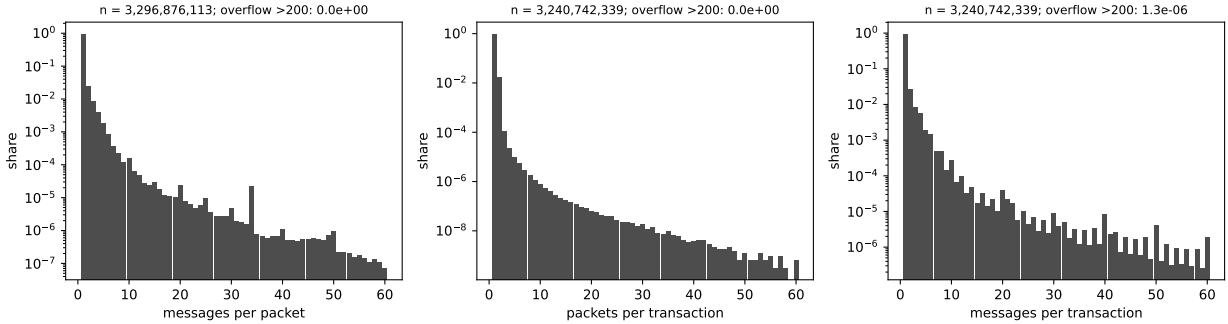


Figure 4. The three size distributions of Table 4 on a logarithmic share axis, counts 1–60. Messages per packet and per transaction have the same geometric decay; packets per transaction has almost none beyond two.

Packetisation and the tight gaps. Table 4 gives the three size distributions, and Figure 4 plots them. A split transaction is rare, and when it occurs its packets are not back-to-back: the gap between two packets of the same transaction has a corpus-median p_{50} of $15\mu\text{s}$, 1.6% of such gaps are shorter than $10\mu\text{s}$ and 0.1% shorter than the $7.5\mu\text{s}$ floor; a multi-packet transaction spans $15\mu\text{s}$ at the median and $110\mu\text{s}$ at p_{99} from first to last packet (Figure 5, left, and Figure 6, right). The tight gaps sit between transactions (Table 5).

gap shorter than	share of all packet gaps	of which inside one transaction	Poisson at the same rate
7.9 μs (the floor)	2.15%	0.1%	0.56%
17.8 μs	20.6%	5.0%	1.24%
35.5 μs	34.1%	4.6%	2.46%

Table 5. Tight packet gaps, pooled over the corpus (3.30×10^9 gaps). The rows are the bin edges of the logarithmic grid of Figure 5 nearest to the 7.5 μs floor and to the 16 and 32 μs service times. 2.15% of all packet gaps are shorter than the floor, 20.6% shorter than about 16 μs and 34.1% shorter than about 32 μs ; a Poisson stream at the same rate would carry 0.56%, 1.24% and 2.46%, so the real stream has four, seventeen and fourteen times the Poisson share of gaps below the three service times of interest. Of the gaps below the floor, 0.1% lie between two packets of the same transaction; below 16 and 32 μs the shares are 5.0% and 4.6%. The tight gaps that make a deterministic server queue are overwhelmingly gaps between different matching-engine events, not the exchange’s division of one event into several datagrams, so a receiver that reassembled transactions before decoding would remove at most 5% of the gaps that queue and almost none below the floor; and the surplus of tight gaps over Poisson, which Table 3 shows carries most of the tail at 8–16 μs , is a property of when transactions occur relative to each other. The reason the surplus exists is the self-excitation established in Table 6: a transaction raises the probability of another within tens of microseconds, which puts far more mass at short gaps than a Poisson stream at the same mean rate carries.

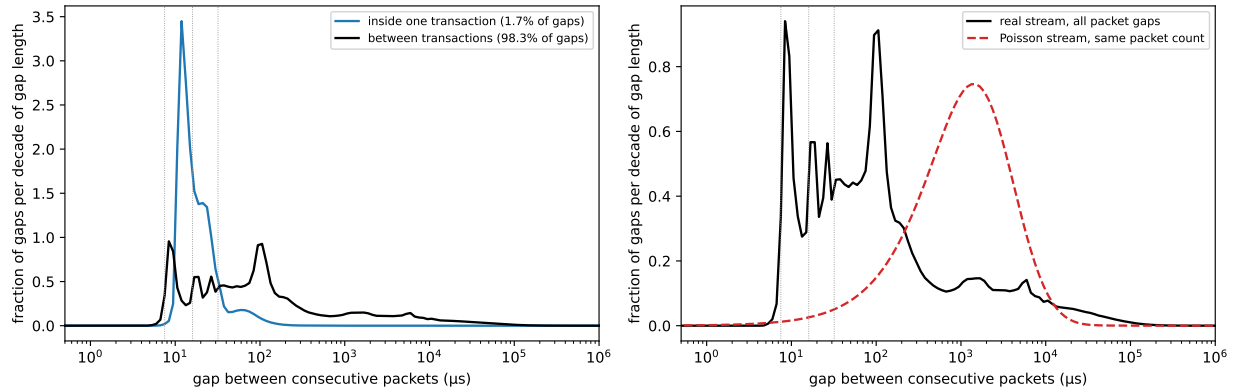


Figure 5. Packet interarrival gaps, pooled over the corpus. Left: the two packets of a split transaction arrive about 15 μs apart (blue: median 15 μs , 54% under 16 μs , 96% under 64 μs ; $n = 5.6 \times 10^7$ gaps), closer than the gap between consecutive packets of different transactions (black: median about 80 μs , spread from the 7.5 μs floor to seconds; $n = 3.24 \times 10^9$ gaps, 98.3% of all gaps) — but almost never as close as the floor (0.1%). Below 10 μs the blue curve holds 1.6% of its gaps and below 7.5 μs 0.1%; the region from the floor to 12 μs is filled by the black curve alone. So the exchange paces the pieces of one event at a steady 15 μs , and the tightest gaps on the feed, the ones that make a receiver queue, are between packets of different events. Right: the gap distribution of the real stream (black; the left panel’s black curve with the blue folded in) against that of a Poisson stream with the same packet count over the same window (red, dashed) — the same null as in Table 2, drawn in gap space. The Poisson density is one hump near 1 ms with a few percent of its mass below 100 μs ; the real density puts a third of its mass below 32 μs , with the edge at the floor and the mode near 100 μs absent from the null. The panel is the visual form of Table 5: the whole region of gaps shorter than an HFT service time is present in the real stream and absent from an independent-events stream at the same rate, which is why the uniform null carries no tail below 32 μs , and the 100 μs mode is the characteristic spacing of successive transactions within a burst, the scale on which the self-excitation of Table 6 acts. Axes: logarithmic gap axis; vertical axis the fraction of gaps per decade of gap length (twenty bins per factor of ten; the area under a curve over a range is the fraction of that curve’s gaps in the range, total area one, so heights compare shapes, not counts). Dotted lines at 7.5, 16 and 32 μs .

Self-excitation of the transaction stream. Treating each transaction as one event at the time of its first packet, the transaction-start process carries the same clustering as the packet process (Table 6): the Fano factor of transaction counts rises from 7.6 at 1 ms to 105 at 1 s, against 5.5 to 17 once the idle gaps between transactions are shuffled and 1 for Poisson, and the exponential-kernel MLE gives a branching ratio of 0.795 on transaction starts against 0.798 on packets, with every window’s fit converged. The transaction-start gap density of Figure 6 (left) is indistinguishable from the packet gap density of Figure 5 and unlike the Poisson curve at every scale. The self-excitation belongs to the transaction stream: the exchange publishes each matching event as it occurs, and the packet process inherits the clustering of the events. Why the events cluster is a question about the market, outside the scope of this paper (Section 12.3).

count window τ	transactions	transactions, idle gaps shuffled	packets
1 ms	7.6	5.5	7.8
10 ms	21.6	10.8	22.2
100 ms	50.5	15.8	51.8
1 s	105.4	17.1	108.4
branching ratio n	0.795	—	0.798

Table 6. Fano factor $\text{Var}[C_\tau]/\text{E}[C_\tau]$ of event counts, corpus median over 3512 windows, for transaction starts, for the same starts with the idle gaps between transactions randomly permuted (transactions kept intact), and for packets; Poisson gives 1 at every τ . Last row: exponential-kernel Hawkes branching ratio, corpus median, all fits converged. The within-window Fano values agree with those of Section 3.2 at the same bin width.

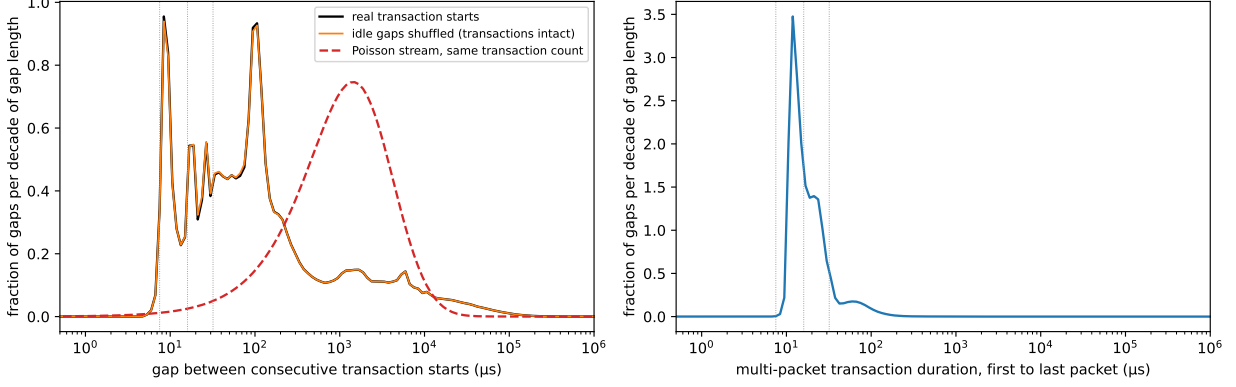


Figure 6. Gaps between consecutive *transaction starts*, to be read against Figure 5, whose gaps are between consecutive *packets*. Left: each transaction is collapsed to one event at the time of its first packet, and the gaps between successive events are shown for the real stream (black; $n = 3.24 \times 10^9$), for the same events with the idle gaps between transactions shuffled into random order (orange), and for a Poisson stream with the same transaction count over the same window (red, dashed). At the resolution of the figure the black curve here is the black curve of Figure 5 (left) — the same edge at the $7.5 \mu\text{s}$ floor, the same plateau to $30 \mu\text{s}$, the same mode near $100 \mu\text{s}$, the same tail — because 98.3% of transactions are one packet and the pieces of the remaining 1.7% are paced at $15 \mu\text{s}$, inside the plateau, so removing the packet structure changes almost nothing. The orange curve lies exactly on the black one: permuting the order of the idle gaps leaves their distribution unchanged, by construction. The Poisson curve is a different shape, one hump near 1 ms with a few percent of its mass below $100 \mu\text{s}$. The packet stream follows the transaction stream. The clustering that Figure 5 shows at the packet level is present, unchanged, at the level of matching-engine events, and it is those events, not the exchange’s delivery of them, that arrive in tight groups; the same conclusion is reached numerically in Table 6, where transaction starts and packets have the same Fano factor at every scale and the same branching ratio (0.795 against 0.798), and by the simulator in Table 7, where merging each transaction into one packet leaves the tail unchanged. Right: duration of a multi-packet transaction from its first to its last packet ($n = 5.5 \times 10^7$), median $15 \mu\text{s}$, p_{99} $110 \mu\text{s}$: the time the exchange takes to deliver a split event, and the spacing that the blue curve of Figure 5 shows between its pieces. Axes as in Figure 5: logarithmic gap axis, fraction of gaps per decade of gap length; dotted lines at 7.5 , 16 and $32 \mu\text{s}$.

The tail under the transaction-level nulls. To separate the parts of the transaction process that build the queue, each window is re-simulated at $N = 1$ with service charged per packet at its measured absolute cost, $S_i = T(1 + r(\sigma_i - 1))$ with σ_i the packet’s message count and r the slope-to-floor ratio of the live decoder (0.043 on NQ book, 0.142 on ZN book, Section 7.3), so that a large packet adds work rather than redistributing it. Five counterfactual streams are built from the real one (H), each removing one ingredient. TG keeps every transaction intact — its packets, their sizes and their spacing — and permutes the idle gaps between transactions, removing the clustering of transactions. TP keeps the transactions intact and redraws their start times uniformly over the window. TS keeps the real start times and permutes the transactions’ shapes among them, decoupling size from timing. TM merges each transaction into a single packet at its start. TW keeps the starts and spreads a split transaction’s packets by the median idle gap instead of their real spacing. Table 7 reports the corpus-median p_{99} of each stream and the share of the real stream’s excess over T that it retains.

T	H	TG	TP	TS	TM	TW
16	34.8	31.0 (80)	20.7 (25)	37.4 (114)	34.9 (101)	35.6 (104)
32	146.8	100.5 (60)	52.9 (18)	155.9 (108)	146.3 (100)	148.1 (101)
64	832	345 (37)	118 (7)	839 (101)	795 (95)	833 (100)
128	5460	1319 (22)	251 (2)	5218 (95)	5035 (92)	5459 (100)

Table 7. Which part of the transaction process builds the queue. Each row is one service time T ; each column is the single-stage p_{99} latency in μs (corpus median over 3512 windows, service charged per packet at the NQ slope ratio, packet-weighted) when the simulator is driven by the real stream H or by one of five rebuilt streams, each identical to the real one except for a single ingredient that has been removed. The number in parentheses is the share, in percent, of the real stream’s tail excess $p_{99} - T$ that the rebuilt stream still produces: 100 means removing that ingredient changed nothing, 0 means it removed the whole tail. The five columns ask: TG — keep every transaction exactly as it was, but scatter the transactions so that they no longer come in bunches: does the tail survive? TP — keep every transaction as it was but place each one at a random moment in the window: does the tail survive? TS — keep the moments at which transactions occur but swap their contents around, so that a big transaction now arrives where a small one did: does the tail survive? TM — deliver every transaction as one packet: does the tail survive? TW — keep the split into packets but spread the pieces of a split transaction apart: does the tail survive? TS, TM and TW are at 92–114% in every row — the tail survives all three, so the size of transactions, their division into packets, and the spacing of those pieces are not what builds the queue. TP is at 25, 18, 7 and 2% — placing transactions at random times removes three quarters of the tail at 16 μs and almost all of it at 128 μs , so the tail is the timing of transactions. TG is at 80, 60, 37 and 22% — scattering the bunches while keeping each transaction intact removes a fifth of the tail at 16 μs and three quarters at 128 μs . The queue is built by *when* transactions happen, and by nothing about *what* they contain or *how* the exchange delivers them. The timing acts in two ways whose weight shifts with the service time. At short service times (16–32 μs) most of the tail is already present in TG, which keeps the number of transactions that land within a service time of each other but not their grouping into bunches: a short server queues whenever the gap between consecutive transactions is shorter than its service time, so what matters is how many such gaps there are, and the real stream has 4–17 times as many as a Poisson stream at the same rate (Table 5). At long service times (64–128 μs) most of the tail is gone in TG: a long server only backs up badly when many transactions arrive in a sustained run, and the runs are what scattering destroys. This is the same profile that the packet-level gap shuffle gave in Table 3 (88, 79, 59, 37, 23%), which identifies the packet-level ordering channel with the bunching of transactions. At $T \leq 8 \mu s$ the real excess is under 1 μs and the shares are not meaningful. The ZN slope ratio and message weighting move every cell by at most a few points. Session-clustered bootstrap over the 276 sessions (1000 draws), 95% intervals: TG 77.0–81.7, 58.7–61.0, 35.7–37.2, 21.4–23.2% at $T = 16, 32, 64, 128$; TP 24.0–25.9, 17.7–18.8, 6.8–7.2, 2.2–2.4%; TS, TM and TW within ± 2 points of the reported values in every cell.

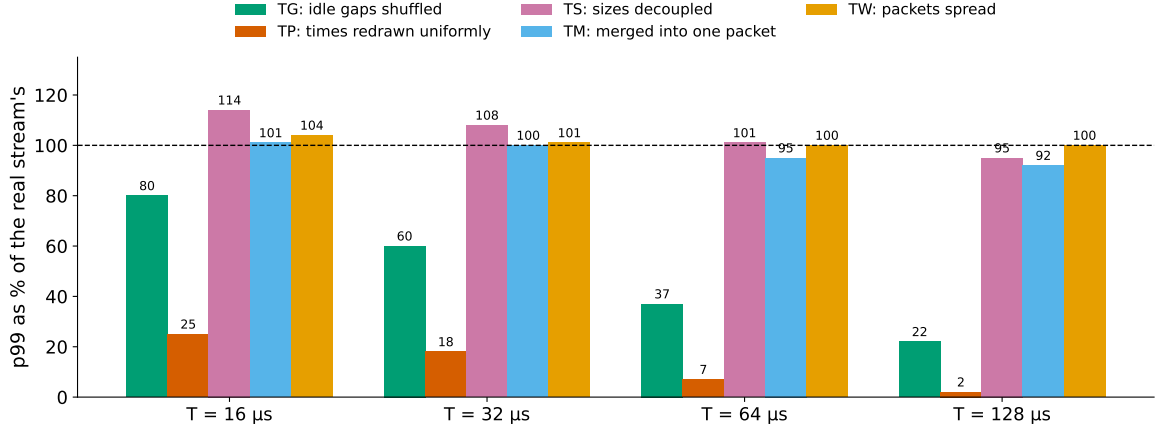


Figure 7. Table 7 as bars: each transaction-level arm’s p_{99} as a percentage of the real stream’s. Redrawing transaction times uniformly (TP, vermillion) removes the tail; shuffling idle gaps (TG, green) removes more of it the longer the service time. Decoupling sizes (TS), merging each transaction into one packet (TM) or spreading its packets (TW) leaves it near 100% from 16 μ s up.

Three things follow. The tail is the transaction process: randomising transaction times removes 75–98% of it. Sizes and packetisation do not enter: decoupling transaction sizes from timing, merging each transaction into one packet, or removing the spacing inside a split transaction leaves the tail within $\pm 15\%$ of its value at every service time, at the ZN slope as at the NQ slope. And the clustering of transactions — the ordering that TG destroys — carries 20% of the tail at 16 μ s, 40% at 32, 63% at 64 and 78% at 128 μ s, the same profile the packet-level gap shuffle gave in Table 3. The remainder, dominant at HFT service times, is the surplus of tight gaps between transactions over a Poisson stream at the same rate (Table 5), which on the publisher’s clock sit at its spacing (Table 12). That surplus says the transaction gap distribution is far from exponential; it does not say what put the mass there, and a renewal stream with the same gaps and no self-excitation (the gap shuffle) reproduces it and most of the tail at 16 μ s. The attribution is therefore the following. The queueing tail behind a single-threaded receiver on this feed is produced by the timing of matching-engine transactions; it is not produced by packetisation, by the message count per packet, or by the packet rate; it acts on a short service time through how often two transactions land within one service time of each other, and on a long one through how long the runs of them last, which is the self-excitation. Why two transactions land a few microseconds apart is a question about the market, not about the receiver, and is left open (Section 12.3).

4.4 Tandem, sharding and dispatch on the same N cores

The tandem of Table 2 runs on N cores and the single thread on one, so part of what it buys is capacity: its throughput is N/T against $1/T$, and more capacity shortens queues under any arrival law. The design question is whether a chain of stages is the right way to spend the same N cores, and the two alternatives are the ones HFT systems use. *Sharding* runs N independent single-thread loops, each owning a disjoint slice of the stream, by instrument or by channel: full service T per message and no hops, so it has the lower median, and its tail depends on whether a burst lands on one shard or spreads over several. *Dispatch*, an M/D/ N queue, puts one queue in front of N identical servers each taking a whole packet at service T : the same capacity N/T as the tandem and no hops, so a burst of k packets drains in about the same time, while each packet’s own latency stays T rather than $T + (N - 1)h$.

Sharding is not available on this corpus: a CME burst is one instrument’s order flow, so sharding by instrument sends the whole burst to one core at service T and relieves nothing, and the corpus is a single instrument. Dispatch is available for any stage that holds no state from one packet to the next — SBE decode is such a stage, order-book application is not — provided its output is put back in order. Two comparisons follow. Table 8 puts dispatch beside the tandem with no ingress hop, no egress hop and no resequencing charged, which is the lowest latency dispatch could reach. Table 9 charges all three.

T	N	tandem		M/D/ N dispatch	
		p_{50}	p_{99}	p_{50}	p_{99}
2	1	2.00	2.00	2.00	2.00
2	2	3.70	3.70	2.00	2.00
2	4	7.10	7.10	2.00	2.00
2	8	13.90	13.90	2.00	2.00
4	1	4.00	4.00	4.00	4.00
4	2	5.70	5.70	4.00	4.00
4	4	9.10	9.10	4.00	4.00
4	8	15.90	15.90	4.00	4.00
8	1	8.00	8.61	8.00	8.61
8	2	9.70	9.70	8.00	8.00
8	4	13.10	13.10	8.00	8.00
8	8	19.90	19.90	8.00	8.00
16	1	16.00	34.59	16.00	34.59
16	2	17.70	18.31	16.00	16.00
16	4	21.10	21.10	16.00	16.00
16	8	27.90	27.90	16.00	16.00
32	1	32.00	146.14	32.00	146.14
32	2	33.70	52.29	32.00	48.04
32	4	37.10	37.71	32.00	32.00
32	8	43.90	43.90	32.00	32.00
64	1	98.88	820.70	98.88	820.70
64	2	65.70	179.84	64.00	171.38
64	4	69.10	87.69	64.00	72.50
64	8	75.90	76.51	64.00	64.00
128	1	364.45	5316.85	364.45	5316.85
128	2	164.58	886.40	128.00	868.98
128	4	133.10	247.24	128.00	223.06
128	8	139.90	158.49	128.00	128.00

Table 8. Real (Hawkes) arrivals, corpus medians in μs : N cores as an N -stage tandem against the same N cores as one queue feeding N whole-packet servers, with no resequencing charged to the latter.

Without its costs, dispatch’s p_{99} is at or below the tandem’s in every cell at $T = 16\text{--}128\ \mu\text{s}$ ($N = 2$: 16.0 against 18.3 μs at $T = 16$, 48.0 against 52.3 at 32, 171 against 180 at 64, 869 against 886 at 128 μs), and its median is T against the tandem’s $T + (N - 1)h$.

T	N	tandem		dispatch, costs charged		reordered packets
		p_{50}	p_{99}	p_{50}	p_{99}	
16	2	17.70	19.59	19.40	20.78	0.00%
16	4	21.10	22.48	19.40	20.78	0.00%
16	8	27.90	29.28	19.40	20.78	0.00%
32	2	33.70	52.84	35.40	51.61	0.00%
32	4	37.10	39.86	35.40	38.16	0.01%
32	8	43.90	46.66	35.40	38.16	0.01%
64	2	65.70	181.08	67.40	175.74	0.01%
64	4	69.10	89.10	67.40	79.99	0.03%
64	8	75.90	81.42	67.40	72.92	0.03%
128	2	166.37	902.82	131.40	883.83	0.08%
128	4	133.10	250.40	131.40	227.94	0.14%
128	8	139.90	162.28	131.40	142.45	0.14%

Table 9. The same N cores as a tandem and as a dispatch pool with every cost charged: an ingress hop to the servers, an egress hop from them, and an in-order resequencer that holds each packet until its predecessors have left. Span-dependent service at the NQ slope; corpus medians over 3512 windows, in μs , $h = 1.7 \mu s$; last column the share of packets that finish before a predecessor and wait in the resequencer (its p_{99} wait is $0.00 \mu s$ in every cell). Constant service gives the same picture. What the table shows: dispatch's median is $T + 2h$ at every N , against the tandem's $T + (N - 1)h$, so dispatch is one hop slower at $N = 2$ and faster from $N = 3$; at $N = 2$ the two are within $1.2 \mu s$ at p_{99} for $T \leq 32 \mu s$ and dispatch is lower above; at $N \geq 4$ dispatch is lower on both quantiles in every cell. Resequencing costs almost nothing, because a packet that waited in the shared queue behind a burst leaves in the order it arrived. What follows: for a stage that carries no state, dispatching whole packets to three or more cores is at least as good as cutting it into the same number of stages; at two cores the two are level, with the tandem one hop cheaper on the median. For a stage that carries state, the order book above all, dispatch is not available and the tandem is the arrangement that spends extra cores on it.

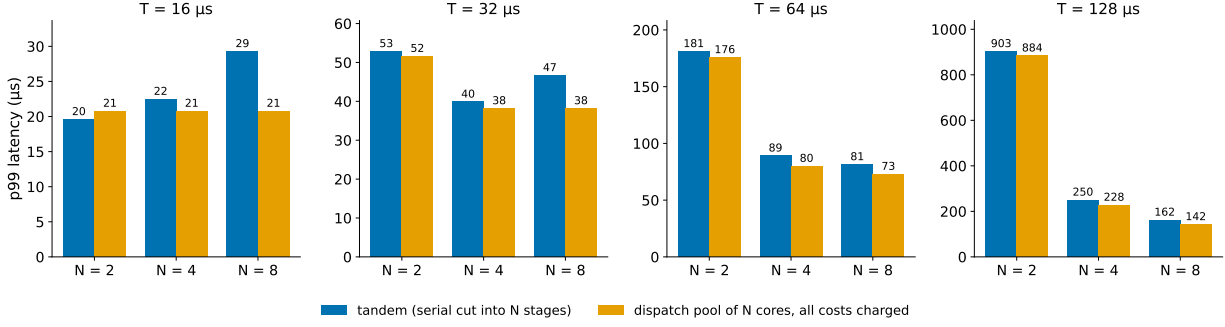


Figure 8. Table 9 as bars: p_{99} of a serial cut into N stages (blue) against a pool of N cores with the ingress hop, egress hop and resequencing charged (orange). At $N = 2$ the two are level; from $N = 4$ the pool is lower at every service time, and at $16 \mu s$ its p_{99} does not grow with N while the tandem's does.

With every cost charged (Table 9), dispatch pays two hops on every packet — in and out — and nothing else measurable: the resequencer holds at most 0.14% of packets, for a p_{99} wait of zero. Its median is therefore $T + 2h$ whatever N is, one hop more than a two-stage tandem and less than any tandem of three or more stages, and its tail is lower than the tandem's from $N = 4$ in every cell. The design consequence is a division by state. A stateless stage above the floor — decode is the natural candidate — is best served by a dispatch pool of three or more cores; at two cores a cut and a pool are level. A stateful stage above the floor is served by cutting the chain, which is the case the rest of the paper analyses.

4.5 Median cost of an N -stage tandem

This section looks at the typical message rather than the slow one. Its point is that splitting a task into more stages makes the typical latency worse by a fixed, predictable amount, exactly one mailbox hop per added stage, and nothing more. The reason is that the typical message arrives when the queue in front of it is empty, so it pays only the service time plus the hops. The simulation confirms this to the reported precision in almost every cell of the sweep. The few exceptions are at the longest service times, where the queue never empties and even the typical message waits; those sit outside the range the paper recommends, and splitting cures them anyway.

The p_{50} columns of Table 2 match the single-message formula of Proposition 1

$$p_{50}(N) = T + (N - 1)h \quad \text{with } h = 1.7 \mu s,$$

in all 28 cells of the Poisson arm and in 25 of the 28 Hawkes cells — for instance 8.00, 9.70, 13.10, 19.90 at $T = 8$. Under Poisson the identity holds because the queue is empty; under Hawkes it holds because the median message arrives in a quiet inter-cluster stretch where the queue is also empty. The hop tax is therefore exactly h per boundary on the median, with no hidden component, as Theorem 1 says it must be at every quantile.

The three exceptions are at the long-service end of the sweep: $(T, N) = (64, 1)$, $(128, 1)$ and $(128, 2)$, where the Hawkes p_{50} is 98.88, 364.45 and $164.58 \mu s$ against floors of 64.00, 128.00 and 129.70. There the service time is long enough that the median message no longer arrives in a quiet stretch: clusters merge into a persistent backlog and the typical message waits. This is the regime boundary appearing in the median rather than in the tail, and it lies outside the range the paper's recommendations address. Splitting restores the identity once the per-stage service is below the threshold; at $T = 128$ the median is on the floor by $N = 4$.

4.6 The design equation: hop cost against tail excess

This section turns the measurements above into a rule for choosing how many stages to split a task into. The idea is simple. Splitting a task across more threads has a fixed price: every added stage costs one mailbox hop, and that hop is paid on every message, fast or slow. It also has a benefit: when packets arrive in bursts, messages queue behind one another, and shorter stages drain those queues faster, so the slow tail shrinks. The question is when the benefit outweighs the price.

The section does four things. It writes the median and the tail latency as two short formulas, a hop-cost term plus a tail term. It tabulates the tail term from the corpus and shows that it depends only on how long the slowest stage is, and that it disappears entirely once every stage is short enough. It combines cost and benefit into one expression the operator can minimise, with a table of where the optimum falls. And it draws the practical conclusion: the goal is not many threads but a short slowest stage, and a task already shorter than the feed’s tight-gap floor, about seven and a half microseconds here, should not be split at all.

From the corpus:

$$p_{99}^{\text{Hawkes}}(N) = T + (N - 1)h + \Delta(N), \quad (11)$$

$$p_{50}^{\text{Hawkes}}(N) = T + (N - 1)h, \quad (12)$$

where $\Delta(N) \equiv p_{99}^{\text{Hawkes}}(N) - p_{50}^{\text{Hawkes}}(N)$ is the “clustering tail excess”. Table 10 lists it, together with the ratio $\Delta(N)/\Delta(1)$ that Theorem 1 bounds by $1/N$.

T	$\Delta(N)$ (μs)				$\Delta(N)/\Delta(1)$			γ
	$N=1$	$N=2$	$N=4$	$N=8$	$N=2$	$N=4$	$N=8$	
2	0.00	0.00	0.00	0.00	—	—	—	—
4	0.00	0.00	0.00	0.00	—	—	—	—
8	0.61	0.00	0.00	0.00	0.000	0.000	0.000	—
16	18.59	0.61	0.00	0.00	0.033	0.000	0.000	—
32	114.14	18.59	0.61	0.00	0.163	0.005	0.000	—
64	721.82	114.14	18.59	0.61	0.158	0.026	0.001	3.32
128	4952.41	721.82	114.14	18.59	0.146	0.023	0.004	2.68
bound					0.500	0.250	0.125	≥ 1

Table 10. Corpus-median clustering tail excess $\Delta(N) = p_{99}(N) - p_{50}(N)$ under Hawkes arrivals at $h = 1.7 \mu\text{s}$, its ratio to the single-stage value, and the least-squares exponent γ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$. The last row is the Theorem 1 bound.

Two features of Table 10 follow from the theorems. First, Δ depends on (T, N) only through the per-stage service T/N : every diagonal of constant T/N carries one number — 0.00 at $T/N \leq 4 \mu\text{s}$, 0.61 at $T/N = 8$, 18.59 at 16, 114.14 at 32, 721.82 at 64 — to the reported precision, across up to four different (T, N) pairs, without fitting. This is the reduction identity Equation (4) on real arrivals: the tandem’s tail is the single-server tail at s_{max} , and the table is a table of $\Delta_{\text{single}}(s)$ at $s \in \{0.25, 0.5, \dots, 64, 128\} \mu\text{s}$, read off the diagonals.

Second, every ratio is at or below the bound $1/N$, and the pathwise check of Section 6 finds no message on any window above it. The margin is large, because of the threshold. Because $\Delta_{\text{single}}(s)$ is identically zero for $s \lesssim 7.5 \mu\text{s}$ (Section 4.8), a split that pushes the per-stage service below that floor does not merely divide the tail excess, it removes it. At $T = 32$, Δ falls $114.14 \rightarrow 18.59 \rightarrow 0.61 \rightarrow 0.00$

as N goes $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$: a ratio of 0.005 at $N = 4$ against a bound of 0.25. The fitted exponent γ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$ is correspondingly steep, 3.32 at $T = 64$ and 2.68 at $T = 128$, well above the $\gamma = 1$ that the bound alone would permit; at $T \leq 32$ the tandem drives $\Delta(8)$ to exactly zero and no power law is fitted, since the least-squares slope would be reading the numerical floor rather than the data. The decay of the excess is therefore not a $1/N$ division on this corpus; it terminates at zero once the per-stage service is below the floor. The design consequence is that the objective is not a large N but an $s_{\max}$ below the feed's tight-gap floor. The utilisation slack described after Theorem 1 contributes as well, and Appendix B reproduces it on synthetic Borel clusters.

The operator's stage-count optimum is

$$N^* = \arg \min_N \left[(N-1)h + \kappa \Delta(N) \right]$$

for the operator's tail weight κ (the number of microseconds of median the operator will spend to remove one microsecond of p_{99} excess). Over the simulated grid $N \in \{1, 2, 4, 8\}$ at $h = 1.7 \mu\text{s}$ the breakpoints are:

T (μs)	$N^* = 1$	$N^* = 2$	$N^* = 4$	$N^* = 8$
2	always	—	—	—
4	always	—	—	—
8	$\kappa < 2.8$	$\kappa \geq 2.8$	—	—
16	$\kappa < 0.095$	$0.095 \leq \kappa < 5.6$	$\kappa \geq 5.6$	—
32	$\kappa < 0.018$	$0.018 \leq \kappa < 0.19$	$0.19 \leq \kappa < 11$	$\kappa \geq 11$
64	$\kappa < 0.0028$	$0.0028 \leq \kappa < 0.036$	$0.036 \leq \kappa < 0.38$	$\kappa \geq 0.38$
128	$\kappa < 0.0004$	$0.0004 \leq \kappa < 0.0056$	$0.0056 \leq \kappa < 0.071$	$\kappa \geq 0.071$

A dash means that stage count is never optimal at any κ : at $T \leq 8 \mu\text{s}$ the tail excess is already zero by $N = 2$, so further splitting can only add hop cost. An operator who values tail and median equally ($\kappa = 1$) lands on $N^* = 1$ at $T = 2, 4$ and 8 , 2 at $T = 16$, 4 at $T = 32$, and 8 at $T = 64$ and $128 \mu\text{s}$, which is also the ordering of the raw p_{99} minima in Table 2. Over the service times an HFT hot path occupies, an operator must weight the tail several times more than the median before splitting pays; above the threshold the required weight is small.

At real cut points the stages are unequal, and by Remark 2.3 the tail excess depends on the partition P of the servicing chain only through its largest stage, $\Delta(P) = \Delta_{\text{single}}(s_{\max}(P))$, i.e. the single-stage tail excess measured at service $s_{\max}$. The optimum over partitions is then

$$P^* = \arg \min_P \left[\text{hops}(P)h + \kappa \Delta_{\text{single}}(s_{\max}(P)) \right],$$

which Table 10 lets the operator evaluate directly, since its diagonals are $\Delta_{\text{single}}(s)$ on a power-of-two grid of s . Two rules follow. A cut is worth its hop only if it lowers $s_{\max}$ by enough that $\kappa [\Delta_{\text{single}}(s_{\max}) - \Delta_{\text{single}}(s'_{\max})] > h$; and among partitions with the same $s_{\max}$ the one with the fewest hops is preferred, so sub-tasks downstream of the bottleneck should be merged onto one thread until their sum reaches $s_{\max}$.

Design principle (only $s_{\max}$ matters). For a servicing chain cut at fixed boundaries, the tail-latency question is not “how many threads?” but “what is the smallest achievable largest stage $s_{\max}$?” By Theorem 1 the clustering tail excess of any partition is $\Delta_{\text{single}}(s_{\max})$, independent of the stage count; a cut that does not lower $s_{\max}$ compresses no tail and still costs one hop h . Choose the partition with the smallest reachable $s_{\max}$, then the fewest hops that realise it.

Floor on T for splitting. Equations (11)–(12) imply a lower bound on the tasks the recommendation applies to. Splitting a task of total service T into N stages spends $(N - 1)h$ at every quantile to compress a tail excess $\Delta(1)$ that is itself a strongly increasing function of T (0 at $T \leq 4\mu s$, rising to $39T$ at $T = 128$, Table 10). When T is at or below the feed’s tight-gap floor there is no tail excess to remove at all, and the design equation prescribes $N^* = 1$; a hop there adds cost only. The sweep confirms this: at $T = 2$ and $4\mu s$, $\Delta(1) = 0$, so a split removes nothing and p_{99} rises by the full hop, from 2.00 to 3.70 and from 4.00 to 5.70. Above the floor: at $T = 16\mu s$, $\Delta(1) = 18.59\mu s$, of which the split removes 17.98 for the same 1.7, and p_{99} falls from 34.59 to 18.31 on every window.

Two separate necessary conditions apply. The first is a property of the *framework*: a split cannot pay unless the tail excess it removes exceeds the hop cost it adds, which on this framework’s shipping BQueue path ($h \approx 1.7\mu s$) puts a floor at roughly $T \gtrsim 2h \approx 3.4\mu s$. The second is a property of the *feed*: there is no tail excess to remove at all until T exceeds the arrival stream’s tight-gap floor, about $7.5\mu s$ on this corpus (Section 4.9). The binding constraint is whichever is larger, and on CME MDP3 channel 318 with a BQueue mailbox it is the feed’s, by roughly a factor of two; this is why $T = 4\mu s$, although above $2h$, has $\Delta(1) = 0$ and should not be split. A framework with a much more expensive mailbox would find its own floor binding instead. Throughout the rest of the paper, “the floor” means the larger of the two. Sub-microsecond tasks, for instance a single SBE integer decode or a small handler that fires from a strategy fast path, belong on `fast_send` rather than on a cross-thread mailbox, regardless of feed statistics.

4.7 Per-window results

The results so far are for a typical half-hour window. This section asks whether they hold on every window, including the busiest ones, or only on average. The answer matters because a busy window has a far worse single-thread tail than the typical one, so an operator cannot size a pipeline from the average alone. The finding is that the split wins on essentially every window from about ten microseconds up, and on no window at eight microseconds or below, where there is no tail to remove and the split only adds hop cost, with a sharp crossover between. The recommendation is therefore not an artefact of averaging.

The corpus medians understate the tail on the busiest windows. Across the 3512 windows the single-stage p_{99}/T has a median of 1.0, 1.0, 1.1, 2.2, 4.6, 12.8, 41.5 and a 95th percentile of 1.0, 1.0, 1.1, 2.5, 6.1, 19.9, 81.7 at $T = 2, 4, 8, 16, 32, 64, 128\mu s$, so one window in twenty at $T = 32$ has a single-stage p_{99} above $195\mu s$ and at $T = 128$ above 10.5 ms. The tandem’s win is not confined to the median window: $p_{99}(2) < p_{99}(1)$ and $p_{99}(4) < p_{99}(1)$ both hold on 100% of windows at every $T \geq 16$, and on no window at all at $T \leq 8$, where there is no tail to compress and the split pays only hop cost. The crossover is sharp and sits just below $T = 10\mu s$: on the refined sweep of Section 4.8, $p_{99}(2) < p_{99}(1)$ on 0% of windows at $T = 8\mu s$, 67.0% at $T = 9\mu s$ and 99.7% at $T = 10\mu s$. The per-window fitted branching ratio has median 0.798 and a 5th–95th percentile range of 0.761–0.832 (Section 3.2); how the tail excess depends on it, and on the window’s packet rate, is examined in Section 4.9. The intervening p_{95} quantile is recorded per window in the sweep (Section 3.5) and omitted from the tables above only for brevity; it lies between the reported p_{50} and p_{99} in every cell by construction, and under the Poisson null it equals the service floor wherever p_{99} does, since $\rho/N < 1 - 0.99$ implies $\rho/N < 1 - 0.95$. The 3512 windows are treated as evaluation units for the descriptive statistics reported here; they are nested within 276 sessions and are not independent, so any statistical uncertainty on the aggregates would be clustered at the session level rather than treating 3512 as an independent sample size.

4.8 Location of the tail onset

The coarse sweep brackets the onset of the clustering tail between $T = 8$ and $T = 16 \mu\text{s}$, too wide for a design rule. We reran the single- and two-stage arms on the same 3512 windows at $T \in \{5, 6, 7, 8, 9, 10, 12, 14, 16\} \mu\text{s}$:

T (μs)	p_{99} Hawkes	excess over T	windows with a tail	$p_{99}(2) < p_{99}(1)$
5	5.00	0.00	0.0%	0.0%
6	6.00	0.00	0.0%	0.0%
7	7.00	0.00	4.5%	0.0%
8	8.61	0.61	98.6%	0.0%
9	10.85	1.85	100.0%	67.0%
10	13.52	3.52	100.0%	99.7%
12	19.83	7.83	100.0%	100.0%
14	26.68	12.68	100.0%	100.0%
16	34.59	18.59	100.0%	100.0%

Two distinct thresholds are visible. The first is the *tail* threshold: between $T = 7$ and $T = 8 \mu\text{s}$ the fraction of windows with any p_{99} excess jumps from 4.5% to 98.6%. The second is the *splitting* threshold, about $1.5 \mu\text{s}$ later, between $T = 8$ and $T = 10 \mu\text{s}$: a tail must exist *and* exceed the hop cost before a split repays it. At $T = 8 \mu\text{s}$ nearly every window has a tail, but its median excess is $0.61 \mu\text{s}$, well under $h = 1.7 \mu\text{s}$, and splitting loses on every window. By $T = 10 \mu\text{s}$ the excess is $3.52 \mu\text{s}$, above h , and splitting wins on 99.7%.

The first threshold is set by the feed’s tight-gap structure rather than by its mean rate; the next section establishes this.

4.9 Dependence of the tail on packet rate and branching ratio

One would expect the slow tail, and with it the case for splitting, to depend on how busy the market is and on how strongly one transaction triggers the next. If it did, the rule for splitting would have to change with market conditions. This section tests that expectation and finds that, over the task lengths a trading hot path uses, it does not hold: a window with four to five times the packet rate has the same tail, relative to the task length, as a quiet one. The reason is that a busy market produces more bursts, not tighter bursts. The tightest spacing between two packets on this feed is about seven and a half microseconds, the gateway’s minimum separation between successive events; the two packets on either side of such a gap are, almost always, two different transactions (Section 4.3), and neither the gateway’s floor nor the spacing of transactions inside a burst changes with the market state. Since it is the tightest spacing that starts a queue, the point at which a tail appears is a constant of the feed. Market load only matters for very long tasks, where the ordinary effect of a busy server takes over. The practical consequence is that the rule for splitting needs no live estimate of market conditions; the section closes by confirming the result on five-minute windows, and by noting that a rule based on the average gap between packets, rather than the tightest, would be wrong by two orders of magnitude.

The results above are conditioned on the service time T alone. A Hawkes tail depends on two further quantities that the corpus median averages over: the window’s mean packet rate $\bar{\lambda}$, and its branching ratio $n = \alpha/\beta$.

We sort the 3512 windows into quintiles, first by $\bar{\lambda}$ and then by the fitted n , and report the corpus median of the normalised single-stage tail p_{99}/T within each quintile. Normalising by T

removes the service floor, so a value of 1.0 means no tail at all and the quintile-to-quintile ratio isolates the conditioning effect.

quintile	median	p_{99}/T , corpus median				
		$T=8$	$T=16$	$T=32$	$T=64$	$T=128$
<i>by packet rate $\bar{\lambda}$ (pkt/s)</i>						
Q1	202	1.08	2.29	4.74	12.25	31.76
Q3	451	1.07	2.12	4.48	12.68	42.81
Q5	942	1.08	2.11	4.51	13.92	55.67
Q5/Q1		1.00	0.92	0.95	1.14	1.75
<i>by branching ratio n</i>						
Q1	0.769	1.08	2.18	4.51	11.61	30.73
Q3	0.798	1.08	2.16	4.61	13.17	42.83
Q5	0.825	1.07	2.15	4.61	14.65	58.73
Q5/Q1		1.00	0.98	1.02	1.26	1.91

The result divides at $T \approx 32\text{--}64\ \mu\text{s}$. For $T \leq 32\ \mu\text{s}$ the normalised tail is flat in both variables: a $4.66\times$ change in packet rate moves it by 0–8%, and in the wrong direction at $T = 16$ and $T = 32$, where the *busiest* quintile has the slightly *smaller* normalised tail. A $0.769 \rightarrow 0.825$ change in branching ratio moves it by at most 2%. For $T \geq 64\ \mu\text{s}$ both variables have an effect, in the expected direction: by $T = 128\ \mu\text{s}$ the busiest quintile’s tail is $1.75\times$ the quietest and the most critical quintile’s is $1.91\times$ the least. This is the boundary between the clustering regime and the utilisation regime found elsewhere in the paper: for $T \geq 64\ \mu\text{s}$ the Poisson null has a tail too (Section 4.1), utilisation $\rho = \bar{\lambda}T$ is the governing quantity, and ρ is proportional to $\bar{\lambda}$. Intensity matters where utilisation queueing applies and not below it.

Simulator check of the invariance. The Poisson arm, run on the same windows through the same simulator, shows the intensity dependence that utilisation queueing requires: at $T = 16$ and $32\ \mu\text{s}$ its normalised p_{99} is $1.34\times$ and $1.68\times$ higher in the busiest quintile than in the quietest, because $\rho = \bar{\lambda}T$ crosses the 0.01 switch-on of Corollary 2 within the quintile range. The Hawkes arm on the same windows at the same T gives $0.92\times$ and $0.95\times$. The simulator therefore resolves intensity; the clustered tail does not depend on it at these service times. The result is the same unbinned: the per-window Spearman correlation of p_{99}/T with $\bar{\lambda}$ is -0.19 at $T = 16$ and -0.09 at $T = 32\ \mu\text{s}$, and with n is -0.05 and $+0.02$; at $T = 128\ \mu\text{s}$ both are $+0.48$ and $+0.54$. Comparing the top and bottom 2% of windows by rate, a $26\times$ range, gives a p_{99}/T ratio of 1.02 at $T = 32$ and 2.57 at $T = 128$.

The mechanism is that at $\rho \ll 1$ clusters do not overlap. Raising $\bar{\lambda}$ raises the number of clusters per window, each with the same internal spacing and, at fixed n , the same size distribution, so the position of the p_{99} packet within its cluster, and hence p_{99}/T , is unchanged. Intensity enters once clusters begin to overlap, and n enters once T is long enough for many members of one cluster to fall within a single service time; on this corpus both occur at $T \geq 64\ \mu\text{s}$.

Intensity below the crossover. The null result has a mechanical explanation in the arrival stream. Taking the same $\bar{\lambda}$ quintiles and reporting the corpus median of the per-window interarrival-gap quantiles:

quintile	$\bar{\lambda}$ (pkt/s)	p_1 gap	p_5 gap	p_{25} gap	p_{50} gap	mean gap
Q1	202	7.45	8.85	21.30	87.10	4944
Q2	320	7.52	8.75	21.87	84.52	3128
Q3	451	7.53	8.70	22.70	82.16	2215
Q4	620	7.51	8.66	23.36	81.98	1612
Q5	942	7.43	8.58	22.55	76.52	1062
Q5/Q1	4.66×	0.997×	0.969×	1.059×	0.878×	0.215×

All gaps in microseconds. The mean gap falls by exactly the reciprocal of the rate increase, $0.215 \approx 1/4.66$, which is definitional. Every other column is approximately invariant, and the tightest is the most invariant: the p_1 gap moves from 7.45 to 7.43 μs , a ratio of 0.997, while the rate more than quadruples.

Raising the intensity of a self-exciting process raises the rate at which clusters are generated; it does not change the spacing of events inside a cluster. The tight end of that spacing is bounded by the gateway’s minimum separation between successive events, about 7.5 μs on this corpus, a property of the CME gateway rather than of the market state; the events on either side of a floor-level gap are distinct transactions arriving within that separation of each other (Section 4.3, Table 5), and how closely successive transactions within a burst follow one another is likewise invariant to the window’s rate. A busy window contains more bursts, not tighter bursts.

The gap statistics on the publisher’s clock are censored at the floor: two transactions closer together than about 7.5 μs reach the receiver in one packet (Section 5.4). A faster self-excitation would therefore show in the packets rather than the gaps. It does not scale with the branching ratio: from the lowest to the highest branching-ratio quintile the share of packets carrying two or more messages falls by a factor of 0.89, the mean message count per packet by 0.99 and the share of packets carrying more than one transaction by 0.97 (per-window Spearman correlations with n of -0.18 , -0.28 and $+0.00$); across packet-rate quintiles the three move by 1.02, 0.99 and 1.13.

The queueing consequence: a deterministic server of service time T begins to build a queue when it is handed two arrivals separated by less than T , so the service time at which a tail appears is set by the *tight* end of the gap distribution, not by its mean. The tight end is invariant to $\bar{\lambda}$; therefore the threshold is invariant to $\bar{\lambda}$. The two numbers agree: the measured p_1 gap of 7.43–7.53 μs across every intensity quintile brackets the measured tail onset of Section 4.8, between $T = 7$ and $T = 8 \mu\text{s}$, where the fraction of windows carrying any tail jumps from 4.5% to 98.6%.

Robustness to window length. A 30-minute window averages $\bar{\lambda}$ over a long interval, and the quintile split above may be comparing time-averaged rates that have smoothed away the relevant intensity variation. We therefore reran the entire pipeline — packet extraction, per-window Hawkes MLE, Poisson null and the full (T, N) grid — on *five*-minute windows, giving 21,060 windows in place of 3512. Shorter windows do resolve more intensity variation, as expected: the corpus $\bar{\lambda}$ spread widens from $6.8\times$ to $7.3\times$ between the 5th and 95th percentiles, from $26.0\times$ to $30.6\times$ between the 1st and 99th, and the observed maximum rises from 2152 to 2692 packets per second; the 5th–95th range of the fitted branching ratio widens from 0.0717 to 0.0786. The conditioning result is unchanged to two decimal places across the sweep:

	$\bar{\lambda}$ Q5/Q1	$T=8$	$T=16$	$T=32$	$T=64$	$T=128$
30-min windows ($n = 3512$)	4.66×	1.00	0.92	0.95	1.14	1.75
5-min windows ($n = 21,060$)	4.98×	1.00	0.93	0.96	1.15	1.74

Entries are the Q5/Q1 ratio of the normalised single-stage tail p_{99}/T . Resolving intensity six times more finely, over a wider realised range, moves no entry by more than 0.01. The result is a property of the arrival process, not of the window length.

Consequence for the design rule. On NQ the rule can therefore be stated as a constant of the feed rather than as a function of market state.

Design principle (on NQ, the threshold is a constant of the feed). Split a stage whose service time exceeds the feed’s tight-gap floor; do not split one below it. The floor is the tight end of the feed’s interarrival-gap distribution (the p_1 gap, about $7.5\mu\text{s}$ on CME channel 318), not its mean, which is two orders of magnitude larger. The floor is measured once per feed: on this corpus it is $7.43\text{--}7.53\mu\text{s}$ in every packet-rate quintile across a $4.7\times$ range of rates and in every branching-ratio quintile, so the rule requires no runtime estimate of market intensity or clustering and does not change between a quiet and a volatile session. Both the floor and the spacing of transactions inside a burst were measured on one instrument, NQ front-month on CME channel 318; recheck them on any other instrument or feed rather than assume them.

Appendix D constructs a feed on which both dependences appear at $T = 8\mu\text{s}$, and identifies the packed-gap fraction as the quantity that separates it from CME.

The tight-gap floor is not the mean interarrival gap. On this corpus the mean gap is $2215\mu\text{s}$ at the median window, about $300\times$ the p_1 gap. A rule stated on the mean gap would recommend never splitting an HFT stage, and would be wrong by two orders of magnitude.

4.10 Sensitivity to the hop cost h

The rule depends on the cost of a mailbox hop, and this section asks what changes if that cost is different. A cheaper hop makes splitting worthwhile over a wider range of task lengths, but it does not move the point below which there is no tail to remove, since that point is set by the feed and not by the framework. Any other framework can therefore measure its own hop cost, put it into the same equation, and read off its own answer; a framework with a very expensive hop should not split at all. The section also explains why the hop cost used throughout is a cautious overestimate. It was measured with the receiving thread asleep, so it includes the cost of waking it. Inside a burst the receiver never sleeps, because messages arrive faster than it can process them, and the true hop cost there is a few hundred nanoseconds rather than nearly two microseconds. The benefit of splitting during a burst is therefore, if anything, understated.

The design equation is monotone in h : at fixed T and fixed corpus $\Delta(N)$, every κ breakpoint in the table above scales linearly with h , so a smaller h lowers every breakpoint and can only raise N^* at a fixed κ . At $h = 1.7\mu\text{s}$ the p_{99} -minimising N is 1, 1, 1, 2, 4, 8, 8 across the seven service floors; at a hypothetical $h = 0.5\mu\text{s}$ busy-poll mailbox it would be 1, 1, 2, 2, 4, 8, 8 — the change is a single notch, at $T = 8\mu\text{s}$, because below the tight-gap floor no reduction in h can make a split worth taking when there is no tail to remove. Reducing the hop cost widens the band of service times over which splitting pays; it does not move the threshold, which is set by the feed. For any other framework the procedure is the same: measure the one-way asynchronous hop cost h on its own dispatch path, put it into Equation (11) with the corpus Δ of Table 10, and read off N^* ; a framework whose h exceeds the tail excess a split could remove gets $N^* = 1$, and the recommendation reverses.

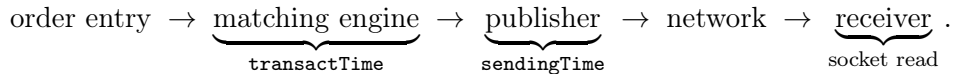
h is load-dependent; $1.7\mu\text{s}$ is a conservative upper bound. The $h = 1.7\mu\text{s}$ figure is the one-way cost of the shipping KASPAR-HFT BQueue mailbox measured at low load, where the receiver is sleeping on a condition variable and the send path must pay the kernel wake latency to bring it back onto CPU. That wake latency is the dominant term in the $\sim 3.4\mu\text{s}$ round-trip of Table 4 in Maciejewski [36], and the general fact that a spinning receiver delivers a message an order of magnitude faster than a wake-from-condvar receiver is standard in the microsecond-latency systems literature: David et al. [62] measure the primitive costs directly on commodity Linux hardware, Belay et al. [63] exploit this in the IX data plane by pinning busy-polling network threads, and Rigtorp’s operational guide [41] formalises the “run one thread per core and busy-wait” rule that flows from it; the `fast_send` paper’s own grouped-send row (Table 4, $\sim 90\text{ ns}$) is the same effect on the same framework — when the receiver never actually sleeps, the send path collapses. Under sustained load, and specifically inside a Hawkes cluster where the receiver’s queue is never empty, the receiver never sleeps: successive messages arrive faster than the receiver’s per-message service time, so it stays awake in a tight run-loop and the enqueue path reduces to a mutex acquire, a queue push, and a condition-variable notify against a receiver that is not waiting (a near-no-op). The effective h in that regime is dominated by the mutex + cache-line-transfer cost, typically a few hundred nanoseconds rather than $1.7\mu\text{s}$. This is the regime in which the tail reduction is largest, so the reported effect underestimates the benefit whenever the pipeline is inside a burst. Characterising the load-conditioned hop cost $h(\rho)$ is left to future work.

5 The exchange side: matching engine, publisher and bursts

The receiver’s arrivals are the output of the exchange. This section looks one step upstream, at the two CME systems that produce them, through the only window the public feed opens on them: the timestamps it carries. It treats the exchange as a queueing system observed from outside and makes no claim about CME’s internal design beyond what the MDP3 specification states. All figures are for NQ front-month on channel 318.

5.1 Three timestamps

Every message on the feed carries two exchange timestamps, and the receiver adds a third. `transactTime` is the start of event processing at the matching engine: when the engine began to match, add or cancel the order behind the message. `sendingTime` is written into each packet by CME’s market-data publisher when it sends the packet. Both come from CME’s synchronised clocks. The receiver’s own timestamp, taken at the socket read, is the third. The path is



The simulator of Part II uses `sendingTime` as each packet’s arrival time, since the receiver sees the publisher’s spacing plus a network delay that is close to constant. The feed does not carry the time an order entered the engine, and nothing in it shows the engine’s or the publisher’s internal queues directly. The MDP3 end-of-event flag marks the last message of each matching event.

5.2 Transactions and events

Section 4.3 grouped packets into blocks whose `transactTime` ranges overlap. Regrouping by the exact `transactTime` value, and checking against the end-of-event flag, on 40 sessions (5.34×10^8 messages, 5.12×10^8 transactions), gives the following. A `transactTime` group contains two or more

end-of-event flags, or straddles one, in 0.002% of cases: one `transactTime` is one matching event. The flag itself cannot serve as the grouping key on a single-instrument tape, since 85% of NQ events end on a message for another security on the channel. Under exact `transactTime` grouping 98.36% of transactions occupy one packet (98.30% under the blocks) and there are 4.8% more transactions than blocks, the difference being transactions that share a packet. The results of Section 4.3 are unchanged at the precision reported.

5.3 Gaps between transactions on the engine and publisher clocks

gap between consecutive transactions	$< 7.5 \mu s$	$< 16 \mu s$	$< 32 \mu s$	p_1 gap
engine clock (<code>transactTime</code>)	17.1%	25.4%	35.4%	$0.18 \mu s$
range across sessions	14.4–22.9	22.3–31.8	31.5–40.7	0.15–0.21
publisher clock (<code>sendingTime</code>)	1.04%	17.8%	32.1%	$7.46 \mu s$
range across sessions	0.44–1.52	15.5–22.2	28.6–36.9	7.23–7.88

Table 11. The same NQ transactions timed by the matching engine and by the publisher; median session and full range over 40 sessions. What the table shows: one consecutive transaction in six leaves the engine less than $7.5 \mu s$ after the previous one, and the tightest one percent are $0.18 \mu s$ apart; almost none leave the publisher that close, and the publisher’s tightest one percent are $7.46 \mu s$ apart. The difference holds in every session. What follows: the $7.5 \mu s$ floor of the gap distribution the receiver sees is the publisher’s spacing, not a property of the market’s event process; events the engine produces faster than that are either held back or packed into one packet.

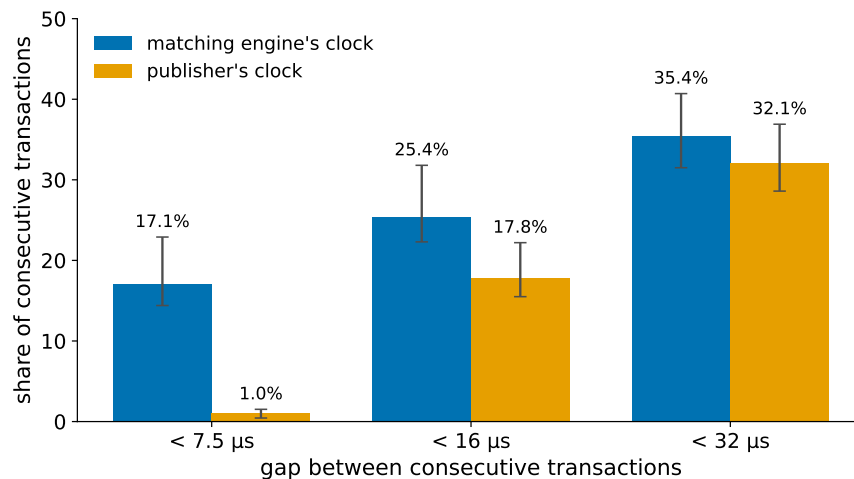


Figure 9. Table 11 as bars: share of consecutive NQ transactions closer than 7.5 , 16 and $32 \mu s$, on the engine’s clock (blue) and the publisher’s (orange); whiskers span the 40 sessions. Below $7.5 \mu s$ the engine has one gap in six and the publisher almost none: the publisher does not send faster than its spacing. By $32 \mu s$ the two clocks nearly agree.

Finding. On NQ, one consecutive transaction in six (17.1%) is processed by the matching engine less than $7.5\mu s$ after the previous one, against 1.04% of consecutive packets sent by the publisher. The tight-gap floor of the receiver’s arrival stream is set by the exchange’s publisher. The engine’s own gaps reach $0.18\mu s$ at the first percentile, which means the engine went directly from one event to the next with the next order already waiting: at those moments it was working through a backlog of orders that reached it within a fraction of a microsecond of each other.

The receiver does not see these transactions together. Table 12 follows every gap between consecutive NQ transactions from the engine’s clock to the publisher’s (the two clocks are defined in Section 4.3): for the same two transactions, the table gives the gap at the engine and the gap between the packets that carry them.

gap on the engine’s clock	share of gaps	gap on the publisher’s clock (row %)					
		same packet	< 7.5	7.5–10	10–16	16–100	> 100 μs
< 1 μs	5.3%	15.7	3.7	43.0	16.1	19.9	1.7
1–7.5 μs	11.8%	10.0	3.8	32.1	20.2	29.9	4.1
7.5–16 μs	8.5%	8.8	2.1	21.5	13.6	48.7	5.3
16–100 μs	31.4%	4.9	0.5	4.9	4.0	71.1	14.7
> 100 μs	43.0%	0.8	0.0	0.1	0.2	15.1	83.7
all gaps	100%	4.6	1.0	9.5	5.7	37.5	41.6
of which engine gap < 7.5 μs		43.3	66.2	64.0	56.4	12.2	1.4

Table 12. The gap between consecutive NQ transactions, measured on two clocks for the same two transactions; 20 sessions, 2.52×10^8 gaps. Rows: the gap on the matching engine’s clock. Columns: the gap between the packets that carry the two transactions, on the publisher’s clock, as a percentage of the row; “same packet” means both were sent in one packet. The last row gives, for each publisher gap, the share of cases in which the engine processed the two transactions less than $7.5\mu s$ apart. What the table shows: when the engine processes a transaction less than $1\mu s$ after the previous one, 16% reach the receiver in one packet and 84% in separate packets, 43% of them 7.5 to $10\mu s$ apart and 22% more than $16\mu s$ apart. Read the other way, 64% of the gaps of 7.5 to $10\mu s$ that the receiver sees separate transactions the engine processed less than $7.5\mu s$ apart. What follows: transactions do not reach the receiver together. The publisher spaces them out to its own interval of about $7.5\mu s$ and packs only a sixth of the transactions that follow their predecessor within $1\mu s$ into the same packet as that predecessor. The tight gaps that load the receiver are, for the most part, near-simultaneous engine events delivered back to back at the publisher’s spacing.

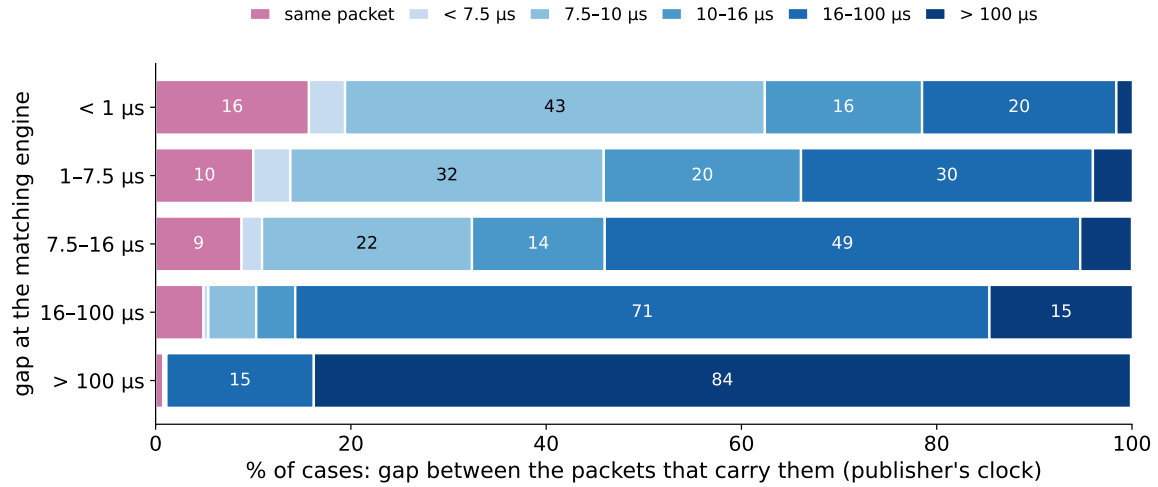


Figure 10. Table 12 as bars. Each bar is the set of consecutive transactions with the engine gap on the left, split by the gap between the packets that carry them; the numbers are percent of the bar. The top bar reads: when the engine processes a transaction less than $1\ \mu\text{s}$ after the previous one, the two share a packet 16% of the time (pink) and otherwise arrive in separate packets, most often $7.5\text{--}10\ \mu\text{s}$ apart (43%). Near-simultaneous at the engine becomes back to back at the publisher's spacing at the receiver.

Finding. Transactions the engine processes a fraction of a microsecond apart do not reach the receiver together: 84% of them arrive in separate packets, most of them 7.5 to $16\ \mu\text{s}$ apart. What the receiver sees is a train of packets at the publisher's spacing. Its queue grows when its service time exceeds that spacing, which is why the threshold of Section 4.8 sits at about $7.5\ \mu\text{s}$. The open market question is why orders reach the engine within a microsecond of each other; what reaches the receiver is set by the publisher.

5.4 The publisher as a queue

If the publisher sent events on a fixed schedule, the time from engine processing to packet send, `sendingTime`–`transactTime`, would not depend on how busy the engine is. On 20 sessions (2.52×10^8 transactions) the delay has a median of $99\ \mu\text{s}$, a p_{90} of $709\ \mu\text{s}$, a p_{99} of $3.6\ \text{ms}$ and a $p_{99.9}$ of $12.6\ \text{ms}$, and it rises steeply with the number of transactions the engine processed in the preceding $50\ \mu\text{s}$ (Table 13).

other NQ transactions in the previous 50 μ s (engine clock)	share of transactions	publisher delay		published sharing a packet with another transaction
		p_{50}	p_{99}	
0	57.9%	86 μ s	2.9 ms	4.2%
1	24.1%	113 μ s	3.5 ms	10.0%
2-3	13.8%	205 μ s	4.4 ms	15.2%
4-7	3.8%	406 μ s	6.8 ms	20.8%
8-15	0.4%	912 μ s	15.0 ms	26.6%

Table 13. Delay from engine processing to packet send, $\text{sendingTime} - \text{transactTime}$, by how busy the engine was in the 50 μ s before each transaction; 20 sessions, 2.52×10^8 NQ transactions. What the table shows: the median delay is 86 μ s for a transaction that follows none within 50 μ s and 912 μ s for one that follows eight to fifteen; the p_{99} rises from 2.9 to 15 ms; the share of transactions sent in a packet with another transaction rises from 4% to 27%. What follows: a fixed publishing schedule would keep the delay flat. It grows about tenfold with engine burst intensity, and coalescing grows with it, which is the signature of a queue: the publisher has a service time of about 7.5 μ s per packet, events the engine produces faster than that wait, and the publisher packs the backlog into fuller packets. The burst measure counts NQ front-month transactions only, while the publisher serves the whole channel, so the table understates the load on the publisher.

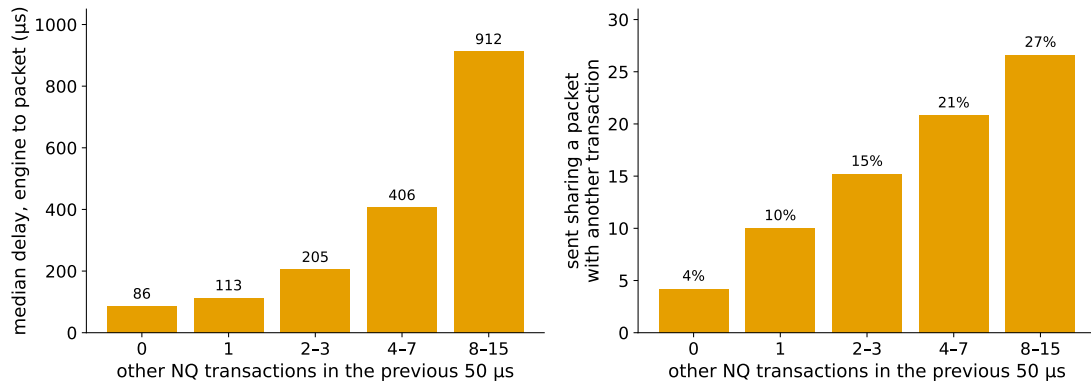


Figure 11. Table 13 as bars. Left: median delay from engine to packet grows tenfold as the engine gets busier. Right: the share of transactions sent in a packet with another transaction grows from 4% to 27%. Both rising together is what a queue at the publisher looks like.

A fixed publishing policy would keep the delay flat. Here it grows roughly tenfold with engine burst intensity, and coalescing grows with it. That is a queue: the publisher has a service time of about 7.5 μ s per packet. When the engine produces events faster than that, they wait, and the publisher packs the backlog into fuller packets.

This accounts for the floor from first principles. The output of a deterministic server cannot contain two departures closer together than its service time, and while the server works through a backlog its departures are spaced by exactly that time. The edge of the receiver's gap distribution at 7.5 μ s (Figure 5) and the mass of gaps just above it are the publisher draining its backlog.

5.5 Packet size under publisher backlog

If the publisher's service time is fixed per packet, a backlog can be drained in two ways: by keeping the spacing and sending many packets, or by packing more into each packet. The end-of-event tapes separate the two. On 20 sessions (2.43×10^8 packets) we take, for each packet, the NQ

front-month messages it carries (its span), the number of distinct transactions in it, the gap to the previous NQ packet on the publisher's clock, and the publisher delay of its first transaction, $\text{sendingTime} - \text{transactTime}$, as the measure of how far behind the publisher is.²

publisher delay of the packet's first transaction	share of packets	mean span (NQ messages)	packets with ≥ 2 transactions
$< 100 \mu\text{s}$	52.3%	1.000	0.02%
100–200 μs	21.3%	1.016	1.2%
200–500 μs	14.9%	1.237	8.9%
0.5–1 ms	5.9%	1.278	11.3%
1–5 ms	5.2%	1.440	18.2%
$> 5 \text{ ms}$	0.4%	1.603	20.6%

Table 14. Packet size against the publisher's backlog; 20 sessions, 2.43×10^8 packets. What the table shows: when the publisher is less than $100 \mu\text{s}$ behind, packets carry exactly one NQ message; when it is more than 5 ms behind, they carry 1.6 on average and one in five holds two or more transactions. What follows: packets get bigger when the publisher falls behind. Table 15 shows which packets.

gap to previous NQ packet	share of packets	mean span (NQ messages)	packets with ≥ 2 transactions	adjacent on the channel	median other packets between
$< 7.5 \mu\text{s}$	1.0%	1.006	0.08%	100%	0
7.5–10 μs (floor)	9.9%	1.007	0.40%	100%	0
10–16 μs	6.8%	1.096	2.8%	95.7%	0
16–32 μs	14.2%	1.141	6.7%	36.2%	1
32–100 μs	25.2%	1.073	3.4%	20.6%	2
0.1–1 ms	26.9%	1.106	4.5%	12.9%	4

Table 15. Packet size against the gap to the previous NQ packet on the publisher's clock; same packets as Table 14. The last two columns count the channel's other packets sent between two consecutive NQ packets, from the packet index of the tape, which numbers every channel packet that carries order or trade records for any instrument; packets carrying only other message types are not counted. What the table shows: packets sent at the floor carry one NQ message (1.007 on average) and in every case follow the previous NQ packet with no other order or trade packet of the channel between them. Bigger packets appear after gaps of 10 to 32 μs , which from 16 μs usually contain at least one packet for another instrument. What follows: the 7.5 μs spacing is the spacing of the channel's order and trade packets, not of NQ alone, and the publisher does not answer a backlog with bigger packets sent at the floor.

²Channel 318 carries other instruments besides NQ front-month, and the publisher packs their messages into the same packets. The span here counts NQ front-month messages only, so it is not the packet's size in bytes, and the publisher's backlog includes work for instruments we do not decode. Doing this properly means measuring the whole channel and modelling how activity in one instrument excites activity in the others (cross-excitation, a multivariate Hawkes process), rather than treating NQ as a stream on its own. On NQ the other instruments are thin. On ZN they are not: the Treasury futures along the yield curve trade against each other, a move in one is likely to trigger quotes and trades in the others within microseconds, and where they share ZN's channel the publisher would carry those reactions in the same packets. Cross-excitation along the curve is therefore a candidate cause of the large ZN packets of Section 7.6. It is untested here and listed among the open questions of Section 13.1.

publisher delay (floor packets only, gap 7.5–10 μ s)	mean span (NQ messages)	packets with ≥ 2 transactions
< 100 μ s	1.000	0.04%
100–500 μ s	1.006	0.50%
0.5–5 ms	1.026	1.0%
> 5 ms	1.114	2.0%

Table 16. Packets sent at the floor, by the publisher's backlog. What the table shows: even more than 5 ms behind, packets sent back-to-back at the floor carry 1.1 NQ messages on average and 98% of them hold a single transaction. What follows: pacing holds under load. The publisher keeps its spacing when it is far behind and does not switch to fuller packets at the same rate.

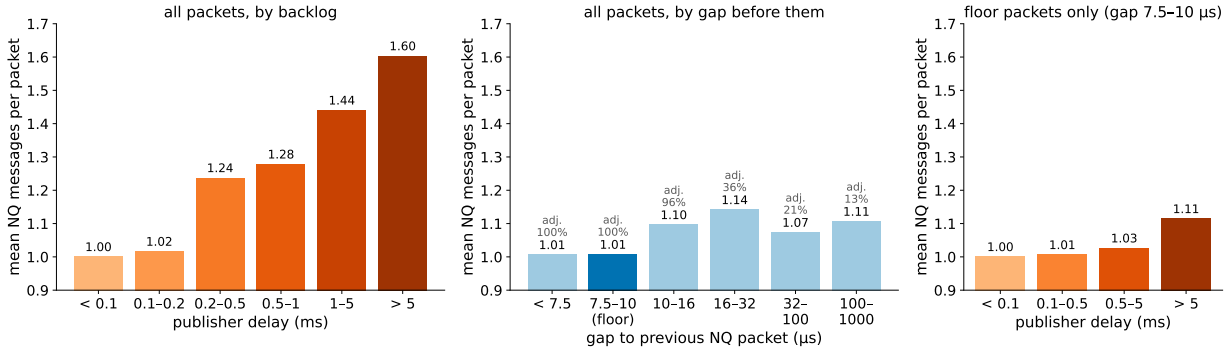


Figure 12. Tables 14–16 as bars: mean NQ messages per packet. Left: packets get bigger as the publisher falls further behind. Middle: packets sent at the floor, 7.5–10 μ s after the previous NQ packet (dark blue), are single-message packets; the grey “adj.” figures are the share with no other order or trade packet of the channel between the two NQ packets, 100% at the floor. Right: floor packets stay single-message even when the publisher is more than 5 ms behind. The publisher drains a backlog with more packets, not bigger ones.

Three things follow. First, pacing holds under load: even when the publisher is milliseconds behind, the packets it sends back-to-back are single-message packets at about 7.5 μ s spacing (Table 16), and that spacing holds across the channel’s order and trade packets (Table 15). Second, coalescing happens elsewhere. Bigger NQ packets come after gaps of 10 to 32 μ s, so when packets are bigger they also leave more slowly, which fits a service time per packet that grows with its size, the span-dependent service of Section 7.6. Third, a receiver sees a burst at the engine as more packets, not bigger ones: a long run of small packets at the floor, with an occasional larger packet after a longer gap. This is the arrival pattern that makes the receiver’s service time, compared with 7.5 μ s, the quantity that sets its tail (Section 5.6).

5.6 The receiver as the second stage of a tandem starting at the exchange

The publisher and the receiver are two servers in series. When a burst at the engine builds a backlog at the publisher, the publisher emits packets back-to-back at its own service interval of about 7.5 μ s. A receiver whose service time is shorter than that keeps up and queues nothing; a receiver whose service time is longer falls behind by the difference on every packet of the drain, and its backlog grows for as long as the drain lasts. This is the threshold of Section 4.8, now explained: the service time at which a receiver’s tail appears equals the service time of the exchange stage upstream of it. It is the tandem of Theorem 1 read in the other direction: a downstream stage slower than the

upstream one is the bottleneck, and it is the receiver’s stage, not the publisher’s, that sets the tail the receiver pays. The two regimes of Section 4.2 follow. At service times just above the floor, a short drain is enough to build a queue, and what matters is how often the publisher drains, which the gap shuffle preserves. At long service times the queue needs a sustained drain, which is a run of engine bursts, and that is what the shuffle destroys.

5.7 Bursts at the matching engine

Section 5.3 shows that the engine often processes events a fraction of a microsecond apart, which means orders were waiting for it; the publisher then delivers them to the receiver about $7.5\mu\text{s}$ apart (Table 12). Why orders reach the engine within a fraction of a microsecond of each other is the market question of Section 12.3. The $0.18\mu\text{s}$ spacing fits many participants sending orders in response to the same public signal, the *races* that Aquilina et al. [33] measured on the London Stock Exchange with order-entry times the CME public feed does not carry. It is consistent with that mechanism and does not prove it: a reaction to the previous transaction that is faster than the engine’s backlog would look the same from here.

Table 17 tests what transactions separated by a tight gap are. It takes every gap between consecutive transactions on the engine clock in 40 sessions (512 million gaps) and compares the two transactions on either side of a tight gap (below $16\mu\text{s}$) with those on either side of a loose gap (0.1 to 1 ms). A transaction is labelled by the first matching kind among trade, delete, new and change. Transactions across a tight gap are ordinary book traffic: their mix is close to that across a loose gap, and only 2.0% of tight gaps follow a trade. After a trade, the two differ: the next transaction is another trade in 11.7% of tight gaps against 2.2% of loose ones, and a delete in 37.4% against 23.5%. A trade followed within microseconds by a further trade or a cancellation is the pattern a race would leave, and it is more frequent across tight gaps. The table cannot measure how much of the engine’s activity is racing: the losers of a race to take liquidity are rejected and never reach the public feed, and a race to cancel appears only as deletes, which dominate across tight and loose gaps alike. What the feed shows is that transactions separated by tight gaps are mostly quote placements and cancellations, and that those following a trade carry the race signature.

Consecutive transactions, engine clock	gap $< 16\mu\text{s}$	gap 0.1–1 ms
share of all gaps	25.8%	27.2%
first is a trade	2.0%	3.5%
first is a delete	41.2%	39.8%
first is a new order	46.0%	39.5%
first is a change	10.8%	17.2%
after a trade: next is a trade	11.7%	2.2%
after a trade: next is a delete	37.4%	23.5%
after a trade: next is a new order	45.5%	68.2%
same side	52.5%	53.9%
same price	20.5%	17.3%

Table 17. The two consecutive transactions on either side of a gap, by gap length on the engine clock. Forty sessions, 512 million gaps; the middle bin ($16\text{--}100\mu\text{s}$, 31.5% of gaps) is omitted. “First” is the earlier transaction, “next” the later. Transactions across a tight gap are mostly quote placements and cancellations, in about the same mix as across a loose gap; only 2% of tight gaps follow a trade. After a trade, a tight follow-up is five times as likely to be another trade and more likely to be a cancellation, the signature of a race. Rejected race orders do not appear on the public feed, so the table does not measure the share of racing.

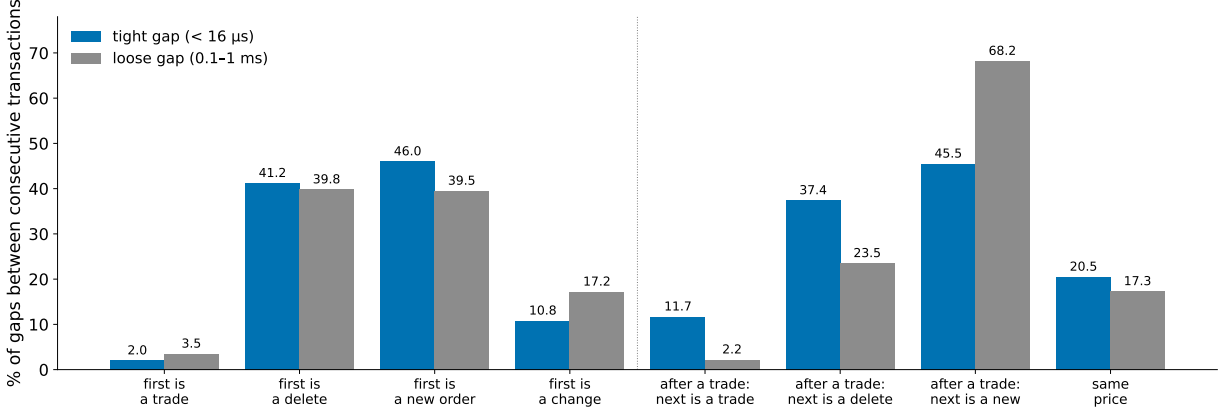


Figure 13. Table 17 as bars: the kind of transaction on either side of a tight gap (blue, under 16μ s) and of a loose gap (grey, 0.1–1 ms). Left of the dotted line the two mixes are alike: tight gaps are ordinary quote traffic. Right of it, after a trade the next transaction is far more often another trade or a delete across a tight gap, the pattern a race would leave.

The fitted self-excitation is a different phenomenon at a different scale. The exponential-kernel Hawkes fit on the engine clock gives a branching ratio of 0.803 and a decay time $1/\beta$ of 155μ s (5th–95th percentile 126–192; on the publisher’s clock 0.801 and 167μ s), so it places 4.7% of its triggering within 7.5μ s and 99.8% within a millisecond. That is the scale of the 100μ s mode of the gap distribution (Figure 5) and of the runs that carry the tail from 64μ s. The self-excitation the fit measures is one transaction making another more likely over the following hundred microseconds or so; it accounts for the runs and does not account for events a fraction of a microsecond apart, which are simultaneous arrivals at the engine, not one event triggering the next.

5.8 Conjecture: the publisher’s capacity as an information-side speed limit

Conjecture. Because the publisher sends at most one packet per 7.5μ s and queues whatever the engine produces faster than that, it bounds the rate at which market information reaches participants, and during bursts it delivers that information 0.4 to 15 ms late (Table 13). Any feedback loop in which participants react to published data therefore cannot run faster than the publisher, and a cascade driven by public information unfolds more slowly than the engine could process it. In this sense the publisher acts as a speed limit on the information side of the market, whether or not it was designed as one, and protects downstream receivers from the engine’s peak rate.

Three qualifications bound the conjecture. First, delaying feedback does not reduce the branching ratio of a self-exciting process: each event still triggers the same number of follow-on events on average, later, so the publisher cannot make a critical cascade subcritical; it bounds the cascade’s speed, not its size. Second, it does not bound simultaneous reactions to a common signal that reach the engine through order entry, which are what the engine-clock gaps of Table 11 suggest; those do not pass through the publisher before they are submitted. Third, whether the publisher’s capacity is a design choice or an engineering limit cannot be read from the feed; the queueing signature of Table 13 points to a limit. The conjecture is related to proposals that slow markets deliberately to blunt speed races, such as frequent batch auctions [29]; the difference is that here the delay is on the information leaving the exchange, not on orders entering it.

6 The analytical framework checked against the simulations

This section checks the closed-form predictions of the analytical framework (Section 2) against the results of the simulations on the recorded corpus (Section 4): each proposition or theorem is stated with the prediction it makes and the simulated numbers that test it.

Proposition 1 (burst-limit throughput). The single-message identity $p_{50}(N) = T + (N - 1)h$ is predicted exactly by Proposition 1 for any arrival law when the median message finds the queue empty. Table 2 confirms it in all 28 (T, N) cells of the Poisson arm and in 25 of the 28 Hawkes cells at $h = 1.7 \mu\text{s}$, to the $0.01 \mu\text{s}$ reporting precision; the three Hawkes exceptions are the longest-service cells, where the median message itself waits (Section 4.5).

Theorem 1 (reduction and scaling bound). The prediction is $\Delta(N) \leq \Delta(1)/N$ on every window, and more strongly $W_j^{(N)} \leq W_j^{(1)}/N + (N - 1)h$ for every individual message. The corpus-median ratios in Table 10 sit at or below $1/N$ in every cell, with substantial slack everywhere: the tightest ratio in the sweep is 0.163 at $T = 32$, $N = 2$, against a bound of 0.5. The slack is largest where a split drives the per-stage service under the tight-gap floor and removes the excess outright rather than dividing it, which is why the fitted γ is 3.32 and 2.68 at $T = 64$ and 128 rather than near 1. The pathwise form was checked directly by the simulator: across all 3512 windows, all seven service floors, all four packet-level arrival arms of Section 3.6 (real, uniform, gap shuffle, 1 s-binned), and $N \in \{2, 4, 8\}$, no message violates the bound (the check allows N nanoseconds for the rounding of T/N ; the largest excess over that tolerance is 0). The reduction identity Equation (4) is visible in Table 10 without any fitting: Δ is constant along every diagonal of constant T/N , so the tandem’s tail is the single-server tail at $s_{\max}$ on real arrivals exactly as on synthetic ones (Appendices B and C).

Corollary 2 (Poisson null). The prediction is $p_q^P(N) = T + (N - 1)h$ wherever the per-stage utilisation satisfies $\rho/N < 1 - q$, and a visible wait wherever it does not. Table 2 matches both halves: Poisson p_{99} equals the floor in 22 of the 28 cells and exceeds it in exactly the six cells whose per-stage utilisation clears 10^{-2} — $(T, N) = (32, 1), (64, 1), (64, 2), (128, 1), (128, 2)$ and $(128, 4)$, at $\rho/N = 0.0144$ to 0.0578 — and in no cell below it. Poisson $p_{99.9}$ likewise exceeds the floor exactly in the cells with $\rho/N > 10^{-3}$ (single stage at $T \geq 4$, two stages at $T \geq 8$, four stages at $T \geq 16$, eight stages at $T \geq 32$) and nowhere else. The null arm is therefore a working queue whose small wait appears where PASTA says it must, and splitting under it costs $(N - 1)h$ at every reported quantile.

The renewal null against the cluster picture. The gap shuffle of Section 4.2 is a renewal stream with the real gap distribution, so under it the probability that k consecutive gaps are all shorter than T is $F(T)^k$, with $F(T)$ the tight-gap fraction, and the mean run of tight gaps is $1/(1 - F(T))$. The cluster picture of Section 2 predicts that the real stream exceeds both, and by more where the service time is long enough for many members of one cluster to fall within it. On 254 windows from 20 sessions the per-window median of the probability that a tight gap is followed by another is 0.120, 0.209, 0.363, 0.534 and 0.725 at $T = 8, 16, 32, 64, 128 \mu\text{s}$, against $F(T) = 0.020, 0.183, 0.325, 0.456$ and 0.655 for the renewal stream, and the mean run of tight gaps is 1.14, 1.26, 1.57, 2.14 and 3.63 against 1.02, 1.22, 1.48, 1.84 and 2.90. The excess over renewal on the run length is 1.12, 1.03, 1.06, 1.16 and $1.25\times$, growing with T from $16 \mu\text{s}$, which is the direction of Table 3: the renewal stream keeps 79% of the tail at $16 \mu\text{s}$, where its runs are within 3% of the real ones and the tail is a one-gap wait, and 23% at $128 \mu\text{s}$, where the real runs are a quarter longer and the backlog

compounds over them. At $T = 8\mu\text{s}$ the conditional probability is six times the renewal value (0.120 against 0.020) — gaps near the floor are strongly bunched — yet the renewal stream keeps 88% of the tail there, because at the floor the tail is $0.6\mu\text{s}$ and consists of single waits that no run is needed to produce.

The cluster representation against Section 4.3. The Hawkes–Oakes picture of Section 2.2 is one of immigrant events each generating a cluster of offspring; Section 4.3 identifies the events as matching-engine transactions. Two of its findings are what the picture predicts if the feed relays events one packet each: the branching ratio fitted on transaction starts equals the one fitted on packets (0.795 against 0.798), and merging each transaction into one packet or decoupling transaction sizes from their timing leaves the tail unchanged (Table 7). The corpus also shows that the exchange does not add clustering of its own, since the pieces of a split transaction are paced at $15\mu\text{s}$, above the floor. What the corpus does not test is the cluster-size law itself: the Borel distribution of Appendix A is a statement about offspring counts, which the data do not label, and its check in Appendix A remains indirect.

Summary. All three analytical statements hold on the full corpus. The pathwise bound of Theorem 1 holds on every message of every window under all four packet-level arrival arms, and its slack is measured: $\gamma = 3.32$ and 2.68 at $T = 64$ and $128\mu\text{s}$, and at $T \leq 32\mu\text{s}$ the tandem removes the excess outright rather than dividing it. The renewal null behaves as the cluster picture says a renewal stream should, keeping the one-gap tail at short service times and losing the run-driven tail at long ones, and the transaction analysis places the clusters in the matching engine’s event stream.

7 Cross-validation against a live production receiver

The results of Part II so far are measurements on recorded data: pcap-derived message tapes replayed offline. That design gives scale and repeatability at the cost of external validity. A replayed tape cannot show whether the fitted clustering survives the network path to a receiver, whether it is specific to the 2025–2026 sample, or whether the service times the sweep assumes correspond to production decode times.

This section addresses all three with instrumentation of the KASPAR-HFT production system. The two measurements were built for different purposes and share only the exchange they observe. (The live measurement was taken seven months after the corpus ends, so it also tests whether the clustering is specific to the period the corpus covers.) One limit is stated at the outset: the live per-message record carries the receive and publish timestamps, the ring depth and the position in the datagram, but not the message’s `transactTime`, so the transaction-level identification of Section 4.3 is a corpus result that the live data cannot repeat; what the live data cross-validate is the packet process, and the transaction result rests on the corpus alone.

7.1 The recorded corpus and the live measurement compared

The live measurement instruments the KASPAR-HFT market-data path in production: a timestamp `t0` when a UDP datagram is handed up from the socket, a timestamp `t1` when the resulting book update is published, and one 16-byte record per admitted message carrying the pair together with the ring-buffer depth and the message’s position inside its datagram. It covers 6.81 million messages on six streams — ES, NQ and ZN futures, book and trade — over 53 minutes of a single afternoon. Its latency results are reported in Maciejewski [36]; we draw on them here, and on the underlying per-message records, as a control on Part II.

	Part II corpus	production measurement
source	pcap tapes, replayed offline	live multicast, production recorder
period	2025-01 to 2026-02	2026-09-16 (7 months out of sample)
instruments	NQ front month	ES, NQ, ZN $\times$ book, trade
arrival clock	CME gateway <code>sendingTime</code>	receiver <code>t0</code> at socket read
estimator	per-window, median of 3512 windows	pooled over one 53-min capture
sample	3512 windows, 276 sessions	6.81 M messages, 6 streams

Table 18. The two measurements compared in this section: different instruments, a period that does not overlap, a different arrival clock, and a different estimator.

The arrival-clock difference is the most significant. Part II keys arrivals on `sendingTime`, the timestamp the CME gateway writes into the UDP header as the packet leaves the exchange. The live measurement keys them on `t0`, taken by our own process after the packet has crossed the network, the NIC and the kernel. If the clustering reported in Section 3.2 were an artefact of exchange-side emission scheduling, the intervening path, with its own queueing, coalescing and interrupt moderation, would be expected to attenuate it. It does not; and Section 4.3 rules out the exchange’s packetisation as the source from the other side, by showing that the clustering is already present in the matching engine’s transaction stream.

7.2 Clustering statistics in the corpus and on the live receiver

This section checks that the bursty arrival pattern the paper is built on is real, and not something produced by replaying recordings. It takes four numbers that each measure how far packet arrivals are from evenly spread. The first is the share of gaps between packets that are tiny, under a tenth of the average gap. The second is how much the variability of packet counts drops when the order of the packets is scrambled, which removes any bunching. The third is a measure of whether busy stretches persist rather than fading at once. The fourth is how strongly one event tends to trigger the next. Each of the four is computed twice, independently: once on the recorded corpus that the rest of the paper uses, and once on the live production receiver, seven months later, on different instruments, with a different clock. Three of the four match directly, to within the spread seen across the live streams. The fourth is measured in two different ways, one an estimate and one an upper bound, and the estimate falls inside the bound. The conclusion is that the bursts are a property of the feed as received, not of replay: they show up at the same strength whether one looks at a recording or at the live wire. Where they come from is the question Section 4.3 answers on the corpus: the matching engine’s transaction stream.

Table 19 puts the four statistics that both measurements compute side by side. The NQ column is the like-for-like comparison, since Part II’s corpus is NQ front month; the third column gives the range across the three live book streams.

	Part II (NQ, 3512 win)	live NQ book	live, three books
$P(\text{gap} < \text{mean}/10)$	73.2%	63.1%	63.1–84.8%
Fano(5 s) collapse under shuffle	14.8×	12.2×	12.2–18.5×
Hurst exponent H	0.740	0.702	0.702–0.743
branching ratio n	0.798 (MLE)	$\leq 0.850\text{--}0.972$ (Fano ceiling)	

Table 19. The same four Poisson-rejection statistics, computed independently on recorded and on live data. The Poisson values are 9.52%, 1×, 0.5 and 0 respectively.

Three of the four are direct comparisons and agree to within the spread across live streams. The fourth compares different estimators. Part II fits the branching ratio by maximum likelihood, per window, using the Ogata recursion (Section 3.2); the live measurement does not fit a Hawkes process, and bounds n from above by inverting the Fano ceiling, $n \leq 1 - 1/\sqrt{F}$. The MLE’s 0.798 lies within every bound the live streams imply, which run from 0.850 on NQ trade to 0.972 on NQ book. The bound is loose, and the agreement is weak evidence on its own; its value is that it is independent evidence and is consistent.

The arrival process modelled in Part II is therefore not an artefact of replay, of the sample period, or of the exchange-side clock; it is present at the same strength in a live receiver seven months later on other instruments.

7.3 The production decode floor and the tail threshold

The live system supplies one number that a recording cannot: how long the production decoder actually takes to handle a message when nothing is queued in front of it. This section measures that time three independent ways and finds they agree, at about seven microseconds. The notable result is where that number falls. Part II found, from the recorded corpus, that a tail appears once a stage takes longer than the tightest gap between packets, about seven and a half microseconds. The production decoder sits just under that, within a few hundred nanoseconds, by two measurements that share nothing. The production system therefore operates right at the edge of the tail. The

section then shows what pushes it over: a packet carrying a single message is handled below the threshold, and a packet carrying two or more at or above it, on every stream measured. Whether a packet’s service crosses the threshold is decided by whether it carries one message or more.

Isolating messages that arrive to an empty ring and first in their datagram, so that neither queueing nor in-packet serialisation contributes, gives the hot path directly. Three estimators agree (Table 20):

method	ES	NQ	ZN
direct median, empty ring and first in packet	7.01	7.24	6.89
fitted intercept of the median position ladder	6.98	7.23	6.83
floor at single-message packets during an FOMC release	7.1	7.5	7.0

Table 20. Three independent estimates of the KASPAR-HFT socket-to-book floor, in μs , on the book streams. The three differ in sample, estimator and load regime. The third was taken while the packet rate rose 4.6 to $9.8\times$ within a second; the floor rose by at most $0.27\mu\text{s}$ against either other estimator.

Across all six streams the fitted floor spans 6.81 to $7.23\mu\text{s}$, a range of $0.42\mu\text{s}$ across six multicast channels on one host and one afternoon.

Compare this with Section 4.9. The service time at which a deterministic stage on this feed begins to accumulate a tail is the feed’s tight-gap floor, its 1st-percentile interarrival gap, measured at 7.43 to $7.53\mu\text{s}$ and invariant across intensity quintiles; the refined sweep of Section 4.8 locates the switch-on between $T = 7$ and $T = 8\mu\text{s}$. The production decoder’s floor is 6.81 – $7.23\mu\text{s}$.

The production system operates at the threshold, to within a few hundred nanoseconds, by two measurements made on different data with different estimators. The agreement is close to definitional for a p_{99} target, since a deterministic server leaves its floor at p_{99} once about one packet in a hundred arrives within T of the previous one; for a $p_{99.9}$ target the relevant floor is the 0.1st-percentile gap, which is lower, and Table 2 already shows a $p_{99.9}$ excess at $T = 8\mu\text{s}$.

The live measurement locates the crossing. Per-packet service is the floor plus one slope for each message beyond the first (Section 7.5), so the service time of a datagram carrying span messages is floor + (span – 1) $\times$ slope. Evaluating that on each stream’s own fitted pair:

	span 1	span 2	span 3
ES book	6.98	7.55	8.11
NQ book	7.23	7.54	7.85
ZN book	6.83	7.80	8.76
ES trade	6.81	7.52	8.24
NQ trade	7.14	7.67	8.19
ZN trade	6.81	7.77	8.74
range	6.81–7.23	7.52–7.80	7.85–8.76

Table 21. Per-packet service time in μs by datagram span, from each stream’s measured floor and slope, against the feed’s tight-gap floor of 7.43 – $7.53\mu\text{s}$ (Section 4.9). Every stream is below the threshold at span 1 and at or above it at span 2; ES trade, at 7.52 , sits inside the 7.43 – 7.53 band.

All six streams are below the threshold when a datagram carries one message and at or above it when it carries two (Table 21). All six crossings fall in the same unit interval of span, from floors and slopes fitted separately on six multicast channels against a threshold measured on a different corpus. For this system, whether a packet’s service is above or below the threshold is determined by

whether it carries one message or more.

7.4 Statistics that differ between the corpus and the live receiver

Two numbers do not match between the recorded and the live measurements, and this section explains why neither is a cause for concern. In both cases the disagreement is about how the number was computed, not about the feed, and in both cases one of the two measurements can diagnose the other. The first is the raw level of burstiness, which reads about four times higher live. That is because the live figure was taken over one uncut stretch that mixes a quiet period and a busy one, and a rate that drifts across the stretch inflates the measure whether or not the arrivals are genuinely bursty; the recorded corpus computes the same quantity inside each half-hour window, which removes that drift. The change in burstiness when the packets are scrambled, which is the effect the paper uses, agrees closely between the two. The second is a control value that should come out at exactly one half after scrambling, and reads a little above it live. That is a known small-sample bias of the estimator on a short capture; on the much larger recorded corpus the same control comes out at one half, as it should. Neither discrepancy touches any quantity the paper’s argument rests on, and for the level of burstiness the recorded corpus gives the more conservative value, which is the one reported.

The absolute Fano factor and the length of the stretch. Two estimators are in play. The *windowed* estimator cuts the data into 30-minute windows, computes the Fano factor of the bin counts inside each window, and reports the median across windows; the *pooled* estimator treats one long uncut stretch as a single series and computes the Fano factor over all of it. (This is a different sense of “pooled” from the histograms of Section 4.3, which pool gap *lengths* across windows into one distribution; a distribution of lengths is indifferent to slow rate drift, a count-variance statistic is not.) Part II, windowed, measures a corpus-median Fano at a 5-second bin of 259.7; the live streams, pooled over the 53-minute capture, give 1024 to 1260. The ratio under the gap shuffle agrees closely (Table 19), so the discrepancy is in the level, not in the effect. The cause is the estimator row of Table 18. The live figure is computed over one uncut 53-minute capture that spans a quiet stretch and a busy one; a rate that drifts between the two inflates the count variance and reads as clustering whether or not the process is self-exciting. Part II computes the Fano factor *within* each 30-minute window and then takes the corpus median, which removes between-window rate variation by construction: a slow drift from a quiet stretch to a busy one adds count variance that the estimator cannot distinguish from bursts, so the pooled level reads roughly four times higher. The gap-shuffle ratio cancels the drift, since the drift is present in the numerator and the denominator alike, which is why the ratio agrees between the two measurements and why the paper uses the ratio, not the level. The windowed estimate is the conservative one and is the value reported.

The shuffled Hurst null and finite-sample bias. A Fisher–Yates shuffle of a gap sequence destroys the ordering while preserving the marginal exactly, so the shuffled series is a renewal process and its Hurst exponent must be 0.5. The live measurement recovers 0.600 to 0.660 instead, and attributes the excess to finite-sample bias in the Hurst estimator at the number of windows available in a 53-minute capture and states its result as the gap between 0.74 and 0.62 rather than the distance from 0.5. Part II confirms the attribution: over 3512 windows the same shuffle returns $H = 0.501$, and the uniform Poisson null P returns $H = 0.496$ (Section 3.2). The bias is finite-sample and vanishes with more, shorter series.

7.5 Per-packet service time, message count and coalesced transactions

The simulator in Part II assumed that every packet takes the same time to process. The live measurement shows that it does not, and this section sets out what that means. A packet on this feed carries a variable number of messages, usually one but sometimes dozens, and they are decoded one after another, so a packet takes longer the more it carries. Large packets are rare but expensive: on the most fragmented stream, ZN trade, one packet in nine carries two thirds of the messages and consumes six sevenths of the decode time. The task length used throughout the paper is therefore the work for a typical packet, not a fixed constant of the system. Two points follow, set out at the end of the section: the results survive, because the cost per message is constant and only the bundling varies; and the number of messages per packet is a second place where the same clustering is paid: transactions that reach the publisher while it is busy can be packed into one packet rather than sent in separate packets (Section 5.5). It acts mainly where bursts across packets do not, on tasks at or just below the tightest gap between packets, which is where production receivers run, and adds a few percent above $16\mu\text{s}$. The next section reruns the sweep with the message count included and measures the share of each channel.

The sweep of Section 3.5 treats each packet as one arrival with a constant service time T . Per-packet service is not constant. A UDP datagram on this feed carries a variable number of SBE messages — call it the packet’s *span* — which are decoded in order, so a message at position k waits for k decodes before its own. Median latency follows

$$\text{median} = \text{floor} + \text{slope} \times \text{position},$$

with the floor of Section 7.3 and a slope of 0.31 to $0.97\mu\text{s}$ per message depending on the stream. Span is not close to constant (Tables 22 and 23):

stream	packets	messages	p_{50}	p_{90}	p_{99}	max
ES book	2,123,042	2,415,610	1	1	7	45
NQ book	2,506,205	3,010,036	1	2	11	36
ZN book	824,536	965,334	1	1	6	45
ES trade	83,140	235,601	3	8	17	83
NQ trade	50,279	83,233	2	7	16	85
ZN trade	22,864	100,659	2	11	71	85

Table 22. Messages per UDP datagram (span), packet-weighted quantiles. The median book packet carries one message; the 99th percentile carries 6 to 11 on the book streams and 71 on ZN trade.

Because service is charged per message, large packets account for a disproportionate share of processing time:

span ≥ 10	% of packets	% of messages	% of decode time
ES book	0.52	4.55	5.34
NQ book	1.68	4.87	2.57
ZN book	0.47	7.29	7.29
NQ trade	5.34	19.31	16.16
ES trade	6.69	26.53	40.10
ZN trade	11.31	66.18	85.54

Table 23. Concentration of work in large datagrams. On ZN trade, 11.3% of packets carry 66% of the messages and consume 86% of decode time.

The constant-service assumption is therefore violated at the packet level, and Part II’s T is the per-packet work at a typical span rather than a constant of the system. Two points follow.

First, at the message level the assumption approximately holds: the slope is fixed within a stream to the precision of the fit, and it is the composition of packets, not randomness in per-message work, that makes per-packet service vary. Theorem 1 is pathwise and arrival-law independent, and Remark 2.3 already handles unequal stages through $s_{\max}$; what the live data adds is a measured magnitude for the variability that Section 2.3 lists as an assumption.

Second, span is a second channel for the same clustering on the same stream, active where the queueing channel is not. Section 7.6 reruns the sweep with span-dependent service. The serialisation channel supplies the tail below the tight-gap floor, where queueing across packets supplies none, and adds a few percent above $16\mu\text{s}$, where queueing across packets supplies nearly all of it.

Spans and transactions. On the corpus, where every message carries its `transactTime`, a multi-message packet is usually not one large transaction but several transactions delivered together. On 20 sessions (2.45×10^8 packets) 4.0% of packets carry two or more messages and 3.1% carry parts of two or more transactions; of the packets with two or more messages, 77% carry more than one transaction, and of those with five or more, 62%. The gateway, that is, coalesces transactions that arrive within its emission interval into one datagram. This is consistent with the packet gap floor of Section 4.9: two transactions closer together than about $7.5\mu\text{s}$ do not appear as two packets with a sub-floor gap but as one packet with a larger span, which is why so few packet gaps on the feed (2.15%, Table 5) fall below the floor. Two consequences follow for the rest of this section. The serialisation tail that a large packet produces is, for the most part, the same clustering of transactions as the queueing tail, delivered inside a datagram instead of across datagrams; and the message count per packet is set by how closely transactions follow one another, not by how large individual transactions are (only 4.6% of transactions carry more than one message, Table 4). The two-mechanism description below — queueing across packets, serialisation within them — stands as a description of where the wait is paid; both originate in the transaction stream.

7.6 The sweep rerun with span-dependent service

The previous section found that a packet takes longer to process the more messages it carries. This section reruns the whole simulation with that fact built in, to find out what it changes. The processing time is modelled three ways: fixed per packet, as in Part II; growing with message count at the rate measured on the live system; and, as a deliberately exaggerated upper bound, fully proportional to message count. All three do the same total work on average, so any difference in the slow tail comes from how the work is spread across packets. There are four findings. First, message count and bursts produce slow messages in different places. Message count matters most for short tasks, at or just above the tightest gap between packets, which is where production receivers run, where bursts build little or no queue; for longer tasks the bursts dominate and message count adds a few percent at $16\mu\text{s}$ and one or two beyond. Second, at the simulated task length nearest the production system’s seven microseconds, eight, message count accounts for roughly two thirds of the slow tail and bursts for one third, and the live system reports the same ordering. Third, the recommendation on how many stages to split a task into is unchanged at every task length. The in-packet part of that tail is the plain cost of decoding several messages one after another, which splitting the work into stages does not remove; the queue a large packet opens behind itself is taken up in Section 7.7. Fourth, on NQ the two channels do not compound at the packet level: a large packet is no more likely to arrive after a tight gap than after a loose one, checked two ways, so the

tables of Part II, which measure the queueing channel alone, stand as they are at service times above the floor. On ZN the live data suggest the opposite, and the test has not been run there.

We re-extracted the per-packet span for all 3512 windows — aligned element-wise against the cached arrival arrays, with every window’s span count checked against the packet count already in the cache — and re-ran the entire sweep with service that depends on it.

Three service models are compared. All three hold the *mean* service per window at T , so the utilisation ρ is identical across them and only the distribution of work across packets differs; any difference in the tail is attributable to the span dependence.

const :	$S_i = T$	(the Part II assumption)
span :	$S_i = A(1 + r(\sigma_i - 1))$, $r = 0.0432$	(measured slope/floor)
prop :	$S_i = T \sigma_i / \bar{\sigma}$	(upper bound)

with A set so that the window mean of S_i is T . The *span* model uses the ratio of slope to floor measured on NQ book in Section 7.3, 0.312/7.23. The *prop* model removes the per-packet fixed cost, making service proportional to the message count; this overstates the dependence, since the measured floor exceeds the slope by more than twenty to one, and it is included as an upper bound. As a correctness check the *const* arm reproduces the published constant-service grid of Table 2 to zero difference in all 28 cells.

Message-weighted quantiles. On this corpus, pooled over its 3.30×10^9 packets, 95.9% of packets carry one message and the mean span is 1.078 (per-window medians 96.0% and 1.077), so a packet-weighted quantile is dominated by singletons and changes little under any service model. Weighting by message, which counts a 20-message datagram twenty times, the in-packet position distribution is concentrated at zero with a long upper tail:

message quantile	p_{50}	p_{90}	p_{95}	p_{99}	$p_{99.9}$	max
in-packet position	0	0	1	4	19	84

89.0% of messages are alone in their packet and pay no serialisation; the remaining 11.0% constitute the tail. The live measurement reports the same shape. The comparison below is message-weighted.

T	const			span			span/const		prop/const	
	p_{50}	p_{99}	$p_{99.9}$	p_{50}	p_{99}	$p_{99.9}$	p_{99}	$p_{99.9}$	p_{99}	$p_{99.9}$
2	2.00	2.00	2.00	1.99	2.34	3.54	1.17	1.77	4.64	23.35
4	4.00	4.00	4.00	3.99	4.67	7.08	1.17	1.77	4.69	26.73
8	8.00	8.56	10.10	7.97	9.63	14.87	1.12	1.47	5.21	24.64
16	16.00	34.00	53.55	15.95	35.08	56.62	1.03	1.06	3.36	11.53
32	32.00	143.74	256.51	31.89	144.41	260.14	1.00	1.01	2.37	5.88
64	98.80	815.22	1557.72	100.02	820.05	1573.97	1.01	1.01	2.37	3.94
128	366.78	5440.22	12844.51	371.91	5550.42	13159.71	1.02	1.02	2.51	3.10

Table 24. Effect of the number of messages per packet (the packet’s *span*, not its byte length) on the single-stage tail. Each row is one service floor T ; every entry is the corpus median over 3512 windows of a message-weighted latency quantile, in μs , under real (Hawkes) arrivals with a single stage ($N = 1$). The *const* block charges every packet the same service T , the Part II assumption. The *span* block charges service that grows with the number of messages in the packet, at the slope measured on the live system. Both do the same mean work per window, so any difference between the two blocks is due to message count alone. The *span/const* columns give that difference directly as a ratio; a value near 1 means message count adds nothing. The *prop/const* columns repeat the ratio for a deliberately exaggerated model in which service is fully proportional to message count, as an upper bound. Read across: message count inflates the tail only at $T \leq 8 \mu\text{s}$, at or just above the tight-gap floor, where clustering builds little or no queue, and is negligible above it.

Regimes of the two tail mechanisms. The *span/const* ratio of Table 24 is largest where the queueing tail is absent and smallest where the queueing tail dominates. At $T = 2$ and $4 \mu\text{s}$ the constant-service model has no tail ($p_{50} = p_{99} = p_{99.9} = T$, Section 4.1) while the span-dependent model has $p_{99} = 1.17T$ and $p_{99.9} = 1.77T$. Below the tight-gap floor there is a tail, but it is not a queueing tail. By $T \geq 32 \mu\text{s}$, where cluster queueing produces a p_{99} of $4.5T$ and rising, span contributes 0–2% on top.

Decomposition at the production operating point. The live system’s measured floor is $6.81\text{--}7.23 \mu\text{s}$ (Section 7.3), so the comparable row is $T = 8 \mu\text{s}$. Decomposing the message-weighted p_{99} excess over the floor into the part the constant-service model already produced (queueing) and the remainder introduced by span-aware service (serialisation):

T	floor	queueing excess	serialisation excess	serialisation share
2	1.99	0.00	0.34	100%
4	3.99	0.00	0.69	100%
8	7.97	0.56	1.09	66%
16	15.95	18.00	1.14	6%
32	31.89	111.74	0.78	1%
64	100.02	716.42	3.61	1%
128	371.91	5073.43	105.07	2%

At $T = 8 \mu\text{s}$ serialisation supplies 66% of the p_{99} excess and queueing 34%. The live measurement reports the same ordering: evaluating the position ladder at each stream’s own p_{99} position accounts for 71% to 80% of the observed p_{99} on four of its six streams, with the remainder left to queueing and to stalls. Applying that definition to the simulator at $T = 8$ gives 97%, higher than the live figure because the only residual the model contains is queueing; it has no stall population. Both

measurements attribute the larger share of the tail at the production service time to in-packet serialisation.

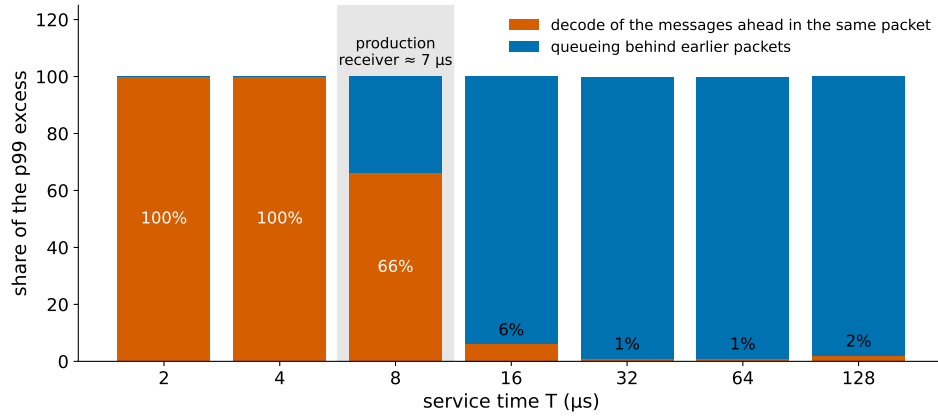


Figure 14. The decomposition above as bars: of the simulated p_{99} excess at each service time, the part due to decoding the messages ahead in the same packet (vermillion) and the part due to queueing behind earlier packets (blue). At and below the floor the decode of long packets is the tail; the shaded bar, $T = 8 \mu s$, is the nearest simulated service time to a production receiver’s $7 \mu s$, and there it is two thirds. From $16 \mu s$ queueing is almost all of it.

This is consistent with Part II. The queueing claim stands: below about $7.5 \mu s$ of service, arrival clustering does not build a queue and splitting the servicing chain removes nothing. The span-dependent run adds that a second mechanism, in-packet serialisation, is active in the same regime; it is deterministic within the packet and a stage cut does not shorten it, while the queue that a large packet opens behind itself is taken up in Section 7.7. N^* , the p_{99} -minimising stage count, is unchanged at all seven service floors when service is made span-dependent: 1, 1, 1, 2, 4, 8, 8 under both models. The design rule of Section 11 is unaffected.

The in-packet ladder. Median latency by in-packet position at $N = 1$ reproduces the linear form of the live measurement. At $T = 8 \mu s$ the corpus-median latency by position fits

$$\text{median} = 7.98 + 0.344 \times \text{position} \quad (\mu s),$$

recovering the $0.345 \mu s$ per-message cost that was input to the model, which confirms that queueing contributes nothing at the median at this service time. The live NQ-book ladder is $7.23 + 0.312 \times \text{position}$; the forms agree and the levels differ by the difference between the $7 \mu s$ measured decode path and the $8 \mu s$ nominal service floor.

Compounding of the two channels on NQ. There are two places a message can be made slow. One is queueing: it waits behind a run of earlier packets. The other is serialisation: it sits deep inside a large packet and waits for the messages ahead of it in that same packet to be decoded first. Section 7.5 showed that a large packet is usually several transactions coalesced by the gateway, so both are deliveries of the same clustering. The question here is narrower: whether a large packet also tends to arrive behind a tight gap, so that one message pays both delays at once. If it did, the two would compound and the Part II tables, which model queueing alone, would understate the tail above the floor. If a large packet arrives after a tight gap no more often than after a loose one,

the tables stand. Two tests find the second case. Both compare a packet’s span with the gap that precedes it, so they cannot see the coalescing itself, which removes a gap rather than tightening one.

The first is a control arm in which each window’s span sequence is randomly permuted across packets, preserving the span marginal and the arrival times exactly and destroying only the association between them. If large packets arrived preferentially inside clusters, breaking that association would lower the tail. It does not: the ratio of the real to the span-shuffled p_{99} is 1.000 at every $T \leq 8 \mu\text{s}$ and lies between 0.99 and 1.02 at every larger T , with no systematic sign. Second, measuring the association directly, the correlation between a packet’s span and the logarithm of the gap that preceded it has corpus median -0.014 ; packets arriving after the tightest decile of gaps have mean span 1.010 against 1.043 for those arriving after the loosest decile, a ratio of 0.97. The association is weakly negative.

The live measurement reports the same instrument dependence: mean span rises with queue depth on ZN, where a quiet book is filled only by genuine bursts, but *falls* with queue depth on ES and NQ, whose rings back up with many small packets. Part II’s corpus is NQ, and it behaves like the live NQ streams. The transaction-level arms of Section 4.3 give the same answer a third way: decoupling transaction sizes from their timing (TS) and merging each transaction into one packet (TM) leave the tail within $\pm 15\%$ at every service time from $16 \mu\text{s}$ up (Table 7). Above the floor, then, the serialisation channel adds a few percent to a tail the queueing channel produces; at the floor, where the production receiver runs, it carries most of the tail, and the constant-service tables of Part II measure that channel; at the floor, where serialisation carries most of the excess, the coalescing of transactions into one datagram described in Section 7.5 means the two channels are two deliveries of the same clustering. On an instrument like ZN, where the far tail is a few large packets, the second delivery dominates; Section 7.7 takes this up.

7.7 The production handler’s tail

The live records also give the production handler’s own tail, on the shipping serial configuration: one thread decoding every packet inline. The question is the one the paper opened with, put to a real handler. On the live feed, why is the slowest message in a thousand so much slower than the typical one? Read with Sections 4.3 and 7.5, the answer is that the clustering of transactions reaches this handler in two forms. Most transactions that the engine processes close together reach it as a train of packets at the publisher’s spacing of about $7.5 \mu\text{s}$ (Table 12); some are packed by the publisher into one packet and reach it as a single packet carrying several messages, which the handler decodes one message after another, so that the packet holds the thread for many times the floor and everything behind it queues. On the corpus, a multi-message packet is several transactions delivered together far more often than it is one large transaction (77% of packets with two or more messages carry more than one transaction, Section 7.5), so a large packet is a transaction cluster in its own right. On the ZN channel the second form dominates the far tail. What follows is argued from the fitted floor and slope of Section 7.3 and the reported percentiles, not from a message-by-message decomposition of the window.

Setup. Channel 344 (ZN), book stream, one 900s window, recovery windows excluded, the shipping serial configuration (`cme_decode_workers 0`). Latency is the leg-1 socket-read-to-book-publish time of Section 7.3, per message. ZN is the instrument on which Section 7.6 left the coupling of span and bursts open. Two populations are reported: every message, and the *hot path*, the messages that arrived to an empty queue and were first in their packet, which is the decode with neither queueing nor in-packet serialisation in it.

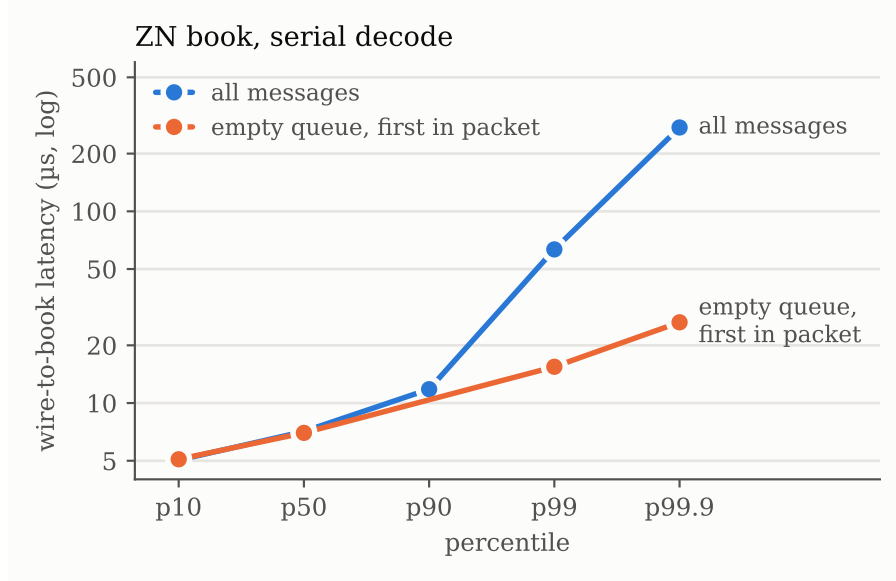


Figure 15. Wire-to-book latency by percentile on the ZN book stream, serial decode, for every message and for the hot path alone (empty queue, first in packet; its p_{90} was not reported). Log latency axis. The two populations share a floor near $5\text{--}7\ \mu\text{s}$ and separate only in the tail: the hot path reaches $26\ \mu\text{s}$ at $p_{99.9}$, all messages $274\ \mu\text{s}$.

population	messages	p_{10}	p_{50}	p_{90}	p_{99}	$p_{99.9}$	mean
all messages	157,699	5.04	7.11	11.81	63.30	273.69	9.95
empty queue, first in packet	122,500	5.09	6.99	—	15.46	26.37	—

Table 25. The data behind Figure 15, in μs . The hot path is 78% of all messages. The window maximum for all messages was $3046\ \mu\text{s}$; the report that records this measurement (`tech_reports/serial_vs_parallel_decode.md` in the KASPAR-HFT repository) flags a few such isolated spikes as possibly residual recovery, so it is not used here.

The spread from p_1 to $p_{99.9}$. The slowest message in a thousand takes $274\ \mu\text{s}$ against $5\ \mu\text{s}$ for the fastest tenth and $7\ \mu\text{s}$ for the median: fifty-four times the one and thirty-eight times the other. On the hot path the same ratio is five. Conditioning on an empty queue and first position in the packet removes nine tenths of the far tail, $274 \rightarrow 26\ \mu\text{s}$, without changing the floor. The difference between the two tails is the contribution of queueing and of position inside the packet. The hot path itself is not at the floor in its tail (15.5 and $26.4\ \mu\text{s}$ at p_{99} and $p_{99.9}$ against a $7\ \mu\text{s}$ floor); that residual, present with neither queueing nor serialisation, is not resolved by these records (Section 13.1).

How the live tail forms. The three findings of this paper meet here. Packets reach the handler as trains about $7.5\ \mu\text{s}$ apart (Section 5.5). The handler’s median service, 6.4 to $7.1\ \mu\text{s}$ on the book streams (Table 26), is just under that spacing, so a train of single-message packets served at the median builds no queue; this is why its p_{10} and p_{50} sit at the floor. Two things push a service over the spacing. A packet with two or more messages costs the floor plus one per-message slope for each further message, which crosses $7.5\ \mu\text{s}$ on every stream (Section 7.3). And a single-message packet sometimes takes far longer than the median for reasons these records do not show: on ZN the hot path reaches $15.5\ \mu\text{s}$ at p_{99} . Either way the packets still arriving behind it queue, and the

queue lasts until the handler catches up with the train. The large-packet mechanism is a matter of arithmetic on the measured floor and slope; the slow-service mechanism is measured only as the hot-path percentiles above. How the $274\mu\text{s}$ at $p_{99.9}$ divides between the decode inside long packets, the queue behind them and slow services was not decomposed, because the per-message records were not retained. What the records do show is that the part of the tail not explained by queueing across packets is large at this service time: the floor plus the decode cost of the messages ahead in the same packet reproduces 71 to 80% of the observed p_{99} on four of the six streams (Section 7.6).

Large packets and the queue behind them. The message count of a packet is set by how closely transactions followed one another, not by how large any one transaction was: only 4.6% of transactions carry more than one message (Table 4), while a packet of ten or more messages is, in most cases, ten or more transactions that the publisher packed into one packet. Section 7.3 gives what such a packet costs this handler: a floor of $6.8\mu\text{s}$ plus about $1\mu\text{s}$ for each further message, so the service time grows in proportion to the number of transactions coalesced, and the largest packet seen, forty-five messages (Section 7.5), holds the thread for some $50\mu\text{s}$, seven times the floor. That is the first delivery of the clustering: the last message of that packet waits some $45\mu\text{s}$ before its own decode, but even so the in-packet serialisation accounts for only a fifth of the $p_{99.9}$. The rest is the queue the packet opens behind it, the second delivery. A deterministic server begins to build a queue when two arrivals are separated by less than its service time (Section 4.9); at one message per packet the service is $7\mu\text{s}$, at the tight-gap floor measured on NQ (ZN’s own gap floor was not measured), and a queue almost never forms, which is the hot path of Table 25. While a forty-five-message packet holds the thread for $50\mu\text{s}$, every packet that arrives waits behind it, and each of those adds its own service to the wait of the ones behind it: the wait of the k -th packet in the queue is the sum of the service times ahead of it, so a large packet followed by any arrivals within its service time produces a wait that grows with the number queued, not with any one packet’s span. The 0.47% of ZN packets carrying ten or more messages are few, but each is a burst of transactions that this handler pays for twice, once inside the packet and once in the queue behind it, and the hundreds of microseconds at $p_{99.9}$ are those queues.

Across the six streams. The same measurement covers three instruments and, on each, a book stream and a trade stream. Figure 16 puts the six serial tails side by side and Table 26 adds, for each stream, its packet rate, the size of its packets and the cost of each extra message. Three things are visible at once. First, the tail does not follow the packet rate: NQ book is by far the busiest stream and has the smallest tail, ZN book carries a third of its rate and a tail an order of magnitude larger, and the trade streams, twenty-five to fifty times quieter than the books, have larger tails than the books on every instrument. Second, on every instrument the trade stream carries more messages per packet than the book stream and has the larger tail: a trade is a transaction that touches several resting orders and is followed by the reaction to it, so it is where transactions arrive closest together and are coalesced most. Third, ZN, with the largest trade packets and the highest cost per extra message, has the largest tails on both streams; its book packets are not large at the 99th percentile, but at three times NQ’s cost per message and a maximum span of 45 the excursions they produce are the longest of any book stream. The clustering statistics are similar across the three instruments (the Fano-implied branching ceilings of Section 7.2 span 0.85–0.97 across all six streams, and the NQ floor is assumed to hold on the others). What separates a twenty-fold spread of tails across streams with similar clustering statistics is not the arrival rate and not those statistics; it is how much of the clustering the publisher delivers inside a packet, the number of coalesced transactions, times the cost this handler pays for each, and the queue that opens behind the packet. On the live system the far

tail is the clustering of transactions delivered as span.

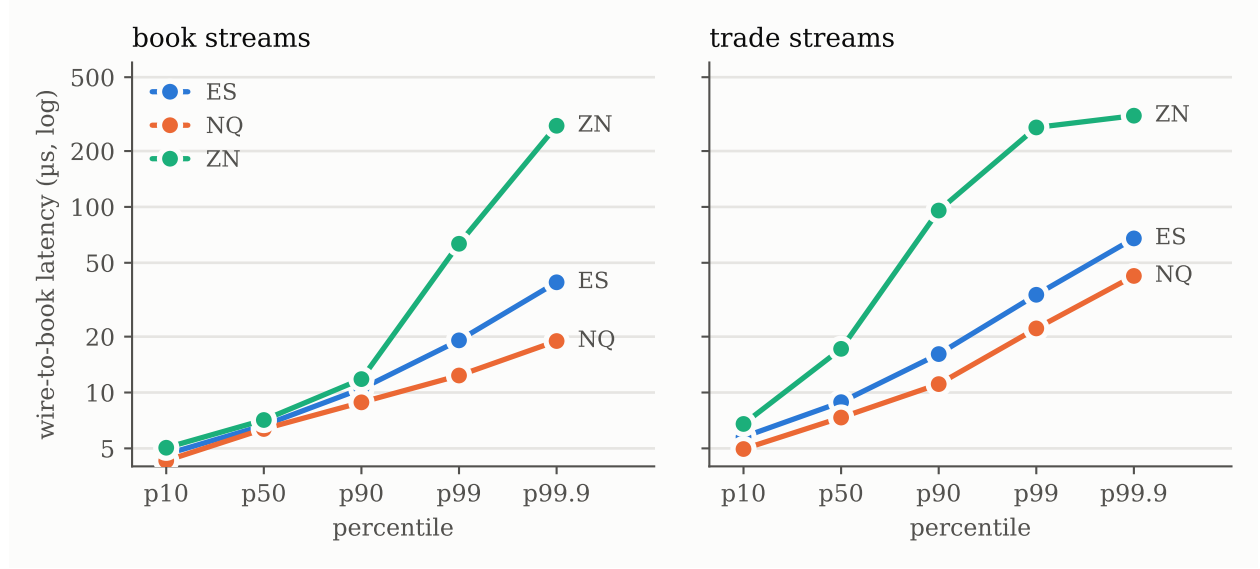


Figure 16. Serial wire-to-book latency by percentile on the six live streams, one 900 s window each, recovery excluded (book streams left, trade streams right, log latency axis). The tails order by messages per packet and cost per message, not by packet rate: the busiest stream (NQ book) has the smallest tail.

stream	packets	span p_{99}	$\mu\text{s}/\text{msg}$	p_{50}	p_{99}	$p_{99.9}$
NQ book	2,506,205	11	0.31	6.36	12.35	18.94
ES book	2,123,042	7	0.57	6.65	19.10	39.25
ZN book	824,536	6	0.97	7.11	63.30	273.69
ES trade	83,140	17	0.71	8.87	33.62	67.65
NQ trade	50,279	16	0.53	7.33	22.15	42.47
ZN trade	22,864	71	0.96	17.19	268.04	310.13

Table 26. The six streams of Figure 16, ordered by packet rate. Packets and span p_{99} are from the 53-minute capture of Section 7.5; the cost per extra message is the fitted slope of Section 7.3; the latency quantiles, in μs , are from the 900 s serial windows. Rate falls down the table while the tail rises; span p_{99} and cost per message, taken together, track the tail.

Consequences for the design rule. The live handler’s tail is the mechanism of Sections 4.3 and 7.5, seen from the receiver’s side. Section 7.5 measured that a packet’s decode time is the floor plus one slope for each message beyond the first, with packets of up to 45 messages on the book streams and 85 on the trade streams, and that such packets are coalesced transactions. Section 7.6 re-ran the sweep with that span-dependent service and found that the message count inflates the tail at service times at or just above the tight-gap floor, where it supplies about two thirds of the p_{99} excess at $T = 8 \mu\text{s}$, adds a few percent above $16 \mu\text{s}$ where the queue across packets takes over, and leaves N^* unchanged; the two are one clustering delivered two ways, inside and across datagrams. The ZN records put the production handler in the first regime: at a $7 \mu\text{s}$ floor the typical packet queues behind nothing and the verdict of Section 8 for it (do not split at the floor) stands, while the far tail is paid on the coalesced packets, whose effective service is several times the floor. That is the regime in which the design rule of Section 4.6 admits a cut, and a cut would shorten the queue

such a packet opens behind it but not the serialisation inside it; the span-aware sweep of Section 7.6, calibrated on NQ's slope, a third of ZN's, found N^* unchanged, so whether a cut pays on ZN is for the measurement to decide. The other lever is the per-message cost itself (Section 11), which scales the whole excursion. Which lever to pull is decided by the end-of-build measurement of Section 9.3.

Part III

Best practices: implications for HFT systems design

8 The reference instance: KASPAR-HFT and its `fast_send` pipeline

This section introduces the system that the rest of Part III works on, and applies the design rule of Section 4.6 to it: cut a chain only where the cut shortens its slowest stage, and never once that stage is already below the feed's tight-gap floor, because below the floor there is no tail left to remove. It first says what KASPAR-HFT is and names the two ways its actors hand messages to one another, one synchronous and one queued, because the choice between those two at each boundary is exactly what the design rule decides. It then lays out how the production pipeline is actually wired: which boundaries are queued and why, and which stages run as one unbroken chain on a single thread. Finally it runs the design equation on that chain. On the chain as it stands, about seven microseconds long, the rule says do not split it. On a longer chain, sixteen microseconds, it says make one balanced cut and no more. The section after this one turns those verdicts into a procedure for changing the real system one boundary at a time and measuring the result.

The recommendations in this Part are stated for HFT systems in general and worked out on one system. KASPAR-HFT [36, 37] is an open-source C++20 trading system for CME futures built on a shared-nothing actor framework, available at <https://github.com/vincent212/kaspar-hft>: every component (socket reader, feed arbitrator, SBE decoder, order book, strategy, order manager, iLink session) is an actor with private state and a mailbox, and components interact only by passing messages. It is at once a production system (MDP3 multicast in, full order books reconstructed order by order, authenticated iLink 3 sessions out) and, with the same actor code driven from recorded packet captures, a deterministic replay simulator in its own right. Part II does not run on that simulator: its measurements are made on a standalone tandem-Lindley model of the pipeline (Section 3.3), driven by packet-arrival timestamps extracted from the same class of capture. The framework supplies the architecture under study and the calibration of h ; it is not in the measurement loop. The `fast_send` paper documents the four extensions that make the actor model usable at HFT latencies: `fast_send` itself, a synchronous delivery in which the sending thread runs the receiver's handler inline and takes the reply as a value; actor `Groups`, which co-schedule a set of actors on one thread behind a single mailbox; mailbox queue implementations selectable per actor; and a memory pool that removes per-message allocation and its tail. The defining property of `fast_send` is *receiver transparency*: a handler is written once and has no way to tell whether it was invoked synchronously or asynchronously, on which thread, or whether its sender is blocked. Section 9 builds on that property.

The `fast_send` paper [36] built and measured a production trading pipeline; that pipeline is the design this paper re-examines. It has three sections joined by exactly two cross-thread hops. Two socket-reader actors, one per redundant multicast channel, do nothing but drain their UDP sockets and hand datagrams to a buffer actor by an asynchronous `send`; that hop is asynchronous because a

reader that blocked on its callee would drop packets under a burst. From the buffer, arbitration between the A and B feeds, SBE decode, order-book construction, and signal generation run as a single chain of `fast_sends` on one thread, with no queue and no context switch between stages. When the signal decides to act it hands the order to an order-management actor by a second asynchronous `send`, because the iLink write takes microseconds and blocking the decode thread on it would stall the market-data queue. The measured cost of each delivery path is given in Section 9. Against a servicing chain of $\sim 7 \mu\text{s}$ measured tick-to-book on live ES, NQ, and ZN feeds, the framework’s own per-message contribution is under 1% of the floor, and the same measurement attributed the latency tail to the arrival stream rather than to the framework; the stream was strongly clustered, with a Fano-implied branching-ratio ceiling of 0.85–0.97. That observation motivated the present paper.

Read against Part II, this pipeline is the single-thread servicing chain of Section 1.2 realised with actors: the whole T sits on one thread between two hops that were placed where blocking was unacceptable. The present paper adds that the same logic applies inside the chain once the chain is long enough for the feed to queue behind it, and the recommendations that follow are stated as changes to this pipeline: where to cut it, in what order to place the stages, which boundaries to leave on `fast_send`, and how to decide each by measurement.

The design rule applied to this pipeline. The rule of Section 4.6, cut only where the cut shortens the slowest stage and never below the feed’s tight-gap floor, can be run on the reference chain directly. It needs one input: the tail excess that a single stage of a given length s would carry, $\Delta_{\text{single}}(s)$. Table 10 supplies it. Section 4.6 showed that the excess depends only on the per-stage service T/N , so every entry of that table is the excess of one stage of length T/N , and read that way the table is a lookup: zero at $s \leq 4 \mu\text{s}$, 0.61 at 8, 18.59 at 16, 114.14 at 32 and 721.82 at $64 \mu\text{s}$.

Whatever the split of the $\sim 7 \mu\text{s}$ chain across its natural boundaries (arbitration, SBE decode, book application, signal), every partition has $s_{\text{max}} \leq 7 \mu\text{s}$, where Part II finds $\Delta_{\text{single}} = 0$: 0.00 at $s \leq 4 \mu\text{s}$, and at $s = 7 \mu\text{s}$ a tail on only 4.5% of windows with a median excess of 0.00 (Section 4.8). The chain has no queueing excess to remove, every cut returns exactly its hop tax, and $N^* = 1$. The tail it does carry comes from the few packets that carry many messages: their in-packet serialisation, which a stage cut does not shorten, and the queue their long service opens behind them (Section 7.7).

The design rule applied to a longer chain. The rule applies once the chain exceeds the floor, for instance when a strategy stage is added, or on a feed with a lower gap floor. Take a $16 \mu\text{s}$ chain with the same four boundaries, decode and book now at $6 \mu\text{s}$ each and the two small stages at $2 \mu\text{s}$:

partition of the $16 \mu\text{s}$ chain	s_{max} (μs)	hops	$\Delta_{\text{single}}(s_{\text{max}})$ (μs)	p_{99} excess + hop tax
single thread (16)	16	0	18.59	18.59
8 + 8 (arb+decode book+signal)	8	1	0.61	2.31
8 + 6 + 2 (three stages)	8	2	0.61	4.01
2 + 6 + 6 + 2 (all four split)	6	3	0.00	5.10

Every entry is read off Table 10 and the refined sweep of Section 4.8: $\Delta_{\text{single}}(16) = 18.59$, $\Delta_{\text{single}}(8) = 0.61$, $\Delta_{\text{single}}(6) = 0$. Three points follow. The balanced two-way cut is the best option: it removes $18 \mu\text{s}$ of p_{99} excess for $1.7 \mu\text{s}$ of hop cost, an eightfold improvement in the tail objective on these illustrative stage times. Three stages do not lower s_{max} , since the $8 \mu\text{s}$ stage remains the bottleneck, so the second hop costs $1.7 \mu\text{s}$ and removes nothing (Remark 2.3, third consequence). Splitting all four lowers s_{max} to 6 and removes the remaining $0.61 \mu\text{s}$ of excess at a cost of 3.4, and is worse. The recommendation for a chain of this length is one cut, placed so that the two halves are

as equal as the natural boundaries allow, with the larger half at ingress (Remark 2.4). The numbers are a prior for the end-of-build measurement of Section 9.3, not a substitute for it. The per-stage service times are illustrative and are replaced by the operator’s profile of the chain.

9 Implementation on KASPAR-HFT: setting the send mode of each boundary

The practical claim of this paper is that on Hawkes-clustered feeds the choice of `fast_send` versus asynchronous `send` at each actor boundary is a late-stage optimisation decision, made per boundary from measured end-to-end median and tail latency of the assembled pipeline on the target feed, rather than an early architectural commitment. KASPAR-HFT supports this workflow. This section states the property that makes it possible and the procedure it enables.

The design equation maps onto framework choices in a specific way. The paper’s model of an N -stage tandem assumes each stage’s actor runs on its own thread and sends its output to the next stage through an asynchronous mailbox; the hop cost h is the wall-clock latency of that handoff.

Two hop paths in the shipping KASPAR-HFT framework. Of the delivery mechanisms described in Section 8, two matter for the design equation. The default asynchronous `send` uses `BQueue`, a blocking mailbox built on `std::mutex + std::condition_variable + boost::circular_buffer`; a producer thread enqueues under the mutex and a consumer thread waits on the condition variable. This is the hop path the paper’s Section 3.3 model corresponds to, and halving its measured $\sim 3.4\mu\text{s}$ round trip gives the one-way hop cost $h \approx 1.7\mu\text{s}$ used throughout Part II.

The `fast_send` path is different: it is a *synchronous* dispatch that acquires a per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and returns the reply. There is no queue and no cross-thread wake-up. Its per-call cost is much smaller than the `BQueue` path’s, but it is not a pipelined hop: the sender blocks until the receiver finishes, so a chain of `fast_sends` serialises through all stages on one thread and offers none of the throughput speedup of Proposition 1. The paper’s design equation does not apply to `fast_send` chains.

9.1 Receiver transparency of `fast_send` and `send`

Every actor in KASPAR-HFT registers its handlers through the `MESSAGE_HANDLER(MsgType, handler_method)` macro. A handler is a function of one message argument; nothing in its body can access, observe, or condition on the delivery mode by which it was invoked. This property is called *receiver transparency* in Maciejewski [36] and is what makes the send-mode choice a purely sender-side decision.

Concretely:

- **Synchronous delivery:** `target->fast_send(m, this)`. The sender’s thread acquires `target`’s per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and takes the reply as a return value. No queue, no cross-thread wake-up. Round-trip cost $\sim 30\text{ ns}$ per Table 4 of Maciejewski [36].
- **Asynchronous delivery:** `target->send(m, this)`. The message is enqueued on `target`’s `BQueue` mailbox (`mutex + condition_variable + boost::circular_buffer`); `target`’s own thread wakes up on the condition variable and runs the handler. Round-trip cost $\sim 3.4\mu\text{s}$ across threads, or $\sim 90\text{ ns}$ same-thread within a `Group`, per the same reference.

Both paths deliver the same handler with the same message. Neither the handler’s body nor any actor downstream of it can tell which path fired. Switching a boundary between the two is a one-line change at the sender’s call site — `target->fast_send(m, this) ↔ target->send(m, this)` — with no code change on the receiver side, no header change, no interface change, no test change.

9.2 End-to-end measurement of the boundary choice

The end-to-end latency profile of an HFT pipeline is a joint property of (i) which stages exist and where they are cut, (ii) which pairs of adjacent stages are joined by `fast_send` versus `send`, and (iii) the arrival law of the input feed. Proposition 1 and Theorem 1 both fold across multiple boundaries: the throughput speedup is N -fold and the tail contraction is $N^{-\gamma}$ with $\gamma \geq 1$, so the joint optimum is not deducible from local reasoning at a single boundary. A designer who reasons one boundary at a time underestimates the aggregate tail benefit of adding hops; this is the reasoning behind the single-thread convention of Section 1.2.

Getting the joint optimum therefore requires end-to-end measurement of the assembled pipeline against a representative feed, and iteration over per-boundary flips. The observable objective is a two-tuple (p_{50}, p_{99}) of end-to-end latency under the target feed, with an operator-chosen trade-off between the two (κ of Section 4.6). The design equation of Section 4.6, $N^* = \arg \min_N [(N-1)h + \kappa \Delta(N)]$, or in its partition form the cut with the smallest reachable $s_{\max}$ and then the fewest hops, supplies a prior on where the optimum lies under a Hawkes feed; the measurement decides.

9.3 The measure-and-flip procedure for setting each boundary

The procedure is: draft the pipeline as a chain of actors at the natural cut points; wire each boundary with whichever of `fast_send` or `send` is convenient during development; once the pipeline runs end-to-end, measure the corpus-median p_{50} and p_{99} on a representative tape as a baseline; flip one boundary at a time, re-measure, and accept the flip if (p_{50}, p_{99}) improves under the operator’s trade-off; iterate until no single flip improves the objective. The full design flow, including the steps that precede this, is given in Section 11.

Under the arrival law of a CME MDP3 feed the design equation of Section 4.6 predicts what the iteration converges to: async `send` at every boundary whose removal would raise the largest stage, and `fast_send` everywhere else, in particular between short sub-tasks below the floor of Section 4.6 and between sub-tasks downstream of the bottleneck whose combined service stays at or below $s_{\max}$ (Remark 2.3). The measurement decides; the prediction indicates where to expect it to converge.

9.4 Thread mapping and send mode as deployment decisions

A consequence of Section 9.3: in KASPAR-HFT two distinct topology decisions are not knowable at design time. The first is the *actor-to-thread mapping* — which actors share a thread by being placed in a common `Group`, and which run on their own thread. The second is the *per-boundary send-mode choice* — whether each pair of adjacent actors is joined by synchronous `fast_send` or asynchronous `send`. Neither decision is deducible from the source code alone. Both are functions of the target feed’s arrival statistics and the operator’s median-versus-tail trade-off, both of which are properties of the deployment environment. The optimum can only be measured from the assembled system running against a representative feed.

Under receiver transparency this is not a limitation. Actor code, message-type headers, and the full suite of unit and integration tests can be written and run before either the actor-to-thread mapping or the send-mode choice has been decided. Code construction and testing therefore proceed independently of the latency-optimisation consideration — both decisions are deferred to the end

of the build, informed by measurements that could not have been taken any earlier because the system did not yet exist to measure. Correctness is established first; latency is optimised against the specific feed the system will run on, second. Two developers building two actors on the two sides of a boundary do not need to agree on whether their actors will ultimately live on the same thread or on different threads, nor on whether their handoff will be synchronous or asynchronous; they need to agree only on the message type. The message-type agreement is checked at compile time; the two topology decisions are parameters set at deployment.

This decoupling of correctness from latency optimisation is the practical consequence of receiver transparency, and distinguishes the design cycle from that of frameworks in which either topology decision enters the receiver’s interface (Section 9.5). Design proceeds without a commitment to a topology; the commitment is made once, near deployment, on the basis of measurements the code did not need to know about while it was being written.

9.5 Frameworks in which the send mode is part of the receiver’s type

Most low-latency actor frameworks bake the sender’s choice of delivery mode into the receiver’s type. An Akka actor’s `tell` versus `ask`, a CAF actor’s `send` versus `request`, and an Erlang process’s `cast` versus `call` each expose different receiver-side surfaces, so flipping a boundary from sync to async is a small refactor rather than a one-line change. Under those frameworks the design flow of Section 9.3 is not tractable, and the operator’s incentive is therefore to lock in the send-mode choice early — reasoning locally — rather than to defer it to an end-of-build measurement pass. This incentive is a plausible contributing cause of the single-thread convention: frameworks that penalise deferral push designers toward the topology that is least sensitive to a wrong early guess, which is the collapsed single-thread servicing chain.

KASPAR-HFT’s joint mailbox API of `fast_send` and `send`, with receiver transparency, is what makes measurement-driven send-mode optimisation practical. The design rule depends on this property operationally.

10 Pre-spinning receivers and pre-warming on self-exciting feeds

The paper’s central claim is that the single-thread hot-path recommendation does not hold above a service-time threshold on Hawkes-clustered feeds (Section 1.2, Section 4). Two adjacent conventions, pre-spinning receivers and pre-warming pipelines, are also affected by the arrival law, and this section examines them. Neither intervention is measured here; both are future work. The argument is that the arrival law provides much of the benefit these interventions are intended to deliver, on the tail quantile.

Pre-spinning receivers. The busy-poll recommendation of Rigtorp [41], Belay et al. [63], and the Aeron/Chronicle lineage (Section 1.2) pins a spinning thread per receiver so the wake path from a sleeping condition variable is never taken. On a Poisson feed, any arrival can find the receiver asleep, and the wake latency is paid on average; pre-spinning eliminates it. On a Hawkes feed with branching ratio $n \approx 0.8$, sleep/wake transitions happen only at cluster boundaries. Every p_{99} message is intra-cluster (Section 4.1: the p_{99} message waits behind a cluster, since at these service times the Poisson null produces no wait at all), so by the time a tail-setting message arrives the receiver is busy: at $8\text{--}16\mu\text{s}$ it is still serving the predecessor the publisher sent one spacing earlier, and from $64\mu\text{s}$ it has been processing a run for many multiples of the per-message service time (Section 4.2). In either case it is effectively spinning as a natural consequence of the load. Under

the model used here, pre-spinning therefore saves little or no wake latency at the tail; this is a prediction of the queueing model, not a measurement of a production receiver’s state, and should be evaluated directly. It saves wake latency for the median and for the minimum-latency floor (inter-cluster arrivals which do find the receiver asleep after seconds-scale gaps), but the saving is bounded by $\sim 1.4 \mu\text{s}$ per hop (the $1.7 \mu\text{s}$ wake-path cost less the few-hundred-nanosecond mutex and cache-line floor of Section 4.10) and is much smaller than the tail win from splitting. There is also an operational cost: in the present author’s production experience, pinning spinning threads across the servicing chain has caused overall system degradation — steady core burn at 100%, cache thrash between the spin loop and the actor’s own working set, and pressure on shared resources like the LLC and memory bandwidth — that offset the median-latency gains; this is anecdotal and not measured here.

Pre-warming. Pre-warming is a subset of the wake-up cost discussed above. The $h = 1.7 \mu\text{s}$ one-way hop cost that this paper calibrates already captures the entire process-wake-up path on the receiver side: the kernel context switch, the reload of the receiver’s instruction cache, the L1/L2 misses on its actor state and message pool, the TLB refill, and the scheduler putting the thread back on a core. To the extent the thread hop is dominated by cluster-wait on Hawkes feeds — because every p_{99} arrival is intra-cluster and the receiver is, under the model, already awake — the cache-warmth components inside that h should follow the same pattern, since they are the same h decomposed. This is a consequence of the model rather than a measured result. Pre-warming interventions that fire synthetic traffic to keep instruction caches, data caches, TLBs, and branch predictors hot are therefore attacking the same h that Section 4 finds is dominated by cluster-wait. What such interventions can meaningfully deliver is confined to the median and the minimum — the inter-cluster first-arrival case where the receiver truly did cool down — and even there the delivery is bounded by the wake-up cost itself at the cost of continuous L1/L2 pollution from the warmup traffic. Under Hawkes the tail is unaffected.

Measurement of both interventions. Pre-spinning and pre-warming are first-order tail-latency techniques when arrivals are uncorrelated. On self-exciting exchange feeds the arrival law itself provides the receiver-awake and cache-warm state on the tail quantile, so both interventions are second-order there and can be counterproductive once their steady costs are included. Both claims are corollaries of the paper’s main result rather than separate contributions, but each deserves a dedicated empirical evaluation on Hawkes-clustered feeds, analogous to the single-thread evaluation this paper carries out. The two open questions are:

- What is the actual ROI of pinned busy-poll receivers on the Hawkes-load $p_{50}, p_{99}, \text{max}$ triple, net of the steady core-burn cost and the cache pressure the spin loop imposes on the actor’s own working set?
- What is the actual ROI of periodic pre-warming traffic on the same triple, net of the L1/L2 pollution it creates and the residual advantage it delivers only on inter-cluster first-arrivals?

Both are direct extensions of the present paper’s methodology (measure under real CME MDP3 tapes; use a matched Poisson null to isolate the arrival-law dependence) and both are candidate follow-ons.

11 Recommendations for HFT system designers

The paper’s results qualify one architectural default (the single-thread hot path as an unconditional choice) and question two adjacent ones (busy-poll receivers, pre-warming). They do not contradict per-message code optimisation. This section separates what the paper leaves unchanged from what it argues against, and states the design flow the design equation implies.

Design principle (measure the crossover on the target feed). Whether a stage should be split is decided by how many packets arrive closer together than its service time, not by the packet rate or the branching ratio as such. That fraction is a property of the transaction stream’s gap distribution and of the gateway’s minimum separation between distinct events. On CME MDP3 it is about 1% at a $7.5\mu\text{s}$ floor and does not vary with rate or with the fitted branching ratio at HFT service times, so a $7\text{--}8\mu\text{s}$ stage is not split. On a feed that places tens of percent of its gaps at the floor at the same fitted branching ratio, the same stage is split (Appendix D). The crossover cannot be read off fitted parameters. Measure the end-to-end latency of the assembled pipeline under the load the system will see, and split where the measured tail excess exceeds the hop cost.

11.1 Recommendations left unchanged

Per-message hot-path optimisation — branch-free decoders, cache-line-aligned actor state, atomic-free single-writer queues, SBE decode in place, memory pools, `fast_send` at $\sim 30\text{ ns}$, kernel-bypass NICs (Solarflare Onload, DPDK, VMA), `isolcpus`, `SCHED_FIFO`, huge pages, RTC counters instead of `clock_gettime`, non-blocking IO on the socket, avoidance of allocation on the hot path, careful use of the branch predictor — remains necessary. All of it is contained in the service floor T , which the paper takes as given. Proposition 1 gives an N -factor throughput speedup at any T , and the $1/N$ bound of Theorem 1 applies to a single-stage wait $\Delta(1)$ that itself scales with T , so the absolute gain from splitting grows with T while the hop cost does not. The splitting rule is therefore complementary to per-message optimisation: a two-stage tandem on a $20\mu\text{s}$ servicing chain has lower latency at every quantile than the same tandem on a $40\mu\text{s}$ chain, and the difference is entirely in the per-message layer; and per-message optimisation that brings a chain below the floor removes the case for splitting it at all (Section 4.8).

11.2 Recommendations the paper argues against

Two conventions in the layer above per-message optimisation. First, the single-thread servicing chain as the unconditional topology default (Section 1.2): on Hawkes-clustered feeds with T above the floor of Section 4.6, the tail is dominated by cluster wait and adding threads along the chain reduces it at a median cost of $(N - 1)h$. Second, pre-spinning receivers and pre-warming pipelines as always-on interventions (Section 10): on Hawkes-clustered feeds the receiver is already busy and warm at the arrival of every p_{99} message, so these interventions affect the median and minimum, not the tail, at a steady CPU and cache cost. The claim that a faster inner loop implies a single thread is preferable conflates two claims; the paper contradicts only the second.

11.3 Recommended design flow

The design flow that follows from the paper’s results, and that we recommend as the canonical HFT infrastructure workflow on self-exciting exchange feeds:

1. **Retain per-message optimisation.** This is orthogonal to the paper’s recommendation and determines absolute latency.
2. **Measure the target feed’s tight-gap floor, and use it as the split/don’t-split test.** Take the corpus of packet arrivals, form the per-window interarrival-gap distribution, and read off a low quantile of it — we use p_1 . This single number is the service time at which a deterministic stage begins to accumulate a tail, and on CME MDP3 channel 318 it is $\approx 7.5 \mu\text{s}$ (Section 4.9). A stage whose service time sits comfortably below it will not develop a queue and must not be split: the split gains nothing and costs h at every quantile. A stage above it should be split, subject to the tail excess exceeding h , which on this corpus happens about $1.5 \mu\text{s}$ further along (Section 4.8). Two cautions. First, use the tight end of the gap distribution, never the mean: on this corpus the mean gap is $2215 \mu\text{s}$, some $300\times$ larger, and a rule stated on it would say never to split anything. Second, measure it under representative and peak-load conditions. On this corpus the floor is the service interval of the exchange’s market-data publisher, which spaces successive packets by at least that much while it drains a backlog (Section 5) — the packets on either side of it are distinct transactions (Section 4.3) — and it did not move between quiet and volatile sessions (Section 4.9), so it was measured once per feed; that invariance is a finding about this feed’s gateway, to be verified on another feed rather than assumed. At longer service times the tail does depend on intensity and branching ratio, and a chain in that regime needs its threshold measured under the load it will actually see.
3. **Draft the pipeline as a chain of actors at natural cut points, then balance it.** On a CME MDP3 handler: `SocketReader`, SBE decoder, order book, strategy actor, order manager — four or five stages, no artificial decomposition (Section 9). The tail factor is $s_{\max}/T$ (Remark 2.3), so the goal of the decomposition is the smallest largest stage, and the largest stage belongs at ingress so that all queueing sits in one mailbox (Remark 2.4).
4. **Divide the stages above the floor by state.** A stage that carries no state from one packet to the next, such as SBE decode, can be served by a dispatch pool of whole-packet servers; with three or more cores for it, a pool is better than cutting it into the same number of stages, at two cores the two are level (Table 9). A stage that carries state, the order book above all, is served by cutting the chain.
5. **Wire boundaries with whichever of `fast_send` or `send` is convenient during development.** Receiver transparency in KASPAR-HFT makes this a one-line-per-boundary decision that can be deferred (Section 9).
6. **Do not commit to an actor-to-thread mapping at design time.** Grouping decisions and per-boundary send-mode decisions are both scalar dials at deployment, not code-structure decisions (Section 9.4).
7. **Measure the assembled pipeline against a representative feed.** Corpus-median p_{50} and p_{99} of end-to-end latency, per operator-chosen trade-off (Section 9.3).
8. **Flip one boundary at a time and re-measure.** Accept the flip if (p_{50}, p_{99}) improves. Iterate until no single-boundary flip helps.
9. **Prior on where step 7 converges.** On Hawkes-clustered feeds, `async send` at every boundary whose removal would raise the largest stage; `fast_send` at every other boundary, in particular between sub-tasks downstream of the bottleneck whose combined service stays at or below

$s_{\max}$, and between short sub-tasks below the floor. A hop that does not lower $s_{\max}$ costs h at every quantile and compresses nothing.

10. **Do not pre-spin or pre-warm by default.** If pre-spinning or pre-warming is used, measure the ROI on a Hawkes-representative tape, net of the steady core-burn and cache-pollution costs. If the measurement improves the tail materially, keep the intervention. The paper’s prior is that on Hawkes feeds it will not (Section 10).

The workflow prioritises measurement over local reasoning because the arrival law of the target feed is a first-order determinant of the optimum and is not a property of the code.

Part IV

Conclusions

12 Conclusion

This paper is about queueing in the software that receives market data: what makes a single-threaded receiver’s latency tail, and how to reduce it. It is not about why the market behaves as it does. Section 12.1 summarises the answer in one place: what causes the receiver’s tail and how splitting the code serially or in parallel reduces it. The remaining subsections give the detail. Section 12.2 states what the paper establishes about the cause of the tail, Section 12.3 what it does not, Section 12.4 how to reduce the tail, and the two remaining subsections revise a prior design context and list the caveats. All empirical results are for one instrument, NQ front-month on CME channel 318.

12.1 Summary: what causes the receiver’s tail and how to reduce it

What causes the tail. The chain starts at the exchange. The matching engine often processes one transaction less than a microsecond after the previous one: on NQ, 17% of consecutive transactions are less than $7.5\mu\text{s}$ apart on the engine’s clock (Table 11). The market-data publisher cannot send that fast. It is a queue that sends about one packet every $7.5\mu\text{s}$, so a burst at the engine leaves it as a train of packets at that spacing, mostly one message each, with an occasional larger packet after a longer gap (Tables 12 and 15). The receiver is the next queue in line. If it handles a packet in less than $7.5\mu\text{s}$, it keeps up with any train the publisher can send and builds no queue from the trains alone. If it needs longer, it falls behind by the difference on every packet of the train, and the messages at the end of the train wait. At service times of $16\text{--}32\mu\text{s}$ the wait is mostly behind the packet or two just ahead; from $64\mu\text{s}$ it is behind runs of transactions spread over a few hundred microseconds, which is the self-excitation a Hawkes fit measures (Table 3, Section 5.7).

A production receiver does not run at $16\mu\text{s}$; it runs at about $7\mu\text{s}$, just under the publisher’s spacing, and there two further mechanisms carry most of its tail. The first is long packets. Decode cost grows linearly with the number of messages, so a packet with two or more messages takes longer than $7.5\mu\text{s}$; its later messages wait for the earlier ones, and the packets arriving behind it queue while it holds the thread. In the simulation at $8\mu\text{s}$ the decode inside packets supplies two thirds of the p_{99} excess, and on the live receiver the floor plus the decode cost of the messages ahead in the same packet reproduces 71–80% of the observed p_{99} on four of six streams (Section 7.6); on ZN, where each extra message costs three times as much as on NQ, a few long packets set the far tail (Section 7.7). The second is service time that varies from packet to packet: on the live ZN receiver, messages with nothing queued ahead of them and first in their packet still take $15.5\mu\text{s}$ at p_{99} against a $7\mu\text{s}$ floor, and any such service in a train can start a queue. The simulation models the first mechanism only through NQ’s per-message cost and the second not at all, so its tails at the production service time are lower than a real receiver’s. The packet rate does not cause the tail at HFT service times: utilisation is below 0.02 there (Section 4.9). Why orders reach the engine within a microsecond of each other is a question about the market and is left open (Section 12.3).

How to reduce it: split serially. Cut the servicing chain into N stages on separate threads, each passing its output to the next through a hop of cost $h \approx 1.7\mu\text{s}$. Every message pays $(N - 1)h$ more at the median, and the queueing excess shrinks by a factor of N or more, because only the slowest stage queues: the tail is set by s_{max} , the service time of the largest stage (Theorem 1). A stage whose service time is below the publisher’s spacing does not queue at all. On NQ, one cut into

two halves lowers p_{99} in every window at every service time from 16 to 128 μs and cuts the tail excess to 3–16% of its single-thread value; the best number of stages is 1 at $T \leq 8 \mu\text{s}$, 2 at 16, 4 at 32 and 8 at 64 and 128 μs , limited by the natural cut points in the chain (Section 12.4). Serial splitting is the only option for a stage that carries state from one message to the next, such as the order book.

How to reduce it: split in parallel. Give whole packets to a pool of N cores, each running the full stage, and put them back in order on the way out. With the hop in, the hop out and the resequencing all charged, the median is $T + 2h$ for any N , and at most 0.14% of packets wait for resequencing (Table 9). With two cores the pool is level with a two-stage serial cut; with three or more it is better than a serial cut into the same number of stages. It works only for a stage without state, such as SBE decode. A design that decodes in a pool and feeds one order-book thread combines the two and has not been costed here (Section 13).

Design rule (NQ, CME channel 318). Measure the service time T of the servicing chain per packet under real market load, including multi-message packets and bursts, and compare its upper quantiles, not its median or p_{10} , with the publisher’s spacing (about 7.5 μs).

- Upper quantiles below the spacing: keep one thread. Splitting only adds hop cost.
- Median below the spacing but upper quantiles above it, the regime of a production receiver: the tail comes from long packets and slow services. Reduce the per-message decode cost and the spread of service times; a stage cut does not shorten the decode of a long packet.
- Typical service above the spacing, stage carries state: split serially, at the natural boundary that minimises the largest stage, with the number of stages set by T .
- Typical service above the spacing, stage carries no state, three or more cores available: dispatch whole packets to a pool and resequence.

On another feed, measure its publisher’s spacing first; the threshold is that spacing.

12.2 What causes the tail: what the paper establishes

The questions were what produces the queueing tail behind a single-threaded receiver — the packet rate, the clustering of arrivals, or the number of messages per packet — and are two queues better than one. The first is answered by controlled simulation on the real arrival times: each counterfactual stream of Section 3.6 removes one property of the stream and keeps the rest, so the change in the tail measures that property’s contribution. The following hold on NQ.

1. *The tail is produced by the timing of matching-engine transactions and, at production service times, by the number of messages per packet.* Redrawing transaction times uniformly removes 75–98% of the tail (Table 7); the message count is claim 5.
2. *Not by the rate.* Utilisation is below 0.02 at every service time an HFT hot path occupies, and a window with four to five times the corpus-median packet rate has the same tail relative to T (Section 4.9); rate matters only from 64 μs , where ordinary utilisation queueing begins. A stream that keeps every one-second packet count and randomises within the second has no tail at 8–16 μs (Table 3).

3. *Not by the exchange's packetisation.* 98.3% of transactions occupy one packet; the packets of a split transaction arrive $15\mu\text{s}$ apart, above the floor; 99.9% of the gaps below the floor separate different transactions; merging each transaction into one packet or spreading its packets leaves the tail within $\pm 15\%$ (Section 4.3).
4. *Not by how sizes are ordered in time.* Decoupling transaction sizes from their timing leaves the tail within $\pm 15\%$ from $16\mu\text{s}$ up (Table 7), and permuting packet spans across packets changes it by at most 2% (Section 7.6). This says the order in which long packets arrive does not matter; it does not say their length does not matter, which is claim 5.
5. *At the service time production receivers run at, long packets carry most of the tail.* Decode time grows linearly with the number of messages, 0.3 to $1\mu\text{s}$ per extra message depending on the stream, so a long packet delays its own later messages and holds the thread while the packets behind it queue. At $8\mu\text{s}$ this supplies two thirds of the simulated p_{99} excess and, on the live receiver, the floor plus the decode cost of the messages ahead in the same packet reproduces 71–80% of the observed p_{99} on four of six streams; it sets the live ZN far tail (Sections 7.6 and 7.7). From $16\mu\text{s}$ up it adds a few percent. 77% of multi-message packets hold more than one transaction packed together by the publisher (Section 7.5).
6. *The timing acts through two channels whose weight depends on the service time.* Consecutive transactions sent back to back at the publisher's spacing, most of which the engine processed less than $7.5\mu\text{s}$ apart (Table 12): a stream with the real gaps in random order keeps 79% of the tail at $16\mu\text{s}$ and 59% at 32. Runs of transactions: the same permutation keeps only 37% at 64 and 23% at $128\mu\text{s}$ (Table 3). The runs are the self-exciting component of the transaction stream, which a Hawkes fit measures (branching ratio 0.795 on transaction starts, 0.798 on packets, count variance 105 times Poisson at one second; Table 6).
7. *The floor, and the pile-up of gaps just above it, are the exchange publisher's output.* On the matching engine's clock 17.1% of consecutive transactions are under $7.5\mu\text{s}$ apart, against 1.04% on the publisher's clock (Table 11); the delay from engine to packet grows about tenfold with engine burst intensity and coalescing grows with it (Table 13). The publisher is a queue with a service time of about $7.5\mu\text{s}$ per packet, and the receiver is the second stage of a tandem that begins at the exchange (Section 5.6).
8. *The results do not depend on the transaction grouping.* Regrouping by exact `transactTime`, checked against the end-of-event flag, leaves them unchanged at the reported precision (Section 5.2), and the survival shares carry session-bootstrap intervals of a few points (Tables 3 and 7).
9. *The arrival statistics are not an artefact of replay.* A live production receiver seven months later, on three instruments and a different clock, gives the same clustering statistics, and its decoder's service floor coincides with the service time at which the simulated tail appears (Section 7).

12.3 What causes the tail: what the paper does not establish

Why the matching engine bursts. The paper traces the receiver's tight gaps to the exchange's publisher draining a backlog, and the backlog to the matching engine processing events a fraction of a microsecond apart (Section 5). It does not show why orders reach the engine within a fraction of a microsecond of each other. They do not reach the receiver together: when the engine processes a

transaction less than $1\mu\text{s}$ after the previous one, the publisher sends the two in separate packets 84% of the time, most 7.5 to $16\mu\text{s}$ apart (Table 12). The open question is about the engine’s input. Three mechanisms remain open. (a) Reaction: a transaction at time t causes other participants to cancel, replace or trade within microseconds, each as a separate transaction. (b) Common reaction: many participants respond to the same public signal at once, their orders reach the matching engine together, and the engine processes them one after another. (c) Pacing or batching inside the exchange’s order entry, with no market information in the spacing. Separating them needs the time each order *entered* the exchange, not only when it was processed and published: (a) predicts that the second order entered after the first transaction was published, (b) that both entered before either was published. The public MDP3 feed carries processing and publication times only, and the paper’s own order-entry sessions record only its own orders.

The nearest published evidence is Aquilina et al. [33], who had message-level data from the London Stock Exchange with participants’ order-entry times and found that *races* — several participants sending messages at the same price level in response to the same public signal, within microseconds of each other — are frequent and account for a material share of trading volume. That is mechanism (b). The engine-clock gaps of $0.18\mu\text{s}$ are what (b) would produce, and it is likely that the same mechanism is at play on CME; the paper has no data that test it, and states it as a hypothesis.

Whether the publisher’s capacity is designed. The publisher behaves as a queue with a capacity of about one packet per $7.5\mu\text{s}$; whether that capacity is a design choice or an engineering limit, and whether it acts as a speed limit on the information side of the market, is stated as a conjecture in Section 5.8, with its qualifications.

Self-excitation at HFT service times. It is not, and this is now shown rather than argued. The fitted self-excitation acts with a decay time of about $160\mu\text{s}$ and places about 4% of its triggering within $7.5\mu\text{s}$ (Section 5.7); it is the cause of the runs, which carry the tail from $64\mu\text{s}$. At 16 – $32\mu\text{s}$ the carrier is the publisher draining, at its $7.5\mu\text{s}$ spacing, backlogs built by orders reaching the engine within a microsecond of each other, whose origin is the open question above.

Events below the floor. Two transactions closer together than the publisher’s service interval reach the receiver in one packet, so every gap statistic on the publisher’s clock is censored there. On the engine’s clock they are visible (Table 11), and on the publisher’s clock neither the message count per packet nor the share of packets carrying several transactions rises with the fitted branching ratio (Section 4.9), so no faster self-excitation scaling with the branching ratio is hidden in the packets. Faster self-excitation of constant strength would not be seen by that test.

Why a receiver’s service time varies. On the live ZN receiver, messages that met an empty queue and were first in their packet, so that neither queueing nor in-packet decode contributes, still take $15.5\mu\text{s}$ at p_{99} and $26.4\mu\text{s}$ at $p_{99.9}$ against a $7\mu\text{s}$ floor (Table 25). The records do not show why. Because packets arrive $7.5\mu\text{s}$ apart, each such service can start a queue, so this spread is a source of tail in its own right, and the simulation, whose service is constant or linear in message count, does not contain it.

How the live far tail divides. The ZN $p_{99.9}$ of $274\mu\text{s}$ is argued to be long packets and the queues behind them, from the measured floor and slope (Section 7.7). How it divides between the decode inside long packets, the queue behind them, and slow services was not measured, because the live per-message records were not retained.

Instruments other than NQ. One instrument was measured. On ZN the live data show the message count per packet rising with queue depth, where on NQ it falls, and whether the two channels compound there has not been tested; no ZN corpus exists. The paper also treats NQ front-month as a stream on its own, while the publisher serves the whole channel. On NQ the other instruments are thin; on ZN the Treasury futures along the yield curve are likely to excite one another, and where they share a channel the publisher would carry those reactions in the same packets. Cross-excitation between instruments is a candidate cause of ZN’s large packets that the paper does not test (Section 13.1).

12.4 How to reduce the tail

The second question was whether a servicing chain cut into two or more stages on separate threads is better than one thread. Above a service-time threshold, yes; below it, no; and the threshold is a property of the feed, measured here on NQ. On 3512 windows at $h = 1.7 \mu\text{s}$ the corpus-median single-thread p_{99} is 2.2, 4.6, 12.8 and 41.5 times the service floor at $T = 16, 32, 64, 128 \mu\text{s}$, a tail excess of 11 to 2900 times the hop cost, so above the floor the hop is small against what it lets the pipeline compress; below the floor the hop is cost only. Splitting a task into two half-length actors on separate threads joined by one hop lowers p_{99} on 100% of the windows (nested in 276 sessions) at every service floor from 16 to $128 \mu\text{s}$, cutting the corpus-median tail excess to between 0.03 and 0.16 of its single-thread value for a median cost of exactly h ; the trade favours the split for any tail weight $\kappa \gtrsim 0.095$ at $T = 16 \mu\text{s}$ and $\kappa \gtrsim 0.0004$ at $T = 128 \mu\text{s}$, and at $T \leq 8 \mu\text{s}$ the split loses on p_{99} . An N -stage tandem contracts the excess by a factor of N or better on every sample path (Theorem 1), in practice by far more where a split drives the per-stage service under the floor, while paying $(N - 1)h$ at every quantile; only its largest stage matters. The p_{99} -minimising stage count rises from 1 at $T \leq 8 \mu\text{s}$ to 2 at 16, 4 at 32 and 8 at 64 and $128 \mu\text{s}$, up to the number of natural cut points in the decode chain. These results do not depend on what places transactions close together: the design rule needs the gap distribution the receiver sees, not its cause.

The single-thread rule is correct whenever the service time of the chain is below the feed’s threshold, and on this corpus that includes the reference system’s socket-to-book path, $6.8\text{--}7.2 \mu\text{s}$, for which $N^* = 1$ (Section 8). The threshold is the service time of the exchange’s publisher upstream of the receiver, which is the tight end of the gap distribution the receiver sees, and is invariant to packet rate and fitted branching ratio at HFT service times; at longer service times both raise the tail (Section 4.9). Appendix D constructs a feed on which both dependences appear at $T = 8 \mu\text{s}$ and on which splitting pays where on CME it does not. On another feed the threshold, and its dependence on load and on clustering, must be measured.

Spending the same N cores as a dispatch pool that sends whole packets to N servers, with the ingress hop, the egress hop and in-order resequencing all charged, gives a median of $T + 2h$ at every N and resequences at most 0.14% of packets (Table 9). For a stage that carries no state, such as SBE decode, a pool of two cores is level with a two-stage cut, one hop dearer on the median and within about a microsecond at p_{99} at $16\text{--}32 \mu\text{s}$; a pool of three or more cores is better than a cut into the same number of stages on both quantiles. The order-book stage carries state from one update to the next and cannot be dispatched; for it, cutting the chain is the way to spend cores. The design rule is therefore: cut stateful chains above the threshold; dispatch stateless stages above it when three or more cores are available.

The recommendation is that HFT designers measure end-to-end latency of the assembled pipeline under actual market conditions, including peak load, and split the servicing chain wherever the measurement shows a tail excess larger than the hop cost. Splitting the hot path is a design option of equal standing with optimising it. For a chain above the threshold, one cut at a natural boundary

reduces the p_{99} tail by more than any per-message optimisation is likely to: on this corpus a balanced cut of a $16\mu\text{s}$ chain reduces the p_{99} excess plus hop cost from 18.6 to $2.3\mu\text{s}$ (Section 8), whereas per-message work moves the floor, and the per-message slope, by tenths of a microsecond. Below the threshold, for the typical packet, per-message optimisation is the lever; the far tail that a few many-message packets open on a live channel (Section 7.7) is a separate regime, whose remedy the end-of-build measurement decides. Framework designers should measure busy-poll dequeue on pinned isolated cores against the wait/notify mailbox on a representative tape rather than assume it is faster (Section 10), drive the mailbox mutex uncontended, keep memory-pool discipline, and cut the decode pipeline at whichever of its natural boundaries (4–5 on CME) minimises the largest stage, with that stage at ingress and everything downstream merged up to its service time; the tail factor is s_{max}/T , and a hop that does not lower s_{max} is pure cost. Strategy designers of both percentage-of-volume (POV) and race-sensitive algorithms benefit for the same reason.

12.5 Revision of a prior design context

An earlier paper by the present author [36] did not itself argue for the single-thread servicing chain; it took the collapsed single-thread servicing chain as the widely accepted design context of the time and offered **fast_send** as the mechanism that makes actor composition inside it free of context-switch overhead. What the present paper retracts is the design context, not the mechanism: on Hawkes-clustered feeds the single-thread servicing chain should no longer be the unconditional default. The corrected position is that **fast_send** (synchronous, inline, sub-microsecond) and asynchronous **BQueue** handoffs (a real pipelined hop at a few microseconds) are *complementary* tools, and the choice between them is a per-boundary design decision that depends on the service time being chained and the arrival law of the feed the pipeline sits on.

In shipping KASPAR-HFT the **fast_send** versus asynchronous-send choice at each actor boundary is normally left open until the end of the build, and is resolved by measuring the end-to-end median and tail latency of the assembled pipeline under the target feed. A convenient property of the KASPAR-HFT actor API is that the receiving actor’s **MESSAGE_HANDLER** is agnostic to how it was invoked: swapping a synchronous **fast_send** at a call site for an asynchronous **send** (or vice versa) is a one-line change on the sender side and requires no code change on the receiver side. The choice therefore lives entirely at each individual call site and can be flipped per boundary at end-of-build with minimal engineering cost. The paper’s design equation now supplies the arrival-law-aware default value of that end-of-build measurement: on Hawkes-clustered feeds, tasks above the threshold should be split into shorter actors joined by asynchronous sends, on the measured result that splitting reduces the tail by more than the hop cost raises the median. The rule presupposes that the task being split has a service time T comfortably above the floor of Section 4.6; a task of order h or below cannot be usefully split (a $1\mu\text{s}$ task cannot be turned into two 500 ns actors when a single hop alone costs $1.7\mu\text{s}$). Short synchronous chains inside one actor’s servicing of one message therefore continue to use **fast_send** unchanged. The end-of-build measurement decides any specific boundary.

12.6 Caveats

The corpus is a single instrument (NQ front-month, channel 318); a ZN recording is the first replication needed, for the reason given in Section 12.3, ahead of ES and BTC. Transactions are grouped by exact **transactTime**, checked against the end-of-event flag (Section 5.2). The live measurements of Sections 7 and 7.7 come from one host on one afternoon, a 53-minute capture and one 900s window, without replication, and the 3512 windows are nested in 276 sessions, so uncertainty on the aggregates is session-clustered; the headline survival shares carry session-bootstrap

intervals. Constant per-packet service is a simplification, and at the production service time it is the one that matters most. Section 7.6 removes it for message count, at NQ’s per-message cost, a third of ZN’s, and finds the stage count unchanged; service that varies for other reasons, which the live receiver shows at twice the floor at p_{99} with no queue, is not modelled, so the simulated tails at about $7\mu\text{s}$ understate a real receiver’s. The cache-pollution cost of spinning receivers on many cores is bounded but not measured. And Theorem 1 is a bound whose slack depends on the arrival law; the paper measures that slack ($\gamma = 3.3$ and 2.7 where it is defined, against the guaranteed $\gamma \geq 1$) rather than deriving it, and a closed form for the slack on Hawkes input is open.

None of these caveats reverses the two answers on this corpus. The survival shares are ratios of corpus medians with session-bootstrap intervals.

13 Future work

The section first states the open questions the paper raises, then lists the remaining work in two groups: questions about the market, which are outside the paper’s scope, and questions about the receiver, which are within it but were not answered with the data at hand.

13.1 Open questions this paper raises

The paper answers its two questions on NQ front-month. Measuring it raised others that need data or scope beyond this paper, and each is better answered in a shorter, focused study.

1. **Does filtering the channel to NQ front-month change any result?** Channel 318 carries other instruments, and the publisher packs their messages into the same packets as NQ. Every packet statistic in the paper counts NQ front-month messages only: span, transactions per packet, and the packet stream fed to the simulated receiver. Two things could change on the full channel. First, packets are larger in bytes and messages than the NQ span shows, which bears on the span-dependent service of Section 7.6 and the tables of Section 5.5. Second, a receiver subscribed to the channel sees every packet, including those without NQ, so its arrival stream is denser than the one simulated unless those packets are discarded at negligible cost. The raw captures hold the full channel; the measurement is packet bytes and message counts on every packet, joined to the NQ tapes on the publisher’s send time, and the sweep rerun with every channel packet as an arrival. The pacing floor is not in question: Table 15 shows it on the channel’s order and trade packets.
2. **Do instruments excite each other, above all on ZN?** The Treasury futures along the yield curve (two-, five-, ten- and thirty-year and the ultra bond) are priced off one another, so a move in one is likely to trigger quotes and trades in the others within microseconds. Where they share ZN’s channel, those reactions reach the publisher together and leave in the same packets, which would make ZN’s large packets a product of the curve and not of ZN alone. The test is a multivariate Hawkes fit across the channel’s instruments on the engine clock and packet size conditioned on cross-triggering. It needs a full-channel ZN recording.
3. **Why do orders reach the matching engine within a microsecond of each other?** The receiver sees them $7.5\mu\text{s}$ apart, spaced by the publisher; the question is about the engine’s input. Reaction, common reaction to one signal (races), or pacing inside order entry (Section 12.3). It needs order-entry times for all participants.

4. **How much of the engine’s activity is racing?** Table 17 shows the race signature in tight gaps that follow a trade, but rejected race orders do not reach the public feed.
5. **Is the publisher’s capacity a design choice, and does it act as a speed limit on information?** The conjecture of Section 5.8.
6. **Where does the spread in a receiver’s service time come from?** On the live ZN receiver the hot path reaches twice the floor at p_{99} with nothing queued and nothing ahead in the packet. The measurement is the full distribution of per-packet service for single-message packets, with the candidate causes (cache misses, interrupts, rare message templates, book state) recorded alongside.
7. **What does a random service time do to the tail and the design rule?** The sweep rerun with service drawn from the measured live distribution instead of a constant, and the span-aware sweep rerun at ZN’s per-message cost, three times NQ’s.
8. **How does the live far tail divide?** Retaining the per-message live records would allow the ZN $p_{99.9}$ to be split into in-packet decode, the queue behind long packets, and slow services.
9. **Can decoding one packet’s messages in parallel shorten the long-packet tail?** A stage cut does not shorten it. An unmerged experiment in the KASPAR-HFT repository that dispatched every message to parallel decoders was slower than serial at every book percentile; a later one that sent only multi-message packets to parallel decoders lost at the median but improved $p_{99.9}$ on five of six streams, on a build with recovery disabled and without validation of its output. Whether it pays in a production build is open.
10. **Does the result replicate on ZN, ES and BTC?** The design rule is measured on NQ; the threshold equals the upstream publisher’s service time only if each channel is paced the same way.

The lists below give the remaining work in two groups.

Why orders reach the matching engine within microseconds of each other (questions about the market).

- What places two distinct transactions within a few microseconds of each other: reaction to the previous transaction, common reaction of many participants to the same signal, or pacing by the gateway or matching engine (Section 12.3). Separating them needs order-entry times for all participants, the data Aquilina et al. [33] had for the London Stock Exchange. Races of the kind they found are the leading hypothesis. Their data were exchange message data, which, unlike the order-book data of a public feed, include the attempts to trade or cancel that failed, so both the winner and the losers of a race can be seen. On FTSE 100 stocks they found that latency-arbitrage races, in which several firms react to the same public signal and try to take or cancel the same stale quote, are frequent (about one per minute per symbol), extremely fast (the modal race lasts five to ten microseconds) and account for about a fifth of trading volume; that the top six firms account for over 80% of race wins and losses; that races make up roughly a third of the cost of liquidity as measured by price impact and the effective spread; and that a market design removing them would lower that cost by about 17%. The timescale of their races is the timescale of the engine-clock gaps of Table 11, and the failed attempts that reveal a race are what the CME public feed does not carry (Section 5.7).

- Which instruments' transactions sit on either side of a tight gap on the engine clock, across the whole channel. Table 17 gives message type, side and price level for NQ front-month; which other instruments' transactions sit within microseconds of an NQ transaction is not measured.
- Cross-excitation between instruments, above all on ZN (Section 13.1, question 2).
- A power-law Hawkes kernel against the exponential, fitted on the engine clock.
- The publisher queue measured channel-wide rather than on NQ front-month only, and on ES and ZN, so that its service interval and backlog are measured under the full load it serves.

The queue in the receiver.

- A ZN recording (pcap and per-message records) and the full sweep on it, with span-dependent service. On ZN the live data suggest that large packets arrive during queues, so the queueing and serialisation channels may compound, which would add a span term to the design rule. This is the first replication; ES and BTC follow.
- Dispatch under state: a design that dispatches the stateless decode stage and feeds a single order-book stage, costed end to end.
- Closed form for the slack in Theorem 1 under Hawkes input, i.e. the exponent $\gamma - 1$ as a function of $(\rho, n, \beta T)$. By the reduction this is a single-server question about $M^{[K]}/D/1$ with Borel batches at two service levels, which is tractable; the paper measures γ instead.
- Load-conditioned hop cost $h(\rho)$. This paper calibrates $h = 1.7 \mu\text{s}$ from the low-load wake-latency figure in Maciejewski [36]; under sustained-load conditions inside a cluster the receiver never sleeps and the effective h collapses toward the mutex + cache-line-transfer floor of a few hundred nanoseconds (Section 4.10). Measuring $h(\rho)$ directly and rerunning the sweep against it would sharpen the reported tail reduction, which the single- h simulation understates.
- Cache-pollution cost per spinning-receiver stage count, measured.
- Overnight-session regime.
- Own-capture pcap validation of the network sub-stage.

14 Reproducibility

All results in Part II are reproducible from the KASPAR-HFT repository [37], <https://github.com/vincent212/kaspar-hft>, directory `arrival_paper/`. `qsim_run.py` holds the packet extraction, the per-window Hawkes fit, the (T, h, N) tandem Lindley sweep with the P , G and B nulls, the M/D/N dispatch arm of Section 4.4 and the per-window bound check. `qsim_tx.py`, `transaction_stats.py` and `make_tx_figs.py` hold the transaction grouping, the transaction-level arms and the tables and figures of Section 4.3. `qsim_marked.py` and `make_spans.py` (with `-trades` for the trade streams) hold the span-aware rerun of Section 7.6. The synthetic feed of Appendix D and the scripts that regenerate every table and figure from the cache are alongside. Every null and arm is seeded per window from an MD5 of `(session, window_id, salt)` with a fixed salt per arm, so the cache is byte-reproducible.

The live measurement of Section 7 uses the `LatencyProbe` actor in the same repository and the readers under `kaspr/perf/`; its latency results are published in Maciejewski [36].

The 281 session tapes are derived from licensed CME MDP3 channel 318 pcaps (NQ front month, 2025-01 through 2026-02) and are not redistributed; the extraction script and the cached per-window metadata are released, and the per-window arrival-time caches, being derived from the pcaps, are subject to the same restriction as the pcaps.

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Appendices

A The Borel batch-size tail at fixed branching ratio

This appendix records the calculation behind Equation (8) and the numerical check that the critical $k^{-1/2}$ survival tail of the Borel distribution is confined to a finite window at every $n < 1$.

Stirling's formula $k! = \sqrt{2\pi k} k^k e^{-k} (1 + O(1/k))$ in Equation (7) gives

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!} = \frac{n^{k-1} k^{k-1} e^{-nk}}{\sqrt{2\pi k} k^k e^{-k}} (1 + O(1/k)) = \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k (1 + O(1/k)).$$

Writing $n = 1 - \epsilon$, $\log(ne^{1-n}) = \log(1 - \epsilon) + \epsilon = -\epsilon^2/2 - \epsilon^3/3 - \dots$, so the geometric factor is $\exp(-k\epsilon^2/2 (1 + O(\epsilon)))$. The pmf is therefore $\propto k^{-3/2}$ for $k \ll k^* = \epsilon^{-2}$ and geometric beyond; summing, $\Pr(K > k) \asymp k^{-1/2}$ on the same window and $\asymp k^{-3/2} e^{-k\epsilon^2/2}$ beyond it. At $n = 1$ exactly, $k^* = \infty$, $\Pr(K > k) \sim \sqrt{2/\pi} k^{-1/2}$ (Borel–Tanner), and $E[K] = \infty$; that is the unstable case $\rho = \infty$, which no stationary queue attains.

Because the tail is geometric, K has finite exponential moments $E[e^{\theta K}] < \infty$ for $\theta < \epsilon^2/2 (1 + O(\epsilon))$ and is therefore outside the subexponential class $\mathcal{S}$. Heavy-tail queueing results that assume subexponential batch work, including the single-big-batch principle for $M^{[K]}/G/1$ and the monotone-separable tail theorem of Baccelli et al. [14], do not apply, and the stationary waiting time of the batch-arrival queue has a Cramér–Lundberg (exponential) tail at every fixed $n < 1$.

Table 27 reports the local log-log slope of the exact pmf, $\Delta \log \Pr(K = k) / \Delta \log k$ between successive k , which a pure $k^{-3/2}$ law would hold at -1.5 .

n	k^*	$10 \rightarrow 30$	$30 \rightarrow 100$	$100 \rightarrow 300$	$300 \rightarrow 1000$	$1000 \rightarrow 3000$
0.80	25	−1.92	−2.84	−5.71	−14.96	−43.63
0.90	100	−1.59	−1.81	−2.48	−4.62	−11.26
0.95	400	−1.52	−1.57	−1.73	−2.25	−3.85
0.99	10000	−1.50	−1.50	−1.51	−1.53	−1.59

Table 27. Local log-log slope of the Borel pmf, Equation (7), between successive cluster sizes k . The critical slope -1.5 holds while $k \ll k^* = (1 - n)^{-2}$ and steepens geometrically beyond.

The practical reading is that at the branching ratios fitted on the corpus ($n_{\text{pkt}} \approx 0.8$, Section 3.2), $k^* \approx 25$: clusters of order a hundred packets are the largest that occur with non-negligible frequency, and the single-stage wait behind the largest clusters has scale $Tk^* \approx 25T$, against the corpus-median p_{99} tail excess of Section 4, which is zero below the tight-gap floor and rises from $1.2T$ at $T = 16 \mu\text{s}$ to $39T$ at $T = 128 \mu\text{s}$. This is the object that Theorem 1 divides by N , and the reason the $n = 0.80$ row of Table 27 is the relevant one.

B Numerical check of Theorem 1

A direct simulation of the model of Theorem 1 checks both the identity Equation (4) and the bound Equation (6). Immigrants arrive as a Poisson process at rate $\mu = 0.004 \mu\text{s}^{-1}$, each spawning a Galton–Watson cluster with Poisson(0.9) offspring, so $K \sim \text{Borel}(0.9)$ and $E[K] \approx 10$; clusters are placed as instantaneous batches, giving $\rho = \bar{\lambda}T \approx 0.40$ at $T = 10 \mu\text{s}$. The tandem has $N = 4$ equal stages of $2.5 \mu\text{s}$ and $h = 1.7 \mu\text{s}$; 2×10^5 messages are simulated with an explicit per-stage FIFO recursion

(independent of the paper’s sweep code). Over all messages, $\max_j |W_j^{(4)} - (w_j(T/4) + 3h)| < 10^{-9} \mu s$ and Equation (6) is violated by no message. The quantiles are

q	$W_q^{(1)}$	$W_q^{(4)}$	$W_q^{(1)}/4 + 3h$	$(W_q^{(4)} - 3h)/W_q^{(1)}$
0.5	350.0	55.1	92.6	0.14
0.9	2066.6	355.1	521.8	0.17
0.99	5342.1	1127.6	1340.6	0.21
0.999	7586.2	1585.1	1901.7	0.21

in microseconds. The tandem lands below the bound at every quantile, with the ratio in the last column between 0.14 and 0.21 against the bound’s 0.25; the slack is the ρ/N utilisation effect described in the remark after Theorem 1. Under the Poisson null at the same message rate ($\rho = 0.40$, which is far above the corpus utilisation and is used here only to make the Poisson wait visible) the single-stage wait quantiles at $q = 0.9, 0.99, 0.999$ are $10.3, 24.8, 38.5 \mu s$ and the tandem’s are $5.1, 7.5, 9.2 \mu s$, again inside the bound. At the corpus utilisation ($\rho = 0.0045$ at this appendix’s $T = 10 \mu s$) the same two collapse to 0 and $3h$ at $q = 0.9$ and $q = 0.99$, which is Corollary 2; at $q = 0.999$ they do not, since ρ there exceeds $1 - q = 0.001$ and the corollary does not apply.

C Numerical check of the unequal-stage claims

The same generator as Appendix B (Borel(0.9) clusters, $\rho = 0.40$, $T = 10 \mu s$, $h = 1.7 \mu s$), with each cluster’s members spread uniformly over $3 \mu s$ instead of placed as an instantaneous batch and 4×10^5 messages, run through the partitions of Remark 2.3 with an explicit per-stage FIFO recursion. The last two columns are the tail-excess ratio $(\Delta_{99}(P) - \text{hops} \cdot h)/\Delta_{99}(\text{single})$ and the bound $s_{\max}/T$ from Theorem 1.

partition	$s_{\max}$	hops	p_{50}	p_{99}	$p_{99.9}$	ratio	bound
10	10.0	0	388.2	5719.0	8249.3	1.00	1.00
7 + 3	7.0	1	213.6	3429.5	5297.1	0.60	0.70
3 + 7	7.0	1	213.6	3429.5	5297.1	0.60	0.70
5 + 5	5.0	1	134.9	2265.5	3574.2	0.39	0.50
7 + 2 + 1	7.0	2	215.3	3431.2	5298.8	0.60	0.70
4 + 3 + 3	4.0	2	104.6	1746.5	2714.6	0.30	0.40
4 + 3 + 2 + 1	4.0	3	106.3	1748.2	2716.3	0.30	0.40
2.5×4	2.5	3	66.4	1049.8	1562.2	0.18	0.25

Latencies are end-to-end in microseconds. The three claims of Remark 2.3 read off directly: the tail excess depends on the partition only through $s_{\max}$ and sits below the bound at every row; $7 + 3$ and $3 + 7$ are identical to the reported precision; and $7 + 2 + 1$ differs from $7 + 3$, and $4 + 3 + 2 + 1$ from $4 + 3 + 3$, by exactly one hop cost $h = 1.7 \mu s$ at every quantile. The $3 \mu s$ intra-cluster spread, which is above $s_{\max}$ for the last row, is why the equal-stage ratio here (0.18) sits slightly below the batch-input value of Appendix B (0.21): a cluster that arrives spread out lets the faster tandem drain part of it before the next member lands.

D A synthetic feed on which the tail at $8\mu\text{s}$ depends on intensity and branching ratio

Section 4.9 finds the normalised tail on CME MDP3 invariant to packet rate and branching ratio at HFT service times. This appendix constructs, with the same simulator, a feed on which both dependences appear at $T = 8\mu\text{s}$, and identifies what distinguishes it from CME.

Arrivals are generated as a Hawkes process through its cluster representation: immigrants arrive as a Poisson process of rate $\mu = \bar{\lambda}(1 - n)$, and each spawns a Galton–Watson tree with $\text{Poisson}(n)$ offspring at exponential delays of mean $1/\beta$. Three conditions are run. In the first the kernel is fast, $1/\beta = 1\mu\text{s}$, and there is no emission floor. In the second the same fast kernel is followed by a $7.5\mu\text{s}$ emission floor: each arrival is delayed until at least $7.5\mu\text{s}$ after its predecessor, modelling a gateway that serialises packet emission at the spacing measured on CME. In the third the kernel is slow, $1/\beta = 80\mu\text{s}$, with the same floor; this is the condition closest to the CME gap structure. The grid is $n \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$ and $\bar{\lambda} \in \{500, 2000, 8000, 32000, 64000\}$ packets per second, so that $\rho = \bar{\lambda}T$ at $T = 8\mu\text{s}$ runs from 0.004 to 0.51; 2×10^5 events per cell, three seeds, median reported. The last column of Table 28 is the fraction of interarrival gaps at or below the floor, measured at $\bar{\lambda} = 8000$; on the CME corpus that fraction is about 1% (Section 4.9).

	p_{99}/T at $T = 8\,\mu\text{s}$, $N = 1$					$\text{gaps} \leq 7.5\,\mu\text{s}$
n	$\bar{\lambda}=500$	2000	8000	32000	64000	at $\bar{\lambda}=8000$
<i>fast kernel, no floor</i>						
0.50	12.2	12.5	13.2	15.5	22.6	52%
0.70	38.9	40.7	41.2	47.3	66.6	71%
0.80	91.4	100.0	97.1	105.0	140.7	80%
0.90	421.3	429.4	411.3	413.7	620.4	90%
0.95	1544.2	1482.6	1721.3	1802.4	2487.2	95%
<i>fast kernel, $7.5\,\mu\text{s}$ floor</i>						
0.50	1.75	1.75	1.88	2.44	4.38	53%
0.70	3.50	3.62	3.88	5.44	11.87	72%
0.80	6.81	7.38	7.56	10.88	23.12	81%
0.90	28.06	28.50	30.44	41.25	100.19	91%
0.95	98.06	94.75	116.36	193.96	382.18	95%
<i>slow kernel, $7.5\,\mu\text{s}$ floor</i>						
0.50	1.12	1.12	1.19	1.51	3.56	13%
0.70	1.25	1.31	1.50	3.18	10.44	20%
0.80	2.38	2.71	3.12	8.12	19.56	28%
0.90	23.50	22.57	26.25	37.25	95.75	47%
0.95	90.12	86.19	112.19	188.78	364.30	67%
<i>Poisson, matched rate</i>						
—	1.00	1.39	1.87	2.69	4.45	—
ρ at $T = 8$		0.016	0.064	0.256	0.512	

Table 28. Synthetic Hawkes feeds at $T = 8\mu\text{s}$. The ρ row gives $\bar{\lambda}T$ for the last four columns; the first column is $\rho = 0.004$.

Three observations.

Intensity. In every Hawkes condition the tail is flat in $\bar{\lambda}$ at $\rho \leq 0.02$ (the Poisson row rises from 1.00 to 1.39 over the same range) and rises from $\rho \approx 0.06$, steeply by $\rho = 0.5$. The CME corpus at $T = 8 \mu\text{s}$ spans $\rho = 0.002$ to 0.008 between its quietest and busiest quintiles, which is inside the flat region of every synthetic condition. The intensity invariance of Section 4.9 is therefore what a Hawkes model predicts at that utilisation, and it ends where utilisation queueing begins, as the CME sweep also finds at $T \geq 64 \mu\text{s}$.

Criticality. The synthetic feeds depend strongly on n in every condition, by a factor of 55 to 130 between $n = 0.5$ and $n = 0.95$, and by a factor of 1.9 to 2.6 between $n = 0.7$ and $n = 0.8$ alone. The CME corpus shows no dependence over its observed range $n = 0.77$ –0.83. The last column explains the difference. A Hawkes process at $n = 0.8$ places 28% to 81% of its interarrival gaps at the floor, depending on the kernel; the CME corpus places 1% there. On the synthetic feeds the branching ratio controls how many arrivals are packed at the minimum spacing, and that packing is what queues on an $8 \mu\text{s}$ server. On CME the packed fraction is set by the gateway, is about 1%, and does not vary with the fitted n over the observed range; the fitted n describes the ordering of arrivals at longer timescales, consistent with Section 3.2, where the corpus is found to be self-exciting with heavy-tailed gaps rather than a pure exponential-kernel Hawkes process.

Feed dependence of the design rule. At $T = 8 \mu\text{s}$, $n = 0.8$, $\bar{\lambda} = 8000$, a two-stage tandem has the lower p_{99} on every synthetic condition: $97.1 \rightarrow 47.4$ without a floor, $7.56 \rightarrow 1.21$ with the fast kernel and floor, $3.12 \rightarrow 1.21$ with the slow kernel and floor. On CME at the same T , $N^* = 1$ (Section 4.8). The service time at which splitting begins to pay is not a constant; it is a property of the feed.

Summary. Whether splitting a stage reduces its tail depends on how many packets arrive closer together than the stage’s service time, not on the packet rate or the branching ratio as such. On CME about 1% of packet gaps fall at or below the $7.5 \mu\text{s}$ floor, the gateway’s minimum separation between distinct events. That fraction does not change when the market becomes busier, since a busier market produces more clusters with the same spacing inside them, nor when the fitted branching ratio moves within its observed range. At the 7 – $8 \mu\text{s}$ service time of a production decode path there is therefore little to queue behind, and one thread is the right design. On a feed that places a large share of its gaps at the floor, the result inverts: a Hawkes process at the same $n = 0.8$ places 28–81% of its gaps at the floor instead of 1%, the tail depends strongly on the branching ratio, splitting reduces the tail at $T = 8 \mu\text{s}$ where on CME it does not, and at high enough load the packet rate matters as well. The Poisson null shows only the load effect, and only at high utilisation, as a null should. The rule is therefore neither one thread nor two. The crossover is a property of the feed’s packet spacing, it cannot be read off the fitted parameters, and it must be measured under the load the system will see. The CME feed places this corpus below the crossover at $7 \mu\text{s}$; another exchange, or the same exchange after a gateway change, may not. Section 11 states this as a design principle.